

Task on Jenkins-04

1) Setup Jenkins CICD pipeline using freestyle jobs using Docker containers using below code.

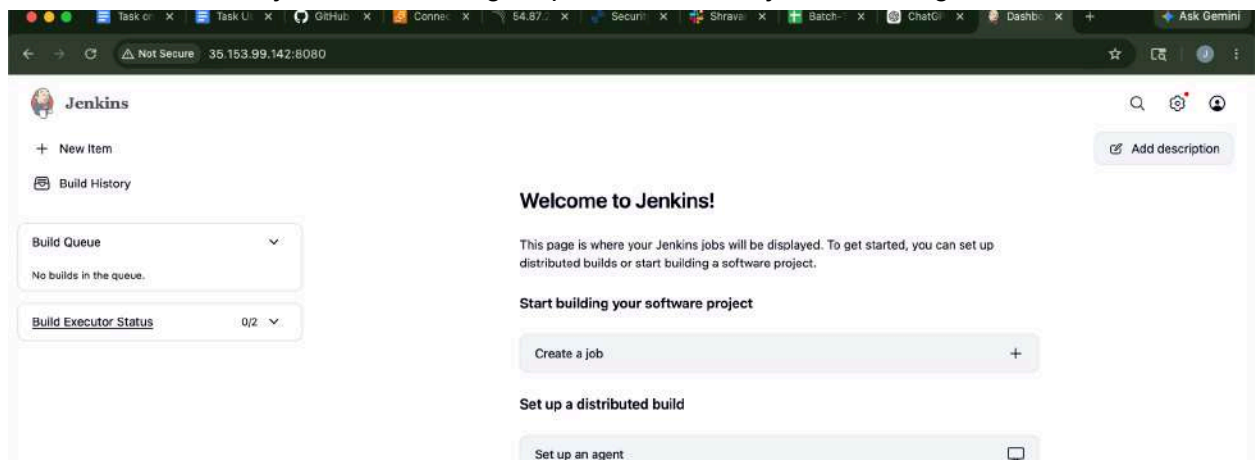
Source Code: <https://github.com/betawins/hiring-app.git>

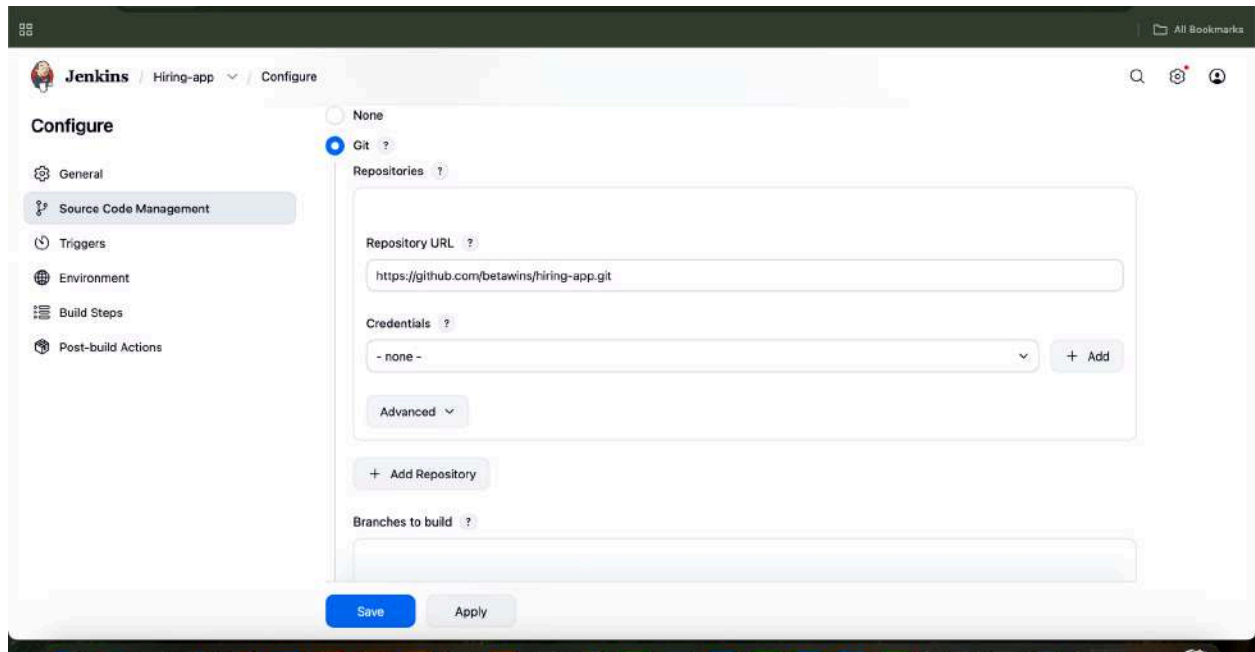
Stages:

- Git Clone
- SonarQube Integration
- Maven Compilation
- Nexus Artifactory
- Slack Notification
- Deploy On tomcat

GIT Clone:

We need to create a job and add the git repo and build the job for cloning





SonarQube Installation, SonarScanner Installation and Integration:

Install Docker using Yum install Docker

```
ds 0.0.0.0:32768->9000/tcp, :::32768->9000/tcp gifted_brattain
[[root@ip-172-31-1-253 ~]# docker --version
Docker version 25.0.14, build 0bab007
[root@ip-172-31-1-253 ~]#
```

Start Docker Using command

Systemctl start docker→To start docker

Systemctl status docker→for checking the status of docker whether its running or not.

```
oot@ip-172-31-1-253 ~]# systemctl start docker
oot@ip-172-31-1-253 ~]# systemctl status docker
docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; disabled; preset: >
   Active: active (running) since Fri 2026-05-15 17:10:32 UTC; 9s ago
   TriggeredBy: ● docker.socket
   Docs: https://docs.docker.com
   Process: 28068 ExecStartPre=/bin/mkdir -p /run/docker (code=exited, status=>
   Process: 28069 ExecStartPre=/usr/libexec/docker/docker-setup-runtimes.sh (c>
   Main PID: 28070 (dockerd)
   Tasks: 8
   Memory: 29.8M
   CPU: 216ms
   CGroup: /system.slice/docker.service
           └─28070 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/con>
```

Jenkins installed using the below command

```
[root@ip-172-31-0-116 ~]# docker rm -f jenkins
jenkins
[root@ip-172-31-0-116 ~]# docker run -d \
  --name jenkins \
  --restart unless-stopped \
  -p 8080:8080 \
  -p 50000:50000 \
  -u root \
  -v jenkins_home:/var/jenkins_home \
  -v /var/run/docker.sock:/var/run/docker.sock \
  jenkins/jenkins:lts
a6c7b11f7be943189c95d056fbbd01bf7c2a8a54f73cf7a9d0888759b8eb7d92
[root@ip-172-31-0-116 ~]# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
a6c7b11f7be9	jenkins/jenkins:lts	"/usr/bin/tini -- /u..."	20 seconds ago	Up 20 seconds

```
0.0.0.0:8080->8080/tcp, :::8080->8080/tcp, 0.0.0.0:50000->50000/tcp, :::50000->50000/tcp jenkins
[root@ip-172-31-0-116 ~]# cat /var/jenkins_home/secrets/initialAdminPassword
```

To check the password of jenkins in docker we will be using the below command

```
[root@ip-172-31-0-116 ~]# docker exec jenkins cat /var/jenkins_home/secrets/initialAdminPassword
d56fe401cce040448ca18df1b64bde53
[root@ip-172-31-0-116 ~]#
```

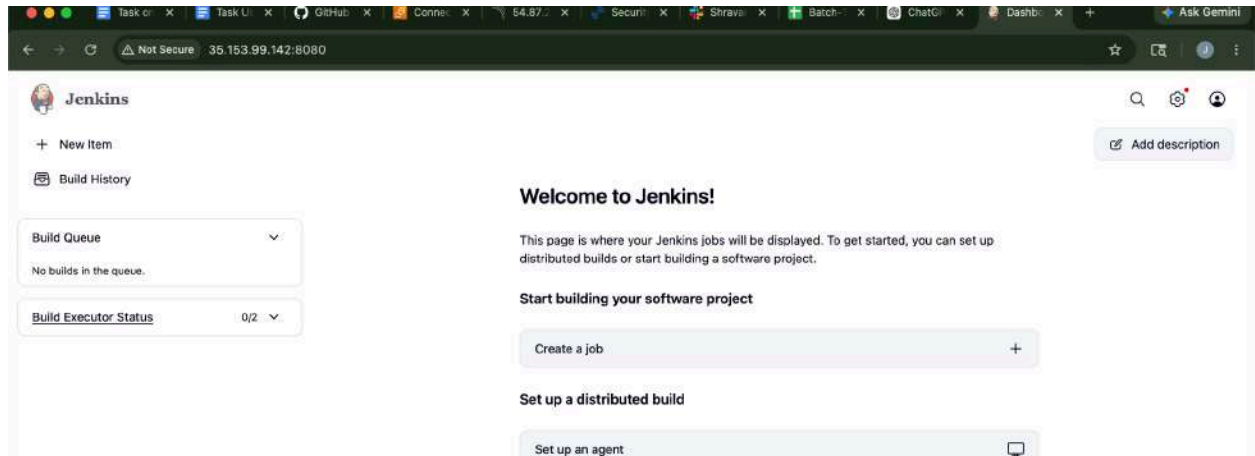
Jenkins is running in docker container and
We can see

To start Jenkins we need to use the below command

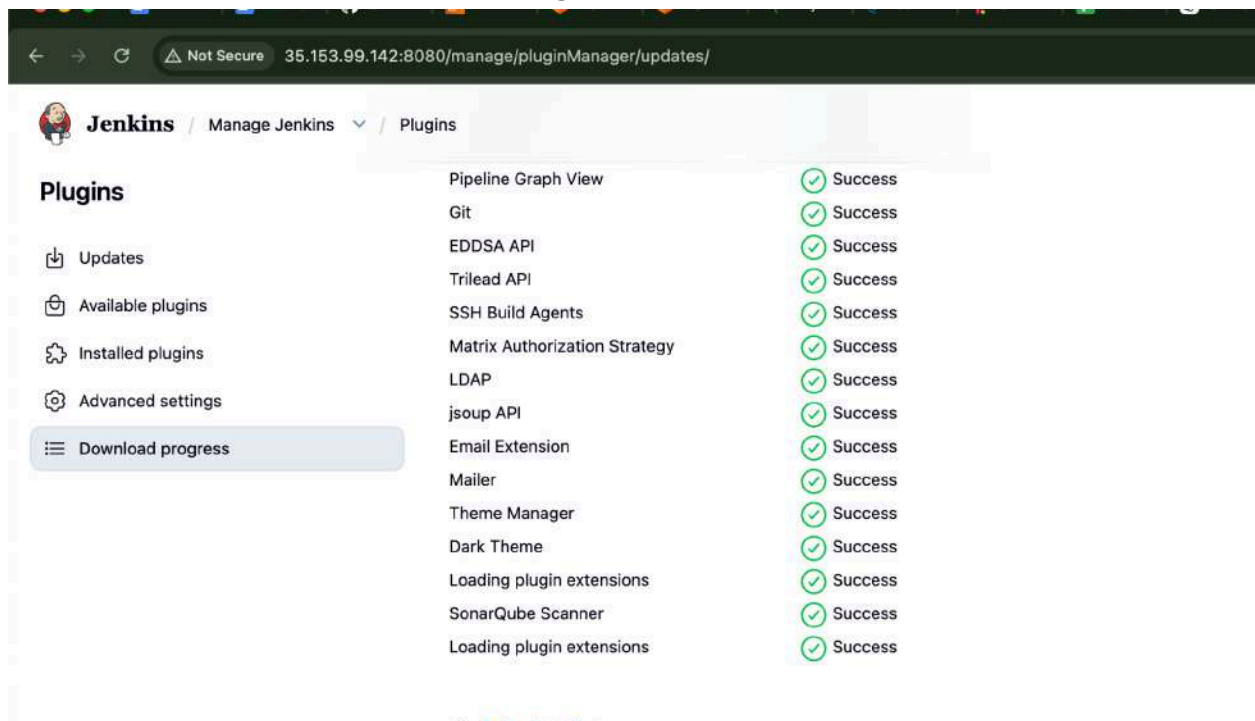
Docker start jenkins

Jenkins is running in docker container and by using <ipaddress>:8080

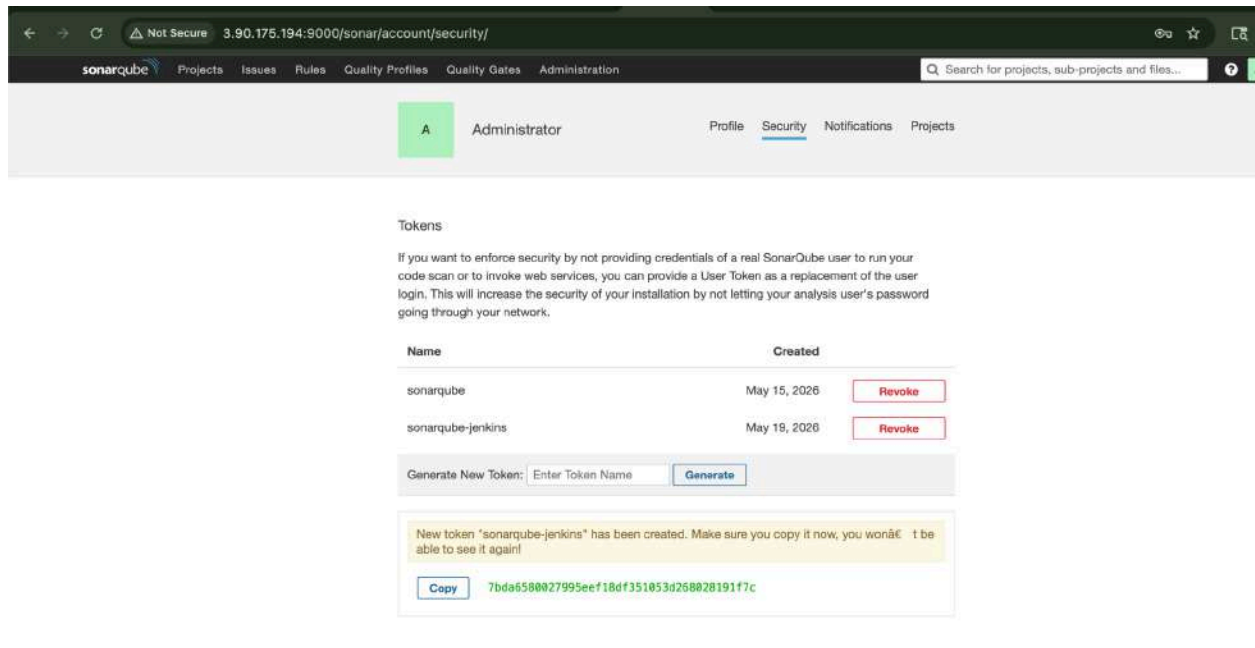
We can see



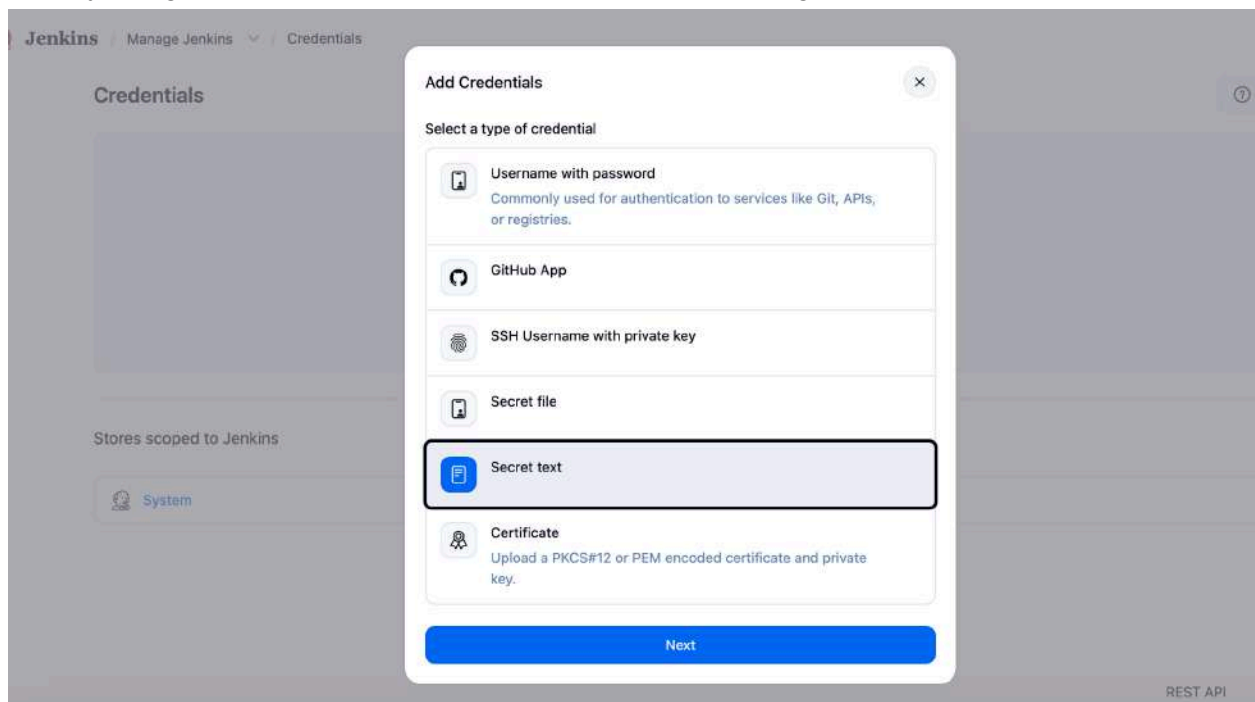
Now install the SonarQube Scanner plugins in Jenkins GUI



Now we add to add credentials to our JEnkins GUI of Sonarqube for that go Sonarqube GUI→account→security→generate token



Now by using the token created in SQ GUI we will be creating a credential in JENKINS GUI



Credential Created successfully



Need to install **SonarQube scanner inside** jenkins using the command below, has its a zip file we will be downloading , we need to unzip again

```
root@a6c7b11f7be9:/opt# ls
java jenkins-plugin-manager.jar
root@a6c7b11f7be9:/opt# curl -fL -o sonar-scanner.zip \
https://binaries.sonarsource.com/Distribution/sonar-scanner-cli/sonar-scanner-cli-5.0.1.3006-linux.zip
  % Total    % Received % Xferd Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100 44.9M  100 44.9M    0     0  354M      0 --:--:-- --:--:-- --:--:-- 356M
root@a6c7b11f7be9:/opt# unzip sonar-scanner.zip
Archive:  sonar-scanner.zip
  creating: sonar-scanner-5.0.1.3006-linux/
  creating: sonar-scanner-5.0.1.3006-linux/jre/
  creating: sonar-scanner-5.0.1.3006-linux/jre/lib/
  creating: sonar-scanner-5.0.1.3006-linux/jre/lib/security/
  creating: sonar-scanner-5.0.1.3006-linux/jre/lib/jfr/
  creating: sonar-scanner-5.0.1.3006-linux/jre/lib/server/
```

We need to edit the sonar-scanner.properties has it does not have the edit option we will download the edit packages using commands below

Apt-get update

apt-get install vim -y

```
root@a6c7b11f7be9:~# apt-get install vim -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package vim
root@a6c7b11f7be9:~# apt-get update
Get:1 http://deb.debian.org/debian trixie InRelease [140 kB]
Get:2 http://deb.debian.org/debian trixie-updates InRelease [47.3 kB]
Get:3 http://deb.debian.org/debian-security trixie-security InRelease [43.4 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 Packages [9671 kB]
Get:5 http://deb.debian.org/debian trixie-updates/main amd64 Packages [5412 B]
Get:6 http://deb.debian.org/debian-security trixie-security/main amd64 Packages [165 kB]
Fetched 10.1 MB in 1s (14.1 MB/s)
Reading package lists... Done
root@a6c7b11f7be9:~# apt-get install vim -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libgpm2 libsodium23 vim-common vim-runtime xxd
Suggested packages:
  gpm ctags vim-doc vim-scripts
The following NEW packages will be installed:
  libgpm2 libsodium23 vim vim-common vim-runtime xxd
0 upgraded, 6 newly installed, 0 to remove and 20 not upgraded.
Need to get 9496 kB of archives.
After this operation, 45.9 MB of additional disk space will be used.
Get:1 http://deb.debian.org/debian trixie/main amd64 vim-common all 2:9.1.1230-2 [421 kB]
Get:2 http://deb.debian.org/debian trixie/main amd64 libgpm2 amd64 1.20.7-11+b2 [14.4 kB]
```

i-050b81a849ad207a2 (Docker)

PublicIPs: 35.153.99.142 PrivateIPs: 172.31.0.116

Will edit and add the path in sonar properties


```
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1

aws Search [Option+S] Ask Amazon Q

#Configure here general information about the environment, such as SonarQube server
#No information about specific project should appear here

#----- Default SonarQube server
sonar.host.url=http://3.90.175.194:9000/sonar

#----- Default source code encoding
#sonar.sourceEncoding=UTF-8

-- INSERT --
```

i-050b81a849ad207a2 (Docker)

PublicIPs: 35.153.99.142 PrivateIPs: 172.31.0.116

Need to give the Path in Jenkins GUI→Tools

← → ↻ ⚠ Not Secure 35.153.99.142:8080/manage/configureTools/

 **Jenkins** / Manage Jenkins ▾ / Tools

SonarScanner for MSBuild installations

+ Add SonarScanner for MSBuild

SonarQube Scanner installations

+ Add SonarQube Scanner

⋮ SonarQube Scanner

Name

sonar-scanner

SONAR_RUNNER_HOME

/opt/sonar-scanner

☐ Install automatically ?

Save Apply App Store

You can get the path from jenkins cli in docker

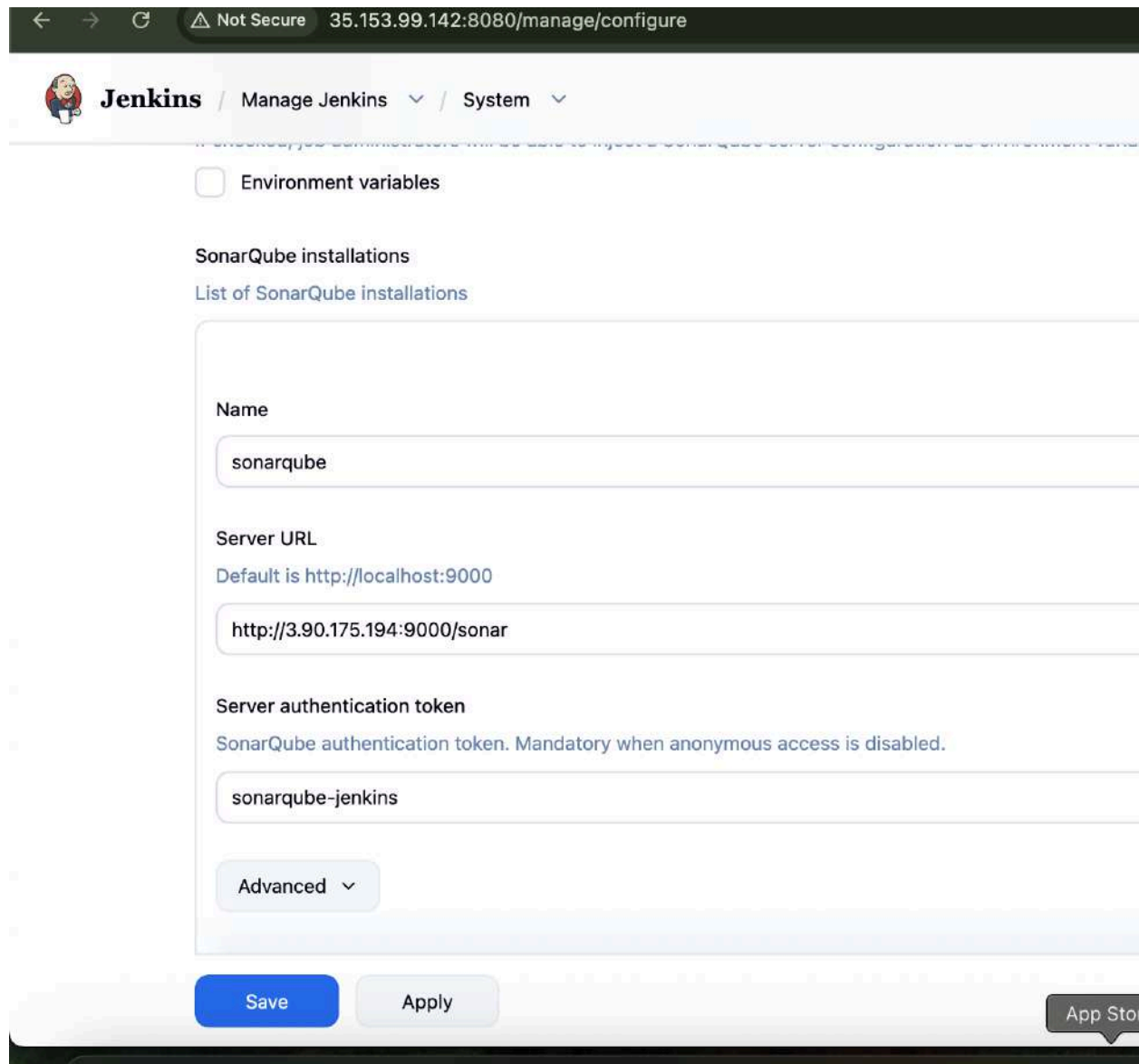
```
root@a6c7b11f7be9:/opt/sonar-scanner/conf# cd
root@a6c7b11f7be9:~# cd /opt/sonar-scanner/bin
root@a6c7b11f7be9:/opt/sonar-scanner/bin# pwd
/opt/sonar-scanner/bin
root@a6c7b11f7be9:/opt/sonar-scanner/bin#
```

i-050b81a849ad207a2 (Docker)

PublicIPs: 35.153.99.142 PrivateIPs: 172.31.0.116

SONARQUBE INTEGRATION

We also need to add sonarqube in managejenkins→system→sonarqube

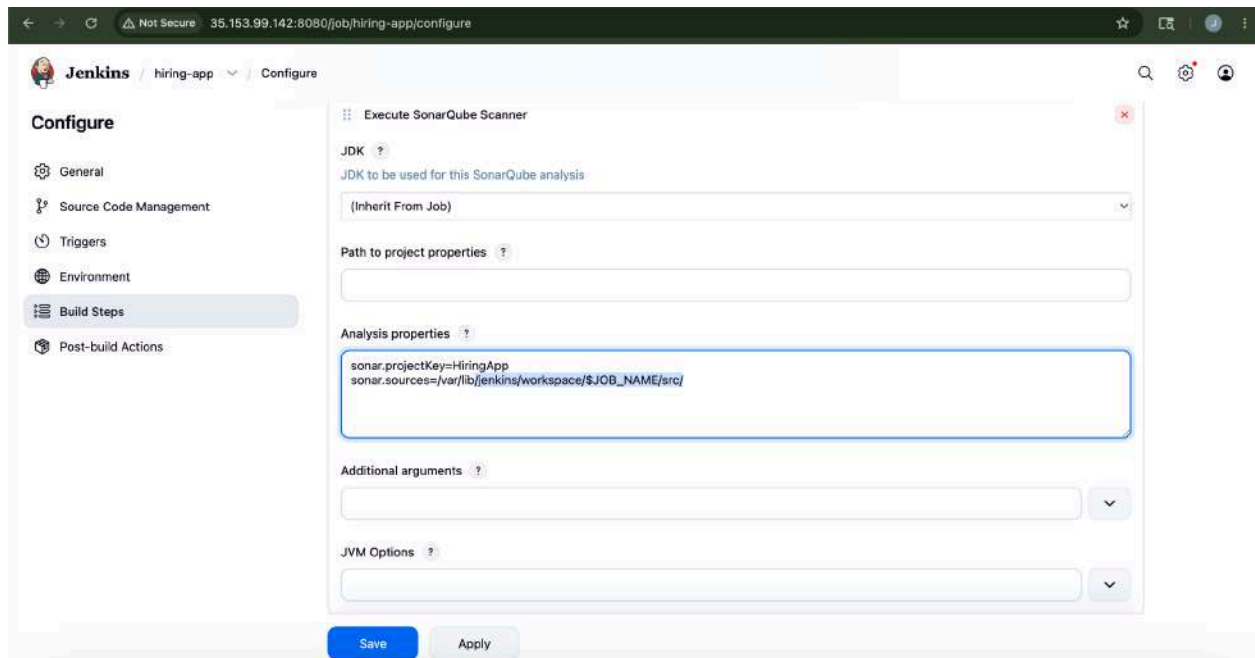


The screenshot shows the Jenkins web interface at the URL `35.153.99.142:8080/manage/configure`. The breadcrumb navigation is `Jenkins / Manage Jenkins / System`. Under the `System` tab, the `Environment variables` checkbox is unchecked. The `SonarQube installations` section includes a link to `List of SonarQube installations`. A configuration form for a new installation is visible with the following fields:

- Name:** `sonarqube`
- Server URL:** `http://3.90.175.194:9000/sonar` (Default is `http://localhost:9000`)
- Server authentication token:** `sonarqube-jenkins` (SonarQube authentication token. Mandatory when anonymous access is disabled.)
- Advanced:** A dropdown menu is currently set to `Advanced`.

At the bottom of the form are `Save` and `Apply` buttons. An `App Store` badge is visible in the bottom right corner of the browser window.

Created a new Job using analysis properties from the git repo,
In the new , we will be doing the **GIT CLONE**,



Now The new job is working and has been executed and we can see the result in SQ Dashboard

Maven Compilation:

Need to install maven in jenkins server using wget

```

[root@ip-172-31-1-127 opt]# wget https://d1cdn.apache.org/maven/maven-3/3.9.16/binaries/apache-maven-3.9.16-bin.tar.gz
--2026-05-19 13:28:31-- https://d1cdn.apache.org/maven/maven-3/3.9.16/binaries/apache-maven-3.9.16-bin.tar.gz
Resolving d1cdn.apache.org (d1cdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to d1cdn.apache.org (d1cdn.apache.org)[151.101.2.132]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9278065 (8.8M) [application/x-gzip]
Saving to: 'apache-maven-3.9.16-bin.tar.gz'

apache-maven-3.9.16-bin.tar.gz      100%[=====] 8.85M  --.-KB
2026-05-19 13:28:31 (466 MB/s) - 'apache-maven-3.9.16-bin.tar.gz' saved [9278065/9278065]

[root@ip-172-31-1-127 opt]# ls
apache-maven-3.9.16-bin.tar.gz  aws  sonar-scan  sonar-scanner
[root@ip-172-31-1-127 opt]# tar xvf apache-maven-3.9.16-bin.tar.gz
apache-maven-3.9.16/README.txt
apache-maven-3.9.16/LICENSE
apache-maven-3.9.16/NOTICE
apache-maven-3.9.16/lib/
apache-maven-3.9.16/lib/aopalliance.license
apache-maven-3.9.16/lib/asm.license
apache-maven-3.9.16/lib/commons-cli.license
apache-maven-3.9.16/lib/commons-codec.license

```

i-0c8b7f0daac46fa4f (Jenkins-Master)

PublicIPs: 54.161.134.10 PrivateIPs: 172.31.1.127

After installing and extracting the files now we need to come home directory(~) and from their go vi .bash_profile as we need to give the variables

```

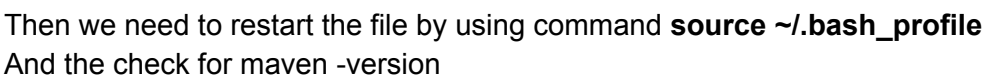
[root@ip-172-31-1-127 ~]# vi .bash_profile
[root@ip-172-31-1-127 ~]#

```

i-0c8b7f0daac46fa4f (Jenkins-Master)

PublicIPs: 54.161.134.10 PrivateIPs: 172.31.1.127

Below are variables to be given inside the above file



Then we need to restart the file by using command **source ~/.bash_profile**
And the check for maven -version


```
[root@ip-172-31-1-127 ~]# mvn -version
Apache Maven 3.9.16 (2bdd9fddda4b155ebf8000e807eb73fd829a51d5)
Maven home: /opt/apache-maven
Java version: 21.0.11, vendor: Amazon.com Inc., runtime: /usr/lib/jvm/java-21-amazon-corretto.x86_64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "6.1.170-213.321.amzn2023.x86_64", arch: "amd64", family: "unix"
[root@ip-172-31-1-127 ~]# vi .bash_profile
[root@ip-172-31-1-127 ~]#
```

echo \$M2_HOME → To check the path

```
[root@ip-172-31-1-127 ~]# vi .bash_profile
[root@ip-172-31-1-127 ~]# echo $M2_HOME
/opt/apache-maven
[root@ip-172-31-1-127 ~]#
```

i-0c8b7f0daac46fa4f (Jenkins-Master)

PublicIPs: 54.161.134.10 PrivateIPs: 172.31.1.127

Maven Integration

Now we add to add the path in Jenkins GUI→Tools→Maven Installation

← → ↻ ⚠ Not Secure 3.82.241.31:8080/manage/configureTools/

 **Jenkins** / Manage Jenkins ▾ / Tools

Maven installations

Maven installations ^

✎ Edited

+ Add Maven

⋮ Maven

Name

M2_HOME

MAVEN_HOME

/opt/apache-maven

☐ Install automatically ?

+ Add Maven

Save

Apply

Now we need to also add in Build step In the job created regarding the maven details
Jenkins GUI→Job→configure→build actions→M2_HOME→clean Install(Will delete if any
existing and create new file)

← → ↻ Not Secure 3.82.241.31:8080/job/Hiring%20App/configure

Jenkins / Hiring App ▾ / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions**

Additional arguments ?

JVM Options ?

Invoke top-level Maven targets ?

Maven Version

M2_HOME

Goals

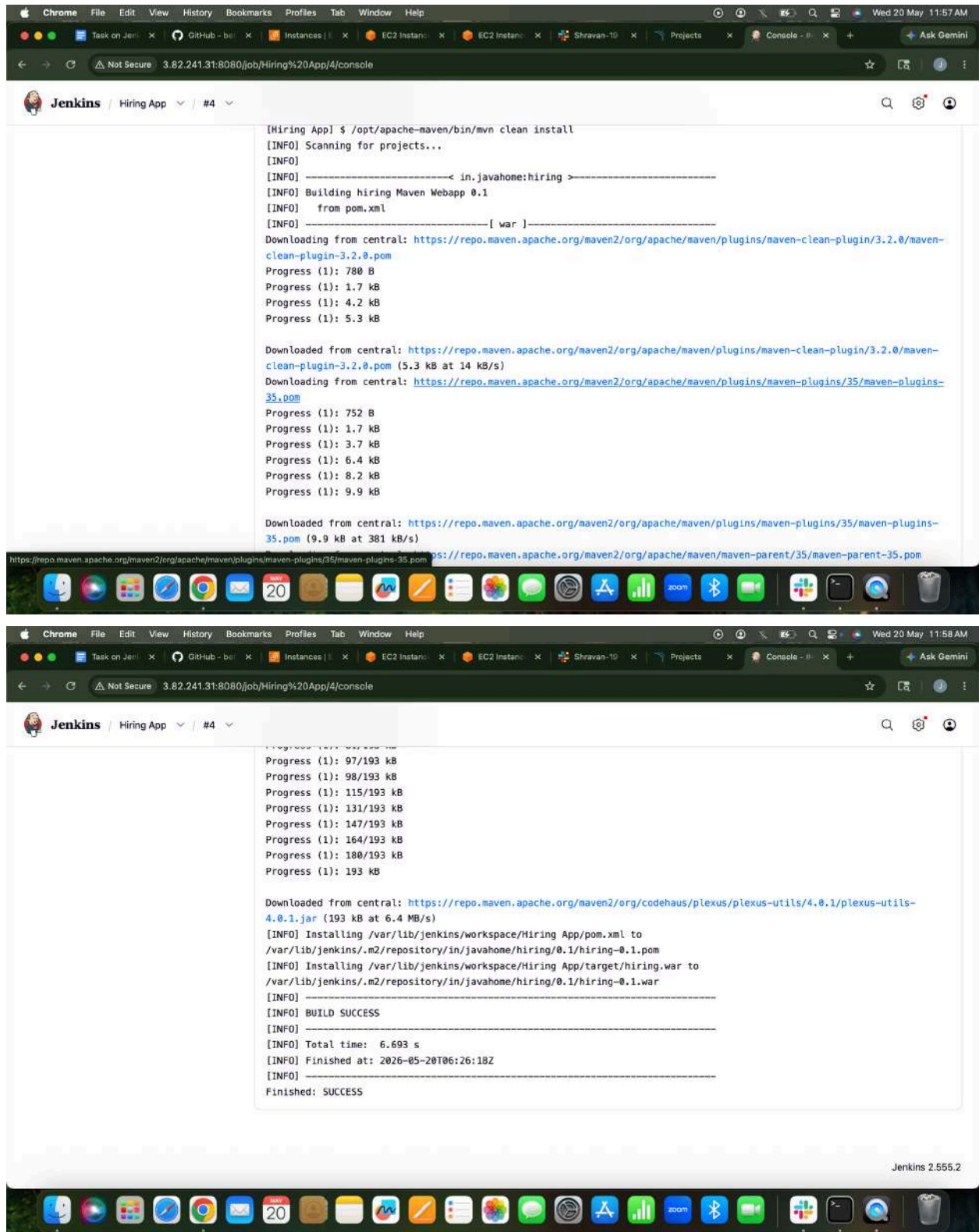
clean_install

Advanced ▾

+ Add build step

Save Apply Fac

Now Build the Job



Job run successfully

Now goto Jenkins CLI and check if the war files have been created or not
The compilation run successfully will create a target file which stores the dependencies/ war packages

Inn Jenkins server

Cd /var/lib/jenkins/workspace/target/.war files → Here target file is created after successfully compilation and inside that war files will be created

```
[root@ip-172-31-1-127 ~]# cd /var/lib/jenkins/workspace/
[root@ip-172-31-1-127 workspace]# ls
'Hiring App'
[root@ip-172-31-1-127 workspace]# cd Hiring\ App/
[root@ip-172-31-1-127 Hiring App]# ls
Dockerfile  Jenkinsfile  README.md  'Untitled Diagram.drawio'  jenkinsfile-cicd  pom.xml  src  target
[root@ip-172-31-1-127 Hiring App]# cd target/
[root@ip-172-31-1-127 target]# ls
Hiring  hiring.war  maven-archiver
[root@ip-172-31-1-127 target]#
```

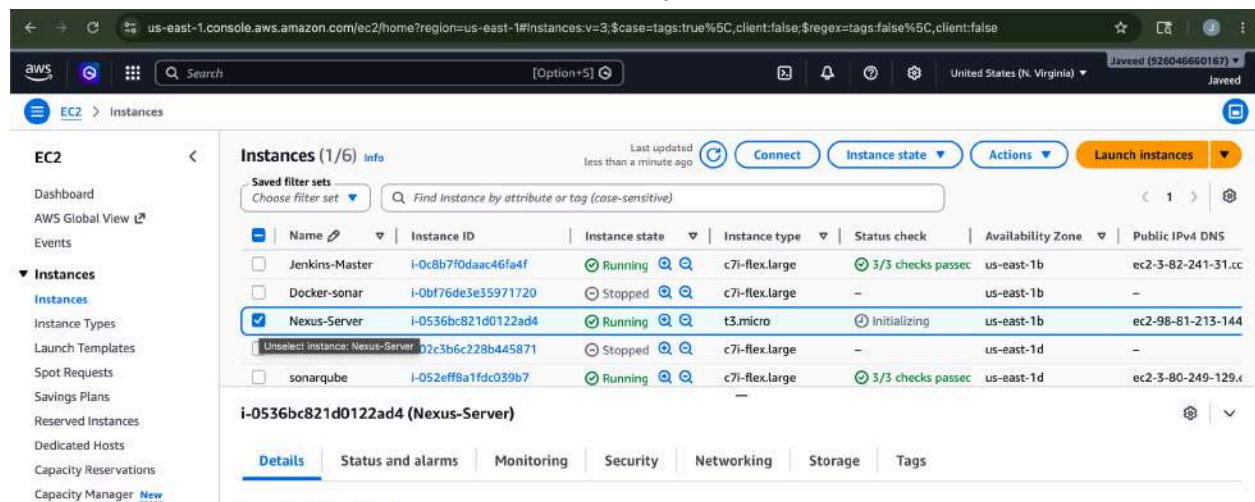
i-0c8b7f0daac46fa4f (Jenkins-Master)
PublicIPs: 3.82.241.31 PrivateIPs: 172.31.1.127

Nexus Installation and Integration: → this is used to save the artifactory

Nexus will only run on ec2-user

Create a new instance

Java is the prerequisite for Nexus so we will install java too



Yum install java-1.8* -y installed successfully

```
[ec2-user@ip-172-31-1-158 ~]$ sudo yum install java-1.8* -y
Last metadata expiration check: 0:00:27 ago on Wed May 20 06:47:11 2026.
Dependencies resolved.
=====
Package                                     Architecture      Version            Repository          Size
-----
Installing:
java-1.8.0-amazon-corretto                 x86_64            1:1.8.0_492.b09-1.amzn2023    amazonlinux          38 M
java-1.8.0-amazon-corretto-debugsymbols    x86_64            1:1.8.0_492.b09-1.amzn2023    amazonlinux          155 M
java-1.8.0-amazon-corretto-devel           x86_64            1:1.8.0_492.b09-1.amzn2023    amazonlinux           63 M
Installing dependencies:
adwaita-cursor-theme                       noarch            47.0-1.amzn2023.0.1          amazonlinux           325 k
adwaita-icon-theme                         noarch            47.0-1.amzn2023.0.1          amazonlinux           285 k
adwaita-icon-theme-legacy                  noarch            46.2-2.amzn2023              amazonlinux           2.2 M
alsa-lib                                    x86_64            1.2.7.2-1.amzn2023.0.3       amazonlinux           503 k
=====

i-0536bc821d0122ad4 (Nexus-Server)
PublicIPs: 98.81.213.144  PrivateIPs: 172.31.1.158
```

```
wireplumber-libs-0.5.7-1.amzn2023.0.1.x86_64
xdg-desktop-portal-gtk-1.15.1-61.amzn2023.x86_64
xml-common-0.6.3-56.amzn2023.0.2.noarch

Complete!
[ec2-user@ip-172-31-1-158 ~]$
```

i-0536bc821d0122ad4 (Nexus-Server)

PublicIPs: 98.81.213.144 PrivateIPs: 172.31.1.158

By usin command

curl -L -O https://download.sonatype.com/nexus/3/nexus-3.92.2-01-linux-x86_64.tar.gz

We will be installing nexus → this creates a tar file (cd /opt) → untar the file

```
[ec2-user@ip-172-31-1-158 opt]$ curl -L -O https://download.sonatype.com/nexus/3/nexus-3.92.2-01-linux-x86_64.tar.gz
% Total % Received % Xferd Average Speed Time Time Time Current
 Dload Upload Total Spent Left Speed
100 138 100 138 0 0 607 0 --:--:-- --:--:-- --:--:-- 610
Warning: Failed to open the file nexus-3.92.2-01-linux-x86_64.tar.gz: Permission denied
0 478.5M 0 0 0 0 0 0 --:--:-- --:--:-- --:--:-- 0
curl: (23) client returned ERROR on write of 1360 bytes
[ec2-user@ip-172-31-1-158 opt]$
```

i-0536bc821d0122ad4 (Nexus-Server)

PublicIPs: 98.81.213.144 PrivateIPs: 172.31.1.158

Now we will untar the file


```
[ec2-user@ip-172-31-1-158 opt]$ sudo curl -L -O https://download.sonatype.com/nexus/3/nexus-3.92.2-01-linux-x86_64.tar.gz
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 138 100 138 0 0 607 0 --:--:-- --:--:-- --:--:-- 607
100 478.5M 100 478.5M 0 0 194.8M 0 0:00:02 0:00:02 --:--:-- 194.1M
[ec2-user@ip-172-31-1-158 opt]$ ls
aws nexus-3.92.2-01-linux-x86_64.tar.gz
[ec2-user@ip-172-31-1-158 opt]$ sudo tar -xvzf nexus-3.92.2-01-linux-x86_64.tar.gz
nexus-3.92.2-01/bin/nexus
nexus-3.92.2-01/bin/nexus.vmoptions
nexus-3.92.2-01/
nexus-3.92.2-01/etc/
nexus-3.92.2-01/etc/ssl/
nexus-3.92.2-01/etc/logback/
nexus-3.92.2-01/etc/jetty/
nexus-3.92.2-01/etc/fabric/
nexus-3.92.2-01/etc/spring/
nexus-3.92.2-01/deploy/
nexus-3.92.2-01/bin/
sonatype-work/
sonatype-work/nexus3/
sonatype-work/nexus3/tmp/
sonatype-work/nexus3/log/
nexus-3.92.2-01/etc/nexus-default.properties
nexus-3.92.2-01/etc/ssl/.placeholder
nexus-3.92.2-01/etc/default-application.properties
nexus-3.92.2-01/etc/logback/logback.xml
```

i-0536bc821d0122ad4 (Nexus-Server)

PublicIPs: 98.81.213.144 PrivateIPs: 172.31.1.158

Now we need to create an ec2-user or else it will throw an error as below

```
[ec2-user@ip-172-31-1-158 opt]$ sudo curl -L -O https://download.sonatype.com/nexus/3/nexus-3.92.2-01-linux-x86_64.tar.gz
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 138 100 138 0 0 607 0 --:--:-- --:--:-- --:--:-- 607
100 478.5M 100 478.5M 0 0 194.8M 0 0:00:02 0:00:02 --:--:-- 194.1M
[ec2-user@ip-172-31-1-158 opt]$ ls
aws nexus-3.92.2-01-linux-x86_64.tar.gz
[ec2-user@ip-172-31-1-158 opt]$ sudo tar -xvzf nexus-3.92.2-01-linux-x86_64.tar.gz
nexus-3.92.2-01/bin/nexus
nexus-3.92.2-01/bin/nexus.vmoptions
nexus-3.92.2-01/
nexus-3.92.2-01/etc/
nexus-3.92.2-01/etc/ssl/
nexus-3.92.2-01/etc/logback/
nexus-3.92.2-01/etc/jetty/
nexus-3.92.2-01/etc/fabric/
nexus-3.92.2-01/etc/spring/
nexus-3.92.2-01/deploy/
nexus-3.92.2-01/bin/
sonatype-work/
sonatype-work/nexus3/
sonatype-work/nexus3/tmp/
sonatype-work/nexus3/log/
nexus-3.92.2-01/etc/nexus-default.properties
nexus-3.92.2-01/etc/ssl/.placeholder
nexus-3.92.2-01/etc/default-application.properties
nexus-3.92.2-01/etc/logback/logback.xml
```

i-0536bc821d0122ad4 (Nexus-Server)

PublicIPs: 98.81.213.144 PrivateIPs: 172.31.1.158

We need to create file and give it the necessary permissions as below
Ccd /opt/nexus/bin→vi [nexus.rc](#)

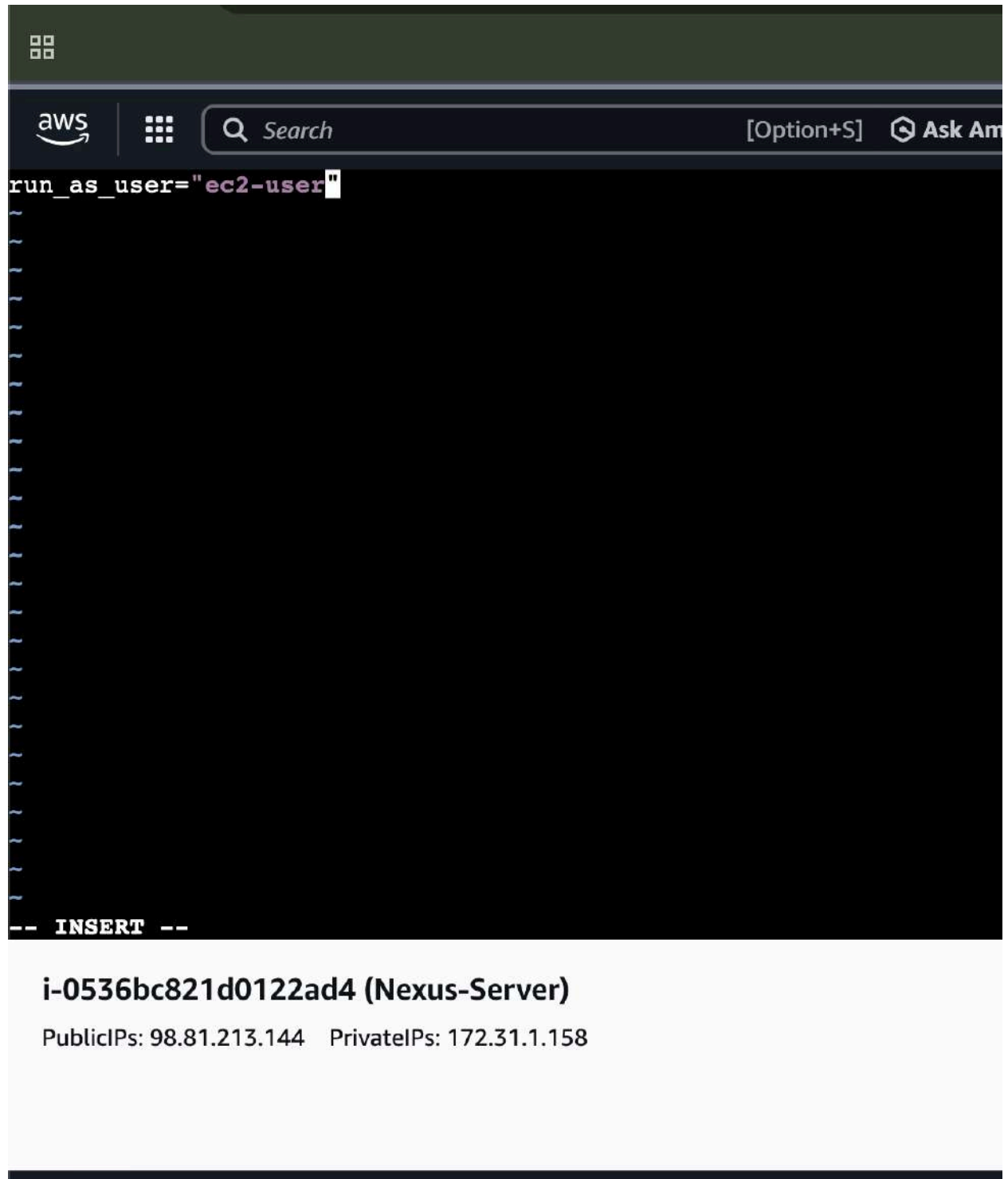
Inside → run_as_user="ec2-user"

```
[ec2-user@ip-172-31-1-158 opt]$ sudo su -
[root@ip-172-31-1-158 ~]# cd /opt/
[root@ip-172-31-1-158 opt]# ls
aws  nexus  nexus-3.92.2-01-linux-x86_64.tar.gz  sonatype-work
[root@ip-172-31-1-158 opt]# cd nexus/
[root@ip-172-31-1-158 nexus]# ls
NOTICE.txt  OSS-LICENSE.txt  bin  deploy  etc  hs_err_pid28145.log  jdk
[root@ip-172-31-1-158 nexus]# cd bin/
[root@ip-172-31-1-158 bin]# ls
nexus  nexus.vmoptions  sonatype-nexus-repository-3.92.2-01.jar
[root@ip-172-31-1-158 bin]# vi nexus.rc
[root@ip-172-31-1-158 bin]#
```

i-0536bc821d0122ad4 (Nexus-Server)

PublicIPs: 98.81.213.144 PrivateIPs: 172.31.1.158

So now lets create a new file and



So now lets create a new file and give it permissions to run on ec2-user

```
[ec2-user@ip-172-31-1-158 bin]$ cd
[ec2-user@ip-172-31-1-158 ~]$ cd /opt/
[ec2-user@ip-172-31-1-158 opt]$ ls -ltr
total 490060
drwxr-xr-x. 3 root root      20 May 13 19:19 sonatype-work
drwxr-xr-x. 4 root root      33 May 15 00:26 aws
-rw-r--r--. 1 root root 501818261 May 20 07:00 nexus-3.92.2-01-linux-x86_64.tar.gz
drwxr-xr-x. 6 root root      121 May 20 07:04 nexus
[ec2-user@ip-172-31-1-158 opt]$
```

i-0536bc821d0122ad4 (Nexus-Server)

PublicIPs: 98.81.213.144 PrivateIPs: 172.31.1.158

CloudShell Feedback Console Mobile App

© 2025, Amazon Web Services, Inc.

Using the command

`sudo chown -R ec2-user:ec2-user sonatype-work`

`sudo chown -R ec2-user:ec2-user nexus`

```
[ec2-user@ip-172-31-1-158 opt]$ sudo chown -R ec2-user:ec2-user sonatype-work
[ec2-user@ip-172-31-1-158 opt]$ sudo chown -R ec2-user:ec2-user nexus
[ec2-user@ip-172-31-1-158 opt]$ ls -ltr
total 490060
drwxr-xr-x. 3 ec2-user ec2-user      20 May 13 19:19 sonatype-work
drwxr-xr-x. 4 root      root          33 May 15 00:26 aws
-rw-r--r--. 1 root      root 501818261 May 20 07:00 nexus-3.92.2-01-linux-x86_64.tar.gz
drwxr-xr-x. 6 ec2-user ec2-user      121 May 20 07:04 nexus
[ec2-user@ip-172-31-1-158 opt]$
```

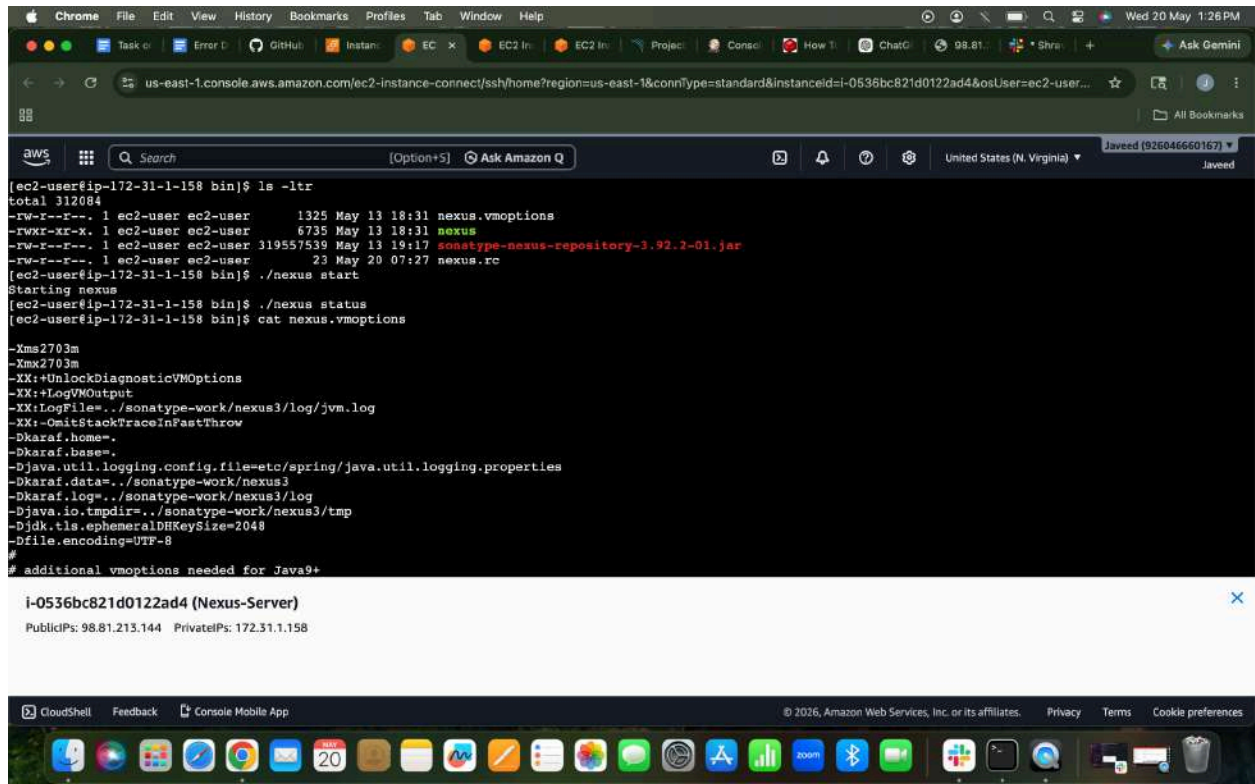
i-0536bc821d0122ad4 (Nexus-Server)

PublicIPs: 98.81.213.144 PrivateIPs: 172.31.1.158

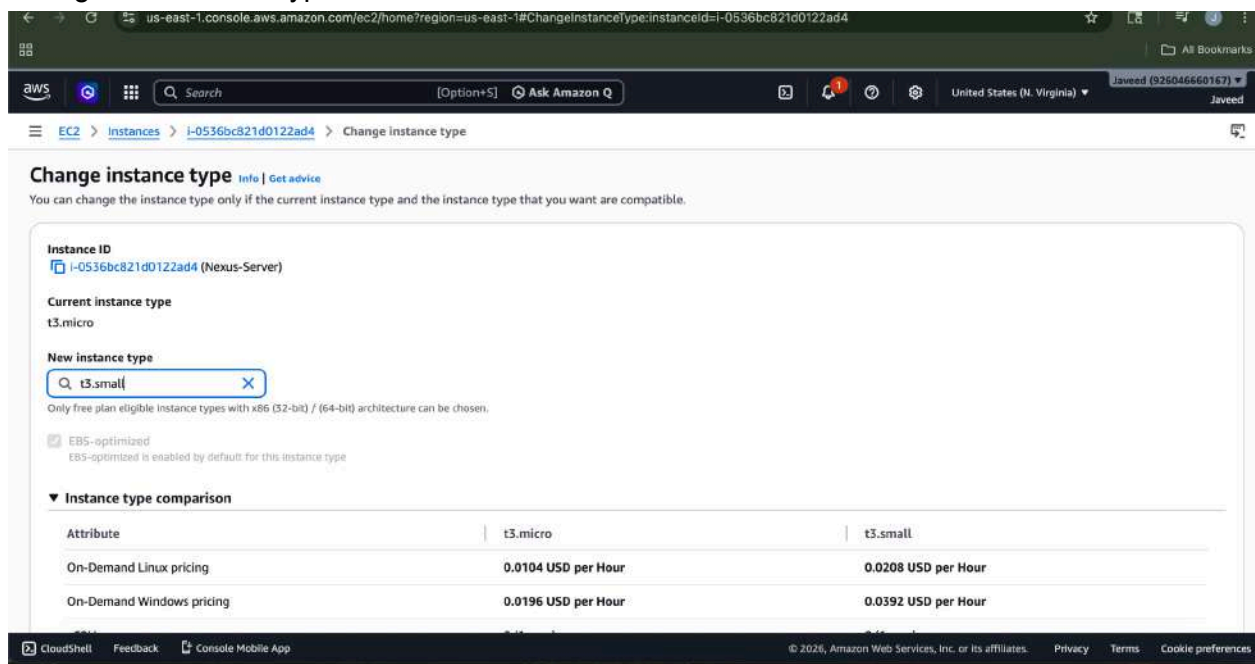
Now lets run

`Cd /opt/nexus/bin → ./nexus start` and `./nexus status` as we can see the service is running and it's getting excited immediately. The reason is due to the ram consumption so lets check the ram size and change it accordingly. To change the ram size we need to got the file called

⇒ **vi nexus.vmoptions**



Change the instance type



Vi nexus.vmoptions

Change -xms2703m ⇒ xms768m

```
← → ↻ us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-  
aws [Option+S] Ask Amazon Q  
-Xms2703m  
-Xmx2703m  
-XX:+UnlockDiagnosticVMOptions  
-XX:+LogVMOutput  
-XX:LogFile=../sonatype-work/nexus3/log/jvm.log  
-XX:-OmitStackTraceInFastThrow  
-Dkaraf.home=.  
-Dkaraf.base=.  
-Djava.util.logging.config.file=etc/spring/java.util.logging.properties
```

```
aws [Option+S] Ask Amazon Q  
-Xms768m  
-Xmx768m  
-XX:+UnlockDiagnosticVMOptions  
-XX:+LogVMOutput  
-XX:LogFile=../sonatype-work/nexus3/log/jvm.log  
-XX:-OmitStackTraceInFastThrow  
-Dkaraf.home=.  
-Dkaraf.base=.  
-Djava.util.logging.config.file=etc/spring/java.util.logging.properties  
-Dkaraf.data=../sonatype-work/nexus3  
-Dkaraf.log=../sonatype-work/nexus3/log  
-Dfile.encoding=UTF-8  
#  
# additional vmoptions needed for Java9+  
#  
--add-reads=java.xml=java.logging  
--add-opens  
java.base/java.security=ALL-UNNAMED
```

Now save the file and delete the tmp files using command in `cd /opt/sona/nexus3/tmp`
Do the following steps by using root user—

```
rm -rf /opt/sonatype-work/nexus3/tmp/*  
rm -rf /opt/sonatype-work/nexus3/log/*.lck  
rm -rf /opt/sonatype-work/nexus3/clean_cache/*  
And also we will be deleting the log files created by JVM using command  
rm -f /opt/nexus/hs_err_pid*.log
```

```

last login: Wed May 20 06:40:23 2020 from 10.200.107.20
[ec2-user@ip-172-31-1-158 ~]$ cd /opt/
[ec2-user@ip-172-31-1-158 opt]$ ls
aws nexus nexus-3.92.2-01-linux-x86_64.tar.gz sonatype-work
[ec2-user@ip-172-31-1-158 opt]$ cd nexus/
[ec2-user@ip-172-31-1-158 nexus]$ cd
[ec2-user@ip-172-31-1-158 ~]$ cd /opt/sonatype-work/
[ec2-user@ip-172-31-1-158 sonatype-work]$ ls
nexus3
[ec2-user@ip-172-31-1-158 sonatype-work]$ cd nexus3/
[ec2-user@ip-172-31-1-158 nexus3]$ ls
blobs clean_cache db downloads etc keystores log nexus.pid restore-from-backup tmp
[ec2-user@ip-172-31-1-158 nexus3]$ sudo rm -rf /opt/sonatype-work/nexus3/tmp/*
[ec2-user@ip-172-31-1-158 nexus3]$ sudo rm -rf /opt/sonatype-work/nexus3/log/*.lck
[ec2-user@ip-172-31-1-158 nexus3]$ cd ..
[ec2-user@ip-172-31-1-158 sonatype-work]$ cd ..
[ec2-user@ip-172-31-1-158 opt]$ ls
aws nexus nexus-3.92.2-01-linux-x86_64.tar.gz sonatype-work
[ec2-user@ip-172-31-1-158 opt]$ cd nexus/
[ec2-user@ip-172-31-1-158 nexus]$ ls
NOTICE.txt OSS-LICENSE.txt bin deploy etc hs_err_pid28145.log hs_err_pid29575.log hs_err_pid30285.log jdk
[ec2-user@ip-172-31-1-158 nexus]$ sudo rm -rf /opt/nexus/hs_err_pid*.log
[ec2-user@ip-172-31-1-158 nexus]$ ls
NOTICE.txt OSS-LICENSE.txt bin deploy etc jdk
[ec2-user@ip-172-31-1-158 nexus]$

```

i-0536bc821d0122ad4 (Nexus-Server)

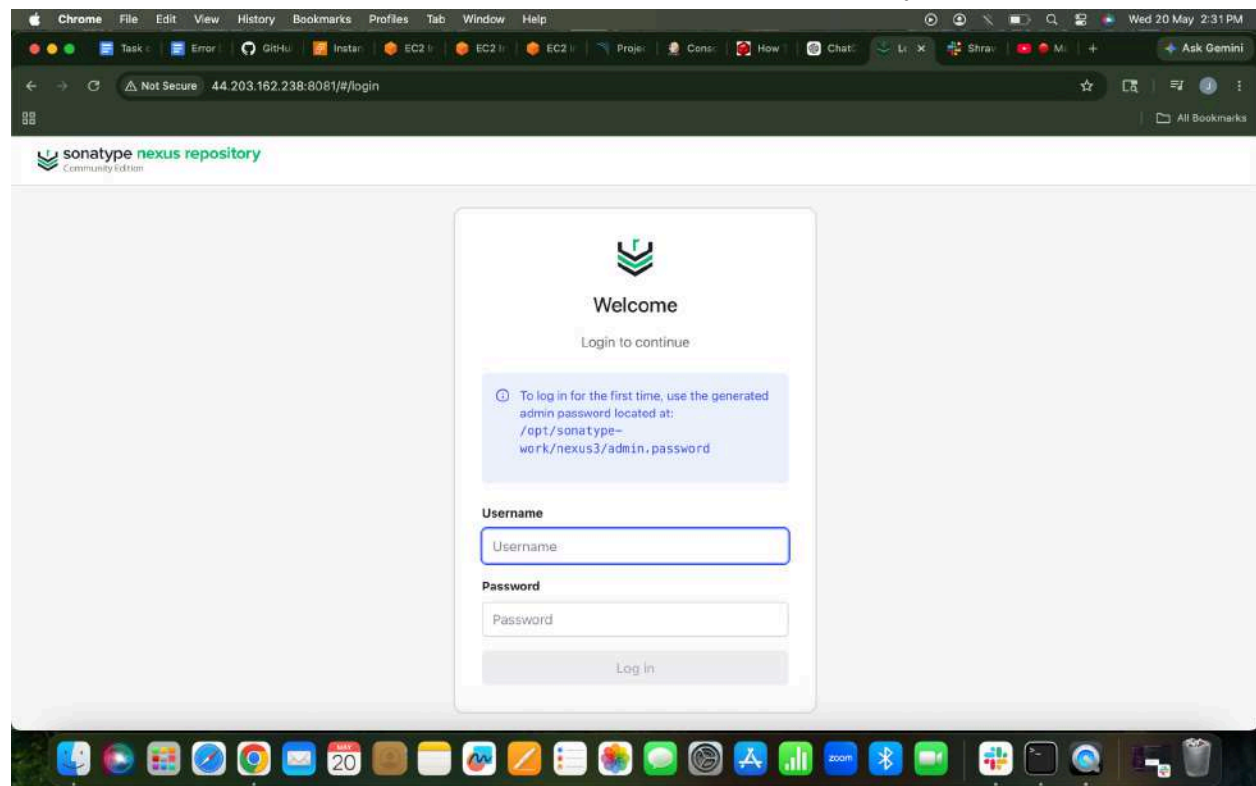
PublicIPs: 44.203.162.238 PrivateIPs: 172.31.1.158

```

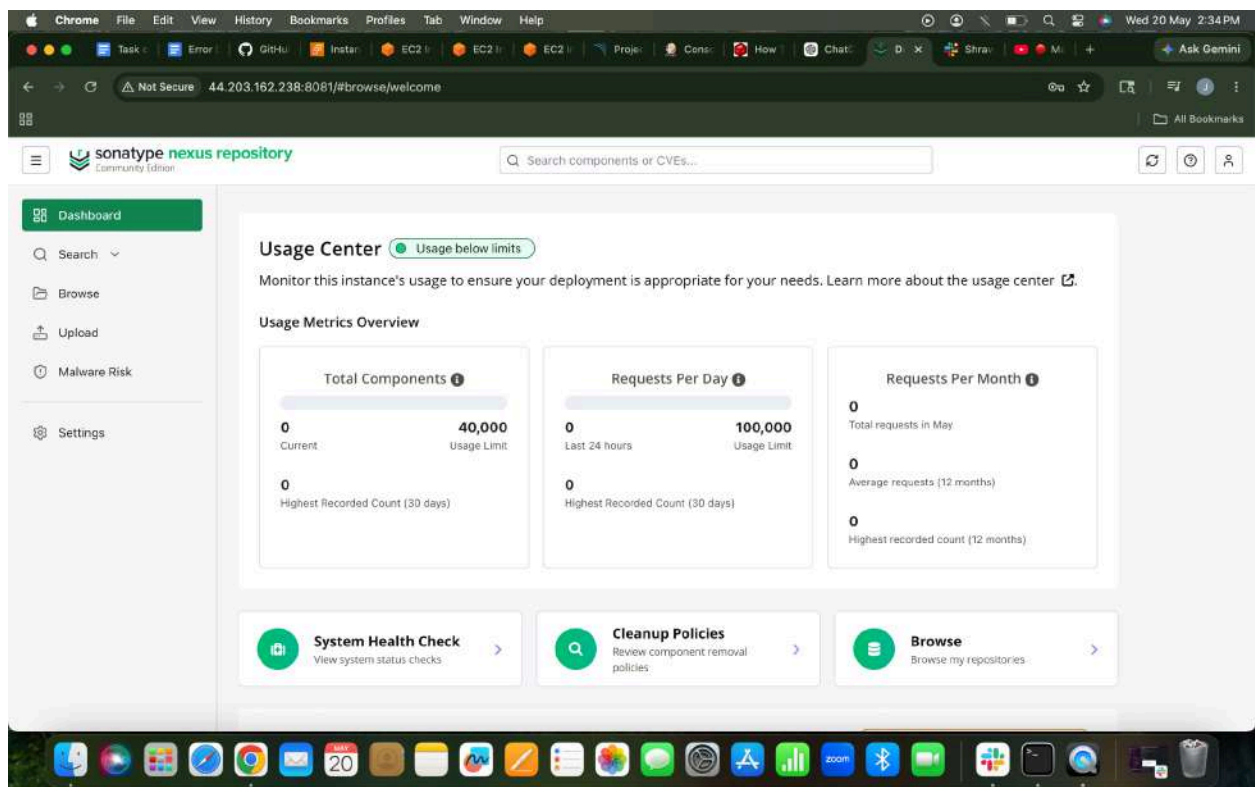
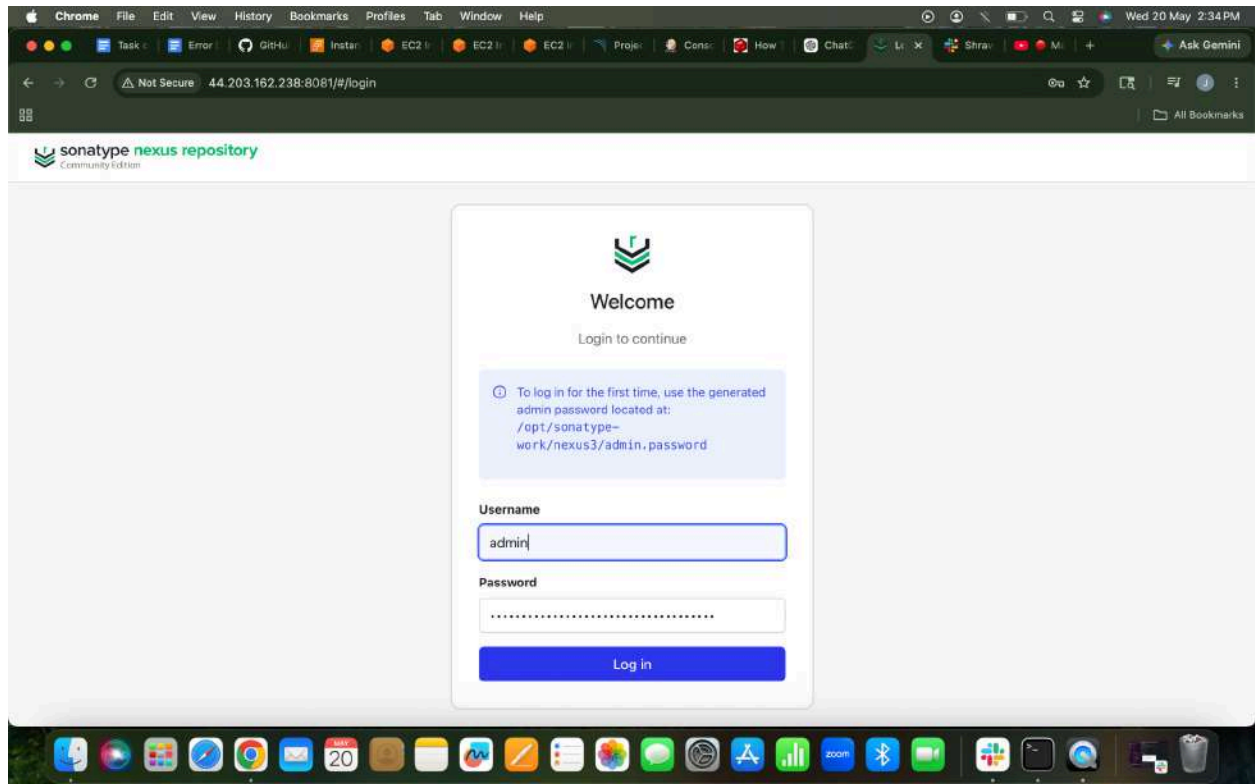
[ec2-user@ip-172-31-1-158 bin]$ cd
[ec2-user@ip-172-31-1-158 ~]$ rm -rf /opt/sonatype-work/nexus3/clean_cache/*
[ec2-user@ip-172-31-1-158 ~]$ ./nexus start
-bash: ./nexus: No such file or directory
[ec2-user@ip-172-31-1-158 ~]$ cd /opt/nexus/bin/
[ec2-user@ip-172-31-1-158 bin]$ ls
nexus nexus.rc nexus.vmoptions sonatype-nexus-repository-3.92.2-01.jar
[ec2-user@ip-172-31-1-158 bin]$ ./nexus start

```

Now run `./nexus start` wait for couple of mins it should run successfully



For password `cat /opt/sonatype-work/nexus3/admin.password` → after logging in change the password and we can check the GUI of NEXus



Need to install the plugins now in Jenkins GUI

The plugin should be **NexusArtifact** download

The first screenshot shows the Jenkins 'Plugins' page with a search for 'nexus'. The 'Nexus Artifact Uploader' 2.16 plugin is listed as installed. A message indicates the plugin is up for adoption and looking for new maintainers. The second screenshot shows the 'Download progress' page, which lists the following steps and their status:

- Preparation
 - Checking internet connectivity
 - Checking update center connectivity
 - Success
- Deploy to container: Success
- Loading plugin extensions: Success
- Nexus Artifact Uploader: Success
- Loading plugin extensions: Success

At the bottom, there are links to 'Go back to the top page' and a checkbox to 'Restart Jenkins when installation is complete and no jobs are running'.

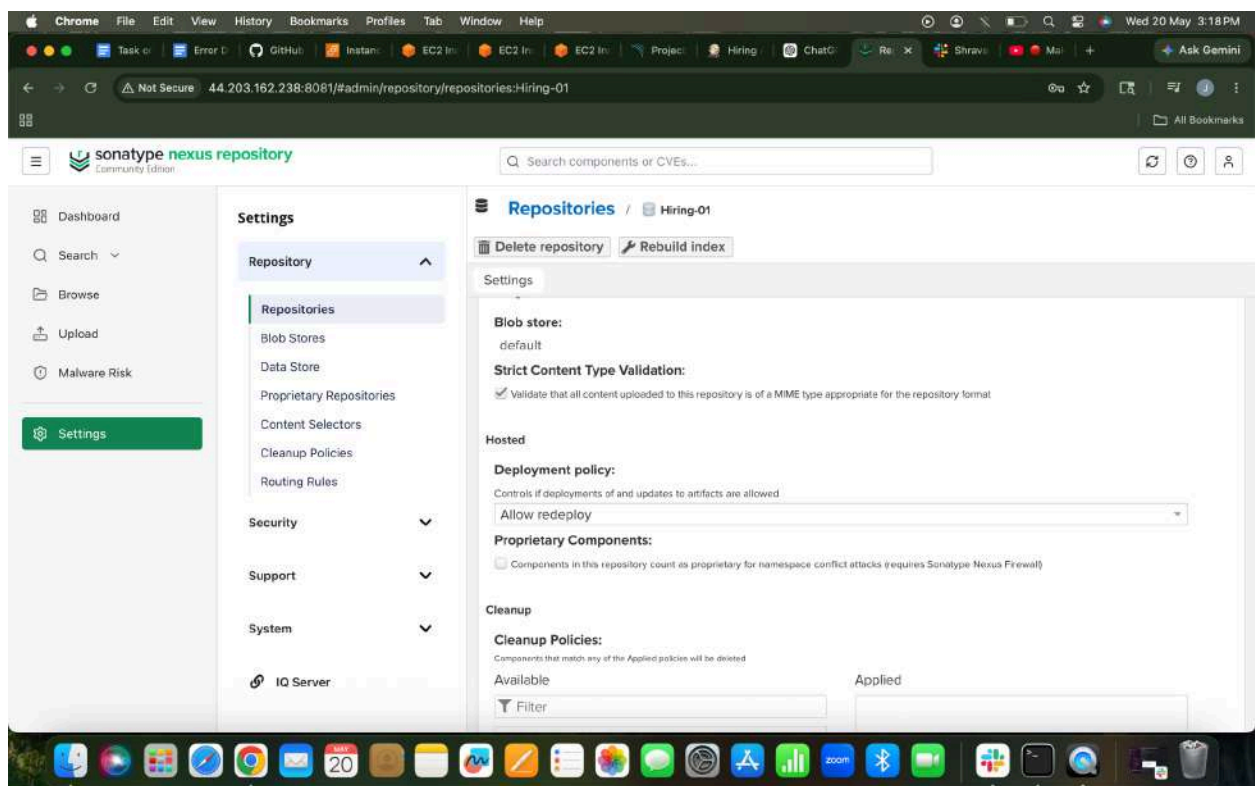
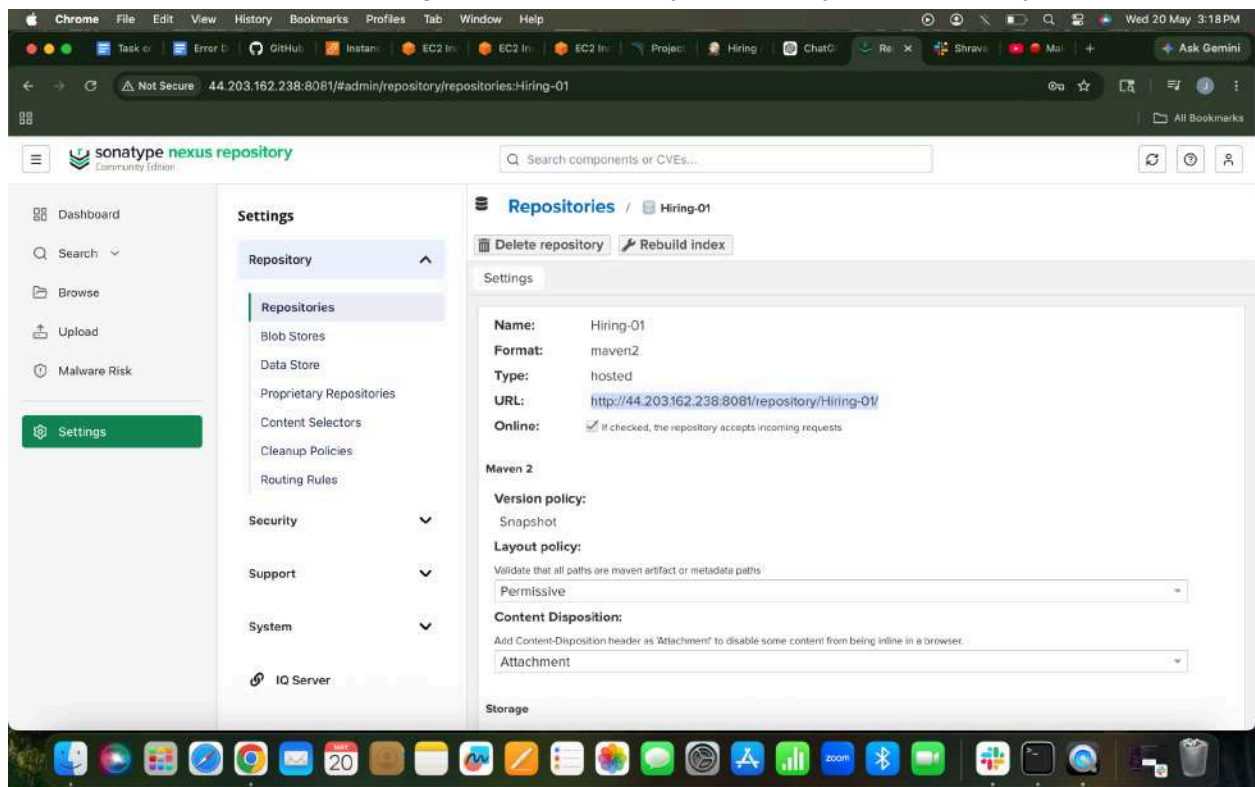
Now lets add nexus artifact details in our Jenkins GUI → Job→Configure→Build actions→Nexus Artifact details→below image

The screenshot shows the 'Configure' page for a Jenkins job named 'Hiring App'. The 'Build Steps' section is selected in the left sidebar. The 'Nexus Details' section contains the following fields:

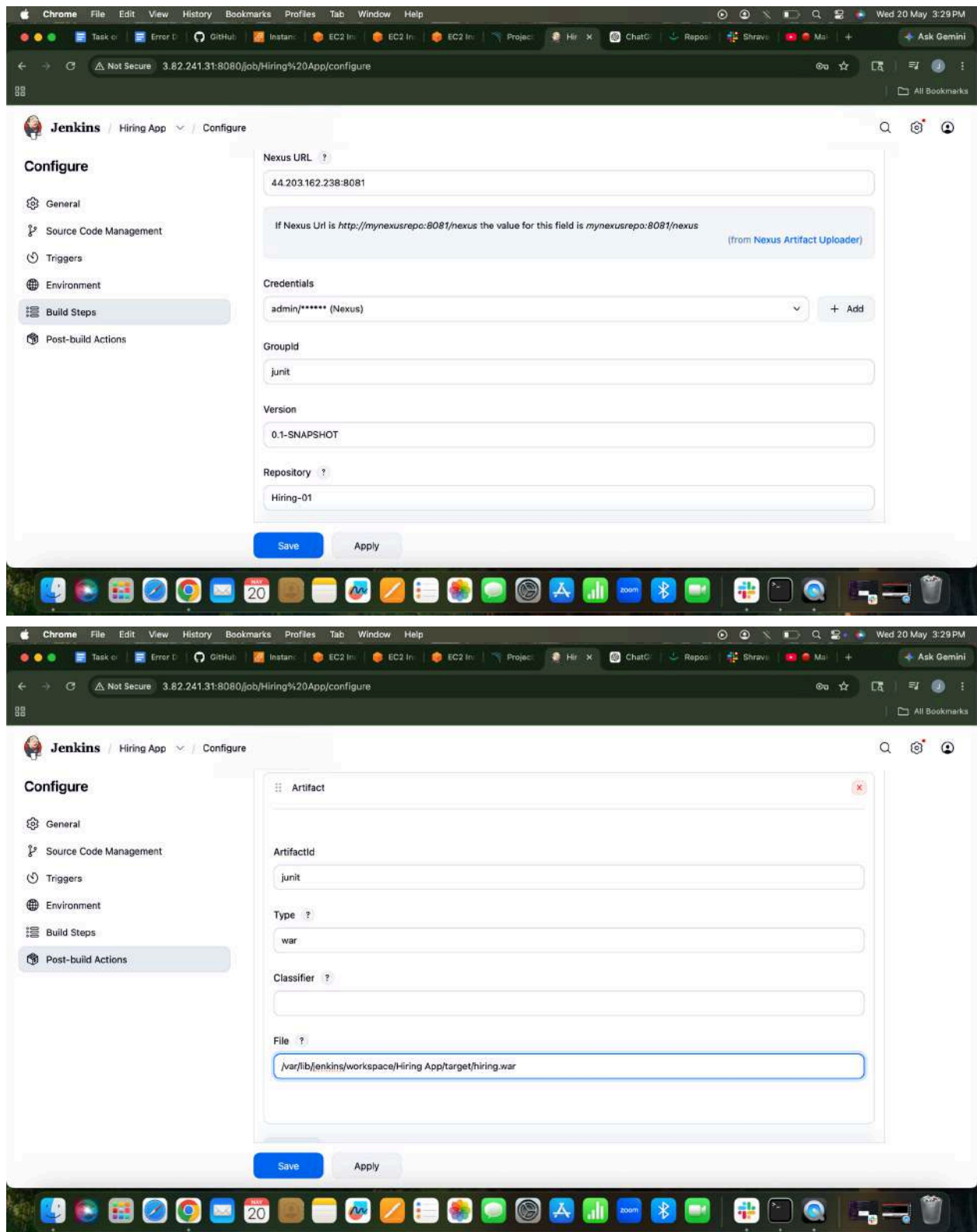
- Nexus Version: NEXUS2
- Protocol: HTTP
- Nexus URL: 44.203.162.238:8081
- Credentials: - none -
- Groupid: (empty field)

A note below the Nexus URL field states: 'If Nexus Url is http://my nexusrepo:8081/nexus the value for this field is mynexusrepo:8081/nexus (from Nexus Artifact Uploader)'. At the bottom, there are 'Save' and 'Apply' buttons.

We need to create a repo in Nexus GUI → Settings → Repository → Create repo → Select Maven2(hosted) → Name → Version Policy(snapshot) → layout policy(permissive) → content Disposition(attachment) → Storage(Default) → Deployment Policy(Allow redeploy) → create



From the above images you can find the URL to give in NEXus artifact in Jenkins GUI –Job
Now we need to give all details in it from the pom.xml file



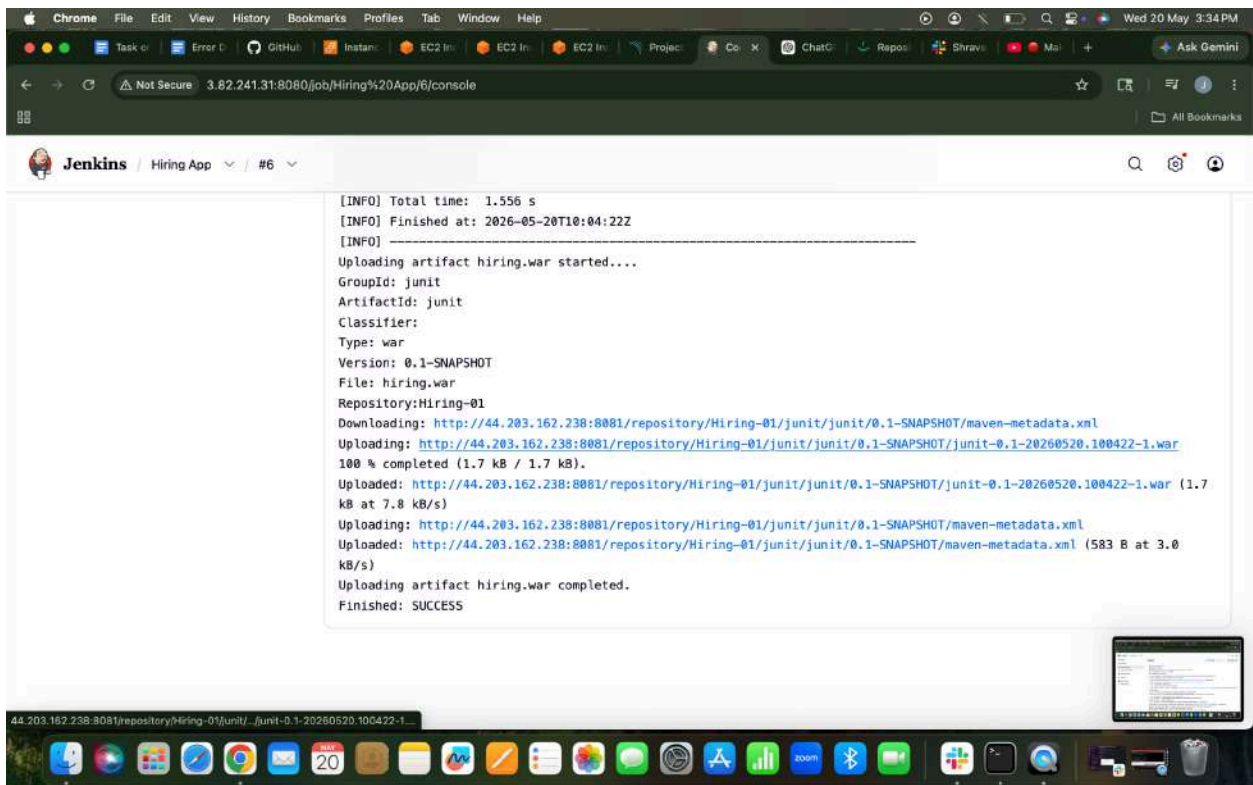
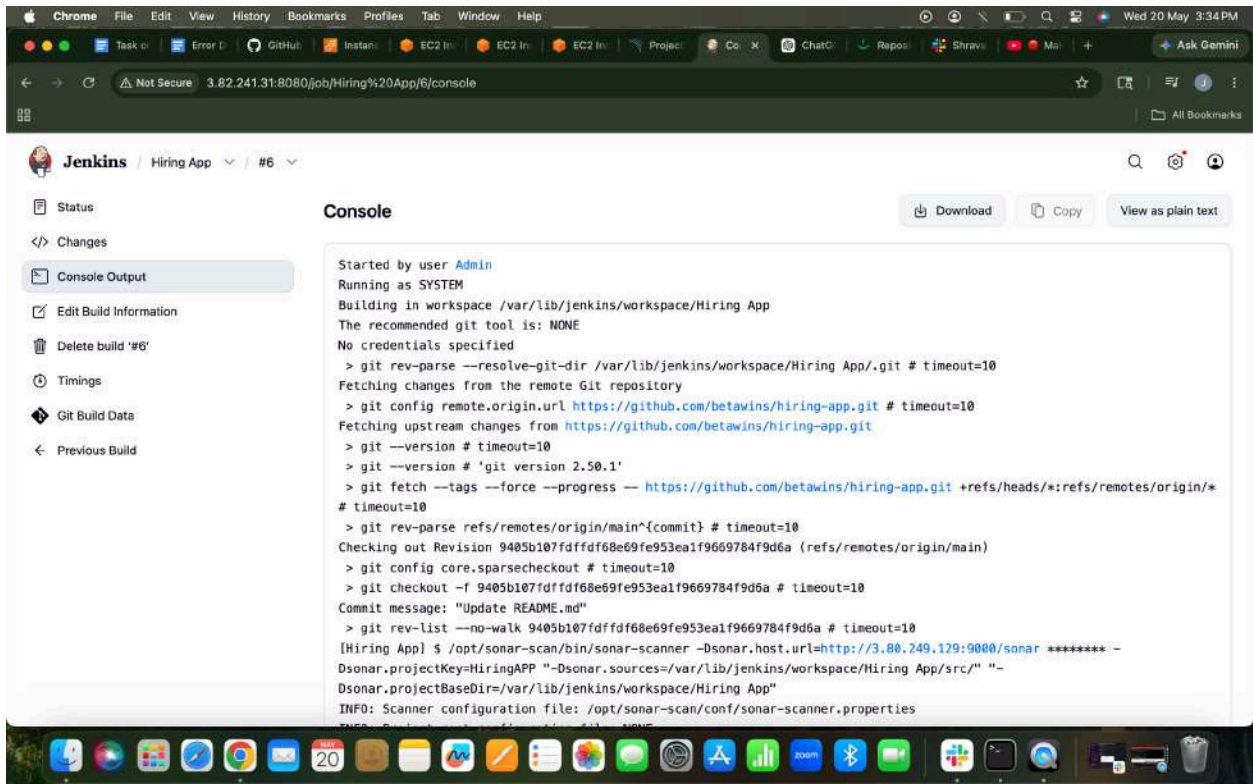
For the file path go to jenkins server and check below image

```
~/lib/
Last login: Wed May 20 06:34:07 2026 from 18.206.107.27
[ec2-user@ip-172-31-1-127 ~]$ sudo su -
Last login: Wed May 20 06:34:13 UTC 2026 on pts/1
[root@ip-172-31-1-127 ~]# cd /var/lib/jenkins/
[root@ip-172-31-1-127 jenkins]# cd workspace/
[root@ip-172-31-1-127 workspace]# ls
'Hiring App'
[root@ip-172-31-1-127 workspace]# cd Hiring\ App/
[root@ip-172-31-1-127 Hiring App]# ls
Dockerfile  Jenkinsfile  README.md  'Untitled Diagram.drawio'  jenkinsfile-cicd  pom.xml  src  target
[root@ip-172-31-1-127 Hiring App]# cd target/
[root@ip-172-31-1-127 target]# ls
hiring  Hiring.war  maven-archiver
[root@ip-172-31-1-127 target]# pwd
/var/lib/jenkins/workspace/Hiring App/target
[root@ip-172-31-1-127 target]#
```

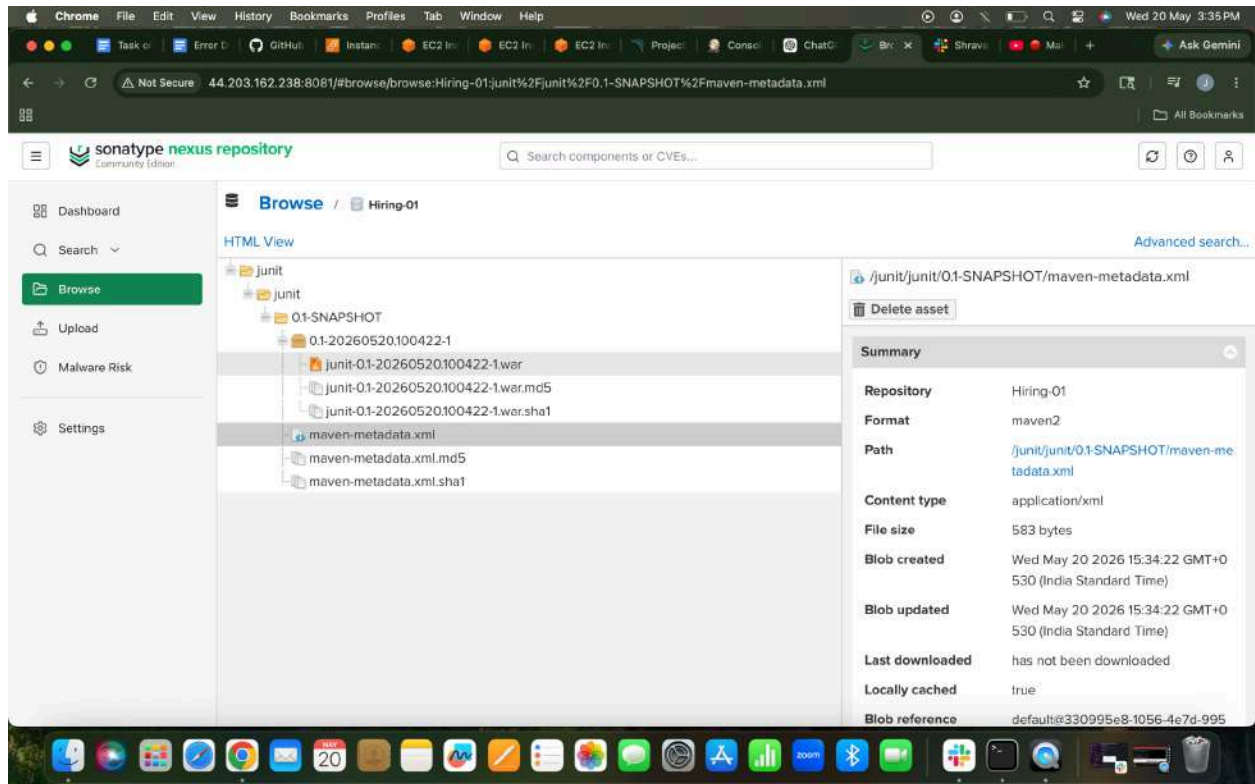
i-0c8b7f0daac46fa4f (Jenkins-Master)

PublicIPs: 3.82.241.31 PrivateIPs: 172.31.1.127

Run The Job and it should be success

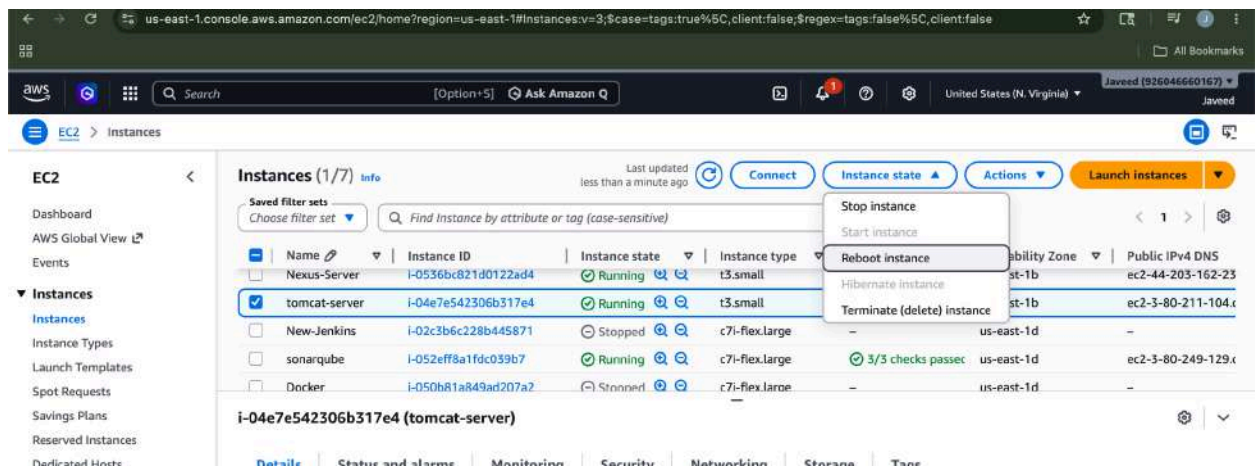


If check in NEXUS GUI we can see in our repo everything has been added



Tomcat Deployment:

Create a new instance for Tomcat



Install tomcat, for tomcat java is the pre requisite

```

root@ip-172-31-1-196 ~]# yum install java-1.8* -y
Last metadata expiration check: 0:00:14 ago on Wed May 20 10:11:06 2026.
Dependencies resolved.
=====
Package                                Architecture                          Ver
=====
Installing:
java-1.8.0-amazon-corretto             x86_64                                1:1
java-1.8.0-amazon-corretto-debugsymbols x86_64                                1:1
java-1.8.0-amazon-corretto-devel       x86_64                                1:1
Installing dependencies:

```

i-04e7e542306b317e4 (tomcat-server)

PublicIPs: 3.80.211.104 PrivateIPs: 172.31.1.196

```

polkit-libs-125-1.amzn2023.0.3.x86_64
poppler-24.08.0-1.amzn2023.x86_64
poppler-glib-24.08.0-1.amzn2023.x86_64
rsvg-pixbuf-loader-2.59.2-318.amzn2023.x86_64
shared-mime-info-2.2-2.amzn2023.0.1.x86_64
tracker-3.7.3-3.amzn2023.0.1.x86_64
upower-1.90.6-139.amzn2023.x86_64
webrtc-audio-processing-0.3.1-6.amzn2023.0.2.x86_64
wireplumber-libs-0.5.7-1.amzn2023.0.1.x86_64
xdg-desktop-portal-gtk-1.15.1-61.amzn2023.x86_64
xml-common-0.6.3-56.amzn2023.0.2.noarch

Complete!
root@ip-172-31-1-196 ~]# █

```

i-04e7e542306b317e4 (tomcat-server)

PublicIPs: 3.80.211.104 PrivateIPs: 172.31.1.196

Wget tomcat tar file
 Untar the file
 Rename the file
 Start by going to cd /opt/tomcat/bin
 Bash shartup.sh

us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-04e7e542306b317e4&osUser=ec2...

```
aws
Search [Option+S]
United States (N. Virginia) Javed (926046650157) Javed

apache-tomcat-9.0.118/bin/startup.sh
apache-tomcat-9.0.118/bin/tool-wrapper.sh
apache-tomcat-9.0.118/bin/version.sh
[root@ip-172-31-1-196 opt]# mv apache-tomcat-9.0.118 apache-tomcat
[root@ip-172-31-1-196 opt]# cd apache-tomcat/
[root@ip-172-31-1-196 apache-tomcat]# ls
BUILDING.txt CONTRIBUTING.md LICENSE NOTICE README.md RELEASE-NOTES RUNNING.txt bin conf lib logs temp webapps work
[root@ip-172-31-1-196 apache-tomcat]# cd conf/
[root@ip-172-31-1-196 conf]# ls
catalina.policy context.xml jaspic-providers.xsd server.xml tomcat-users.xsd
catalina.properties jaspic-providers.xml logging.properties tomcat-users.xml web.xml
[root@ip-172-31-1-196 conf]# cd ..
[root@ip-172-31-1-196 apache-tomcat]# cd bin/
[root@ip-172-31-1-196 bin]# ls
bootstrap.jar ciphers.bat configtest.bat digest.sh setclasspath.sh startup.sh tool-wrapper.sh
catalina-tasks.xml ciphers.sh configtest.sh makebase.bat shutdown.bat tomcat-juli.jar version.bat
catalina.bat commons-daemon-native.tar.gz daemon.sh makebase.sh shutdown.sh tomcat-native.tar.gz version.sh
catalina.sh commons-daemon.jar digest.bat setclasspath.bat startup.bat tool-wrapper.bat
[root@ip-172-31-1-196 bin]# bash startup.sh
Using CATALINA_BASE: /opt/apache-tomcat
Using CATALINA_HOME: /opt/apache-tomcat
Using CATALINA_TMPDIR: /opt/apache-tomcat/temp
Using JRE_HOME: /usr
Using CLASSPATH: /opt/apache-tomcat/bin/bootstrap.jar:/opt/apache-tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[root@ip-172-31-1-196 bin]# find / -name context.xml

i-04e7e542306b317e4 (tomcat-server)
PublicIPs: 3.80.211.104 PrivateIPs: 172.31.1.196
```

CloudShell Feedback Console Mobile App © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

Chrome File Edit View History Bookmarks Profiles Tab Window Help Wed 20 May 3:51 PM

Not Secure 3.80.211.104:8080

Home Documentation Configuration Examples Wiki Mailing Lists Find Help

Apache Tomcat/9.0.118

If you're seeing this, you've successfully installed Tomcat. Congratulations!

 Recommended Reading:

- [Security Considerations How-To](#)
- [Manager Application How-To](#)
- [Clustering/Session Replication How-To](#)

Server Status
Manager App
Host Manager

Developer Quick Start

Tomcat Setup	Realms & AAA	Examples	Servlet Specifications
First Web Application	JDBC DataSources		Tomcat Versions

Managing Tomcat

For security, access to the `manager.webapp` is restricted. Users are defined in:

```
$CATALINA_HOME/conf/tomcat-users.xml
```

In Tomcat 9.0 access to the manager application is split between different users. [Read more...](#)

[Release Notes](#)
[Changelog](#)
[Migration Guide](#)

Documentation

[Tomcat 9.0 Documentation](#)
[Tomcat 9.0 Configuration](#)
[Tomcat Wiki](#)

Find additional important configuration information in:

```
$CATALINA_HOME/RUNNING.txt
```

Developers may be interested in:

- [Tomcat 9.0 Bug Database](#)
- [Tomcat 9.0 JavaDocs](#)

Getting Help

FAQ and Mailing Lists

The following mailing lists are available:

- [tomcat-announce](#)
Important announcements, releases, security vulnerability notifications. (Low volume).
- [tomcat-users](#)
User support and discussion
- [taglibs-user](#)
User support and discussion for [Apache Taglibs](#)
- [tomcat-dev](#)
Development mailing list, including commit messages

Now we need to change context.xml files, like comment in some lines as shown below for that Use command `find / -name context.xml`


```

Using CLASSPATH:      /opt/apache-tomcat/bin/bootstrap.jar:/opt/apache-tomcat/bin
Using CATALINA_OPTS:
Tomcat started.
[root@ip-172-31-1-196 bin]# find / -name context.xml
/opt/apache-tomcat/conf/context.xml
/opt/apache-tomcat/webapps/docs/META-INF/context.xml
/opt/apache-tomcat/webapps/examples/META-INF/context.xml
/opt/apache-tomcat/webapps/host-manager/META-INF/context.xml
/opt/apache-tomcat/webapps/manager/META-INF/context.xml
[root@ip-172-31-1-196 bin]# cat /opt/apache-tomcat/conf/context.xml
<?xml version="1.0" encoding="UTF-8"?>
<!--
Licensed to the Apache Software Foundation (ASF) under one or more
contributor license agreements.  See the NOTICE file distributed with
this work for additional information regarding copyright ownership.

```

In the below files we will be commenting in both the files with the lines starting <!-- **<Value and ending with ip address>** → vim /opt/tomcat/webapps/manager/META-INF/context.xml

vim /opt/tomcat/webapps/host-manager/META-INF/context.xml

```

catalina.policy context.xml jaspic-providers.xsd server.xml tomcat-
[root@ip-172-31-1-196 conf]# cd ..
[root@ip-172-31-1-196 apache-tomcat]# ls
BUILDING.txt CONTRIBUTING.md LICENSE NOTICE README.md RELEASE-NOTES RUNNING.txt
[root@ip-172-31-1-196 apache-tomcat]# cd w
webapps/ work/
[root@ip-172-31-1-196 apache-tomcat]# cd webapps/
[root@ip-172-31-1-196 webapps]# ls
ROOT docs examples host-manager manager
[root@ip-172-31-1-196 webapps]# cd manager/
[root@ip-172-31-1-196 manager]# ls
META-INF WEB-INF css images index.jsp status.xsd xform.xml
[root@ip-172-31-1-196 manager]# cd META-INF/
[root@ip-172-31-1-196 META-INF]# ls
context.xml
[root@ip-172-31-1-196 META-INF]# vi context.xml
[root@ip-172-31-1-196 META-INF]#

```

i-04e7e542306b317e4 (tomcat-server)

PublicIPs: 3.80.211.104 PrivateIPs: 172.31.1.196

```
aws [Search] [Option+S]
<?xml version="1.0" encoding="UTF-8" ?>
<!--
  Licensed to the Apache Software Foundation (ASF) under one or more
  contributor license agreements.  See the NOTICE file distributed with
  this work for additional information regarding copyright ownership.
  The ASF licenses this file to You under the Apache License, Version 2.0
  (the "License"); you may not use this file except in compliance with
  the License.  You may obtain a copy of the License at

      http://www.apache.org/licenses/LICENSE-2.0

  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License.
-->
<Context antiResourceLocking="false" privileged="true" >
  <CookieProcessor className="org.apache.tomcat.util.http.Rfc6265CookieProcessor"
    sameSiteCookies="strict" />
  <!-- <Valve className="org.apache.catalina.valves.RemoteCIDRValve"
    allow="127.0.0.0/8,::1/128" />
  -->
  <Manager sessionAttributeValueClassNameFilter="java\.lang\.(?:Boolean|Integer|Long|Number|string) |
  apache(?:\$1)?|java\.util\.(?:Linked)?HashMap"/>
</Context>
-- INSERT --
```

i-04e7e542306b317e4 (tomcat-server)

PublicIPs: 3.80.211.104 PrivateIPs: 172.31.1.196

```
[root@ip-172-31-1-196 apache-tomcat]# cd w
webapps/ work/
[root@ip-172-31-1-196 apache-tomcat]# cd webapps/
[root@ip-172-31-1-196 webapps]# ls
ROOT docs examples host-manager manager
[root@ip-172-31-1-196 webapps]# cd manager/
[root@ip-172-31-1-196 manager]# ls
META-INF WEB-INF css images index.jsp status.xsd xform.xml
[root@ip-172-31-1-196 manager]# cd META-INF/
[root@ip-172-31-1-196 META-INF]# ls
context.xml
[root@ip-172-31-1-196 META-INF]# vi context.xml
[root@ip-172-31-1-196 META-INF]# cd ../../
[root@ip-172-31-1-196 webapps]# ls
ROOT docs examples host-manager manager
[root@ip-172-31-1-196 webapps]# cd host-manager/
[root@ip-172-31-1-196 host-manager]# ls
META-INF WEB-INF css images index.jsp
[root@ip-172-31-1-196 host-manager]# cd META-INF/
[root@ip-172-31-1-196 META-INF]# vi context.xml
[root@ip-172-31-1-196 META-INF]#
```

i-04e7e542306b317e4 (tomcat-server)

PublicIPs: 3.80.211.104 PrivateIPs: 172.31.1.196

```

WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License.
-->
<Context antiResourceLocking="false" privileged="true" >
  <CookieProcessor className="org.apache.tomcat.util.http.Rfc6265CookieProcessor"
    sameSiteCookies="strict" />
  <!-- <Valve className="org.apache.catalina.valves.RemoteCIDRValve"
    allow="127.0.0.0/8,::1/128" />
  -->
  <Manager sessionAttributeValueClassNameFilter="java\.lang\.(?:Boolean|Integer|Long|Number|Str
apache(?:\$\d)?|java\.util\.(?:Linked)?HashMap"/>
</Context>
-- INSERT --

```

i-04e7e542306b317e4 (tomcat-server)

PublicIPs: 3.80.211.104 PrivateIPs: 172.31.1.196

Now we need to add the user details in ⇒ vi /opt/tomcat/conf/tomcat-users.xml

Details to be added are

```
<role rolename="manager-gui"/>
```

```
<role rolename="manager-script"/>
```

```
<user username="tomcat" password="tomcat123" roles="manager-gui,manager-script"/>
```

```

[root@ip-172-31-1-196 webapps]# ls
ROOT docs examples host-manager manager
[root@ip-172-31-1-196 webapps]# cd host-manager/
[root@ip-172-31-1-196 host-manager]# ls
META-INF WEB-INF css images index.jsp
[root@ip-172-31-1-196 host-manager]# cd META-INF/
[root@ip-172-31-1-196 META-INF]# vi context.xml
[root@ip-172-31-1-196 META-INF]# cd ../../
[root@ip-172-31-1-196 webapps]# ls
ROOT docs examples host-manager manager
[root@ip-172-31-1-196 webapps]# cd ..
[root@ip-172-31-1-196 apache-tomcat]# ls
BUILDING.txt CONTRIBUTING.md LICENSE NOTICE README.md RELEASE-NOTES RUNNING.txt bin conf lib logs temp webapps work
[root@ip-172-31-1-196 apache-tomcat]# cd conf/
[root@ip-172-31-1-196 conf]# ls
Catalina catalina.properties jaspic-providers.xml logging.properties tomcat-users.xml web.xml
catalina.policy context.xml jaspic-providers.xsd server.xml tomcat-users.xsd
[root@ip-172-31-1-196 conf]# vi tomcat-users.xml
[root@ip-172-31-1-196 conf]#

```

i-04e7e542306b317e4 (tomcat-server)

PublicIPs: 3.80.211.104 PrivateIPs: 172.31.1.196

```
See the License for the specific language governing permissions and
limitations under the License.
-->
<tomcat-users xmlns="http://tomcat.apache.org/xml"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://tomcat.apache.org/xml tomcat-users.xsd"
  version="1.0">

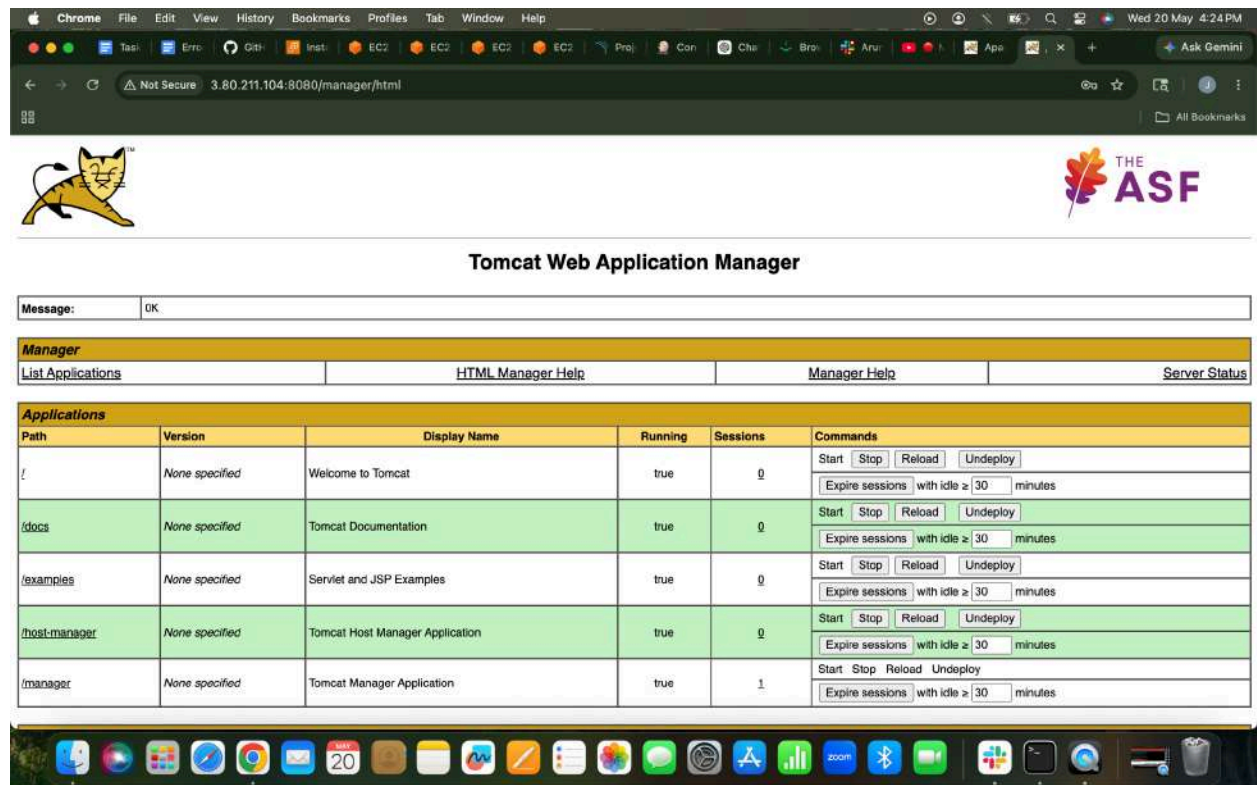
  <role rolename="manager-gui"/>
  <role rolename="manager-script"/>
  <user username="tomcat" password="tomcat123" roles="manager-gui,manager-script"/>

<!--
By default, no user is included in the "manager-gui" role required
to operate the "/manager/html" web application. If you wish to use this app,
you must define such a user - the username and password are arbitrary.
-->

i-04e7e542306b317e4 (tomcat-server)
PublicIPs: 3.80.211.104 PrivateIPs: 172.31.1.196

CloudShell Feedback Console Mobile App zoom
```

Once every file is edited shutdown the tomcat service and startup the service from
Cd /opt/tomcat/bin/[shutdown.sh](#) or [startup.sh](#)
Once after starting the service we can check using <public IP>:8080/Manager/html ⇒ it will ask
for username and password, we need to give the user name and password which we added in
above file and it will run successfully.



Now lets go to Jenkins GUI and add the plugins ⇒ **Deploy to Container**

The top screenshot shows the Jenkins 'Available plugins' page. A search for 'deploy' has been performed, and the 'Deploy to container' plugin (version 1.17) is selected. The description states: 'This plugin allows you to deploy a war to a container after a successful build. Glassfish 3.x remote deployment'. The 'Install' button is visible.

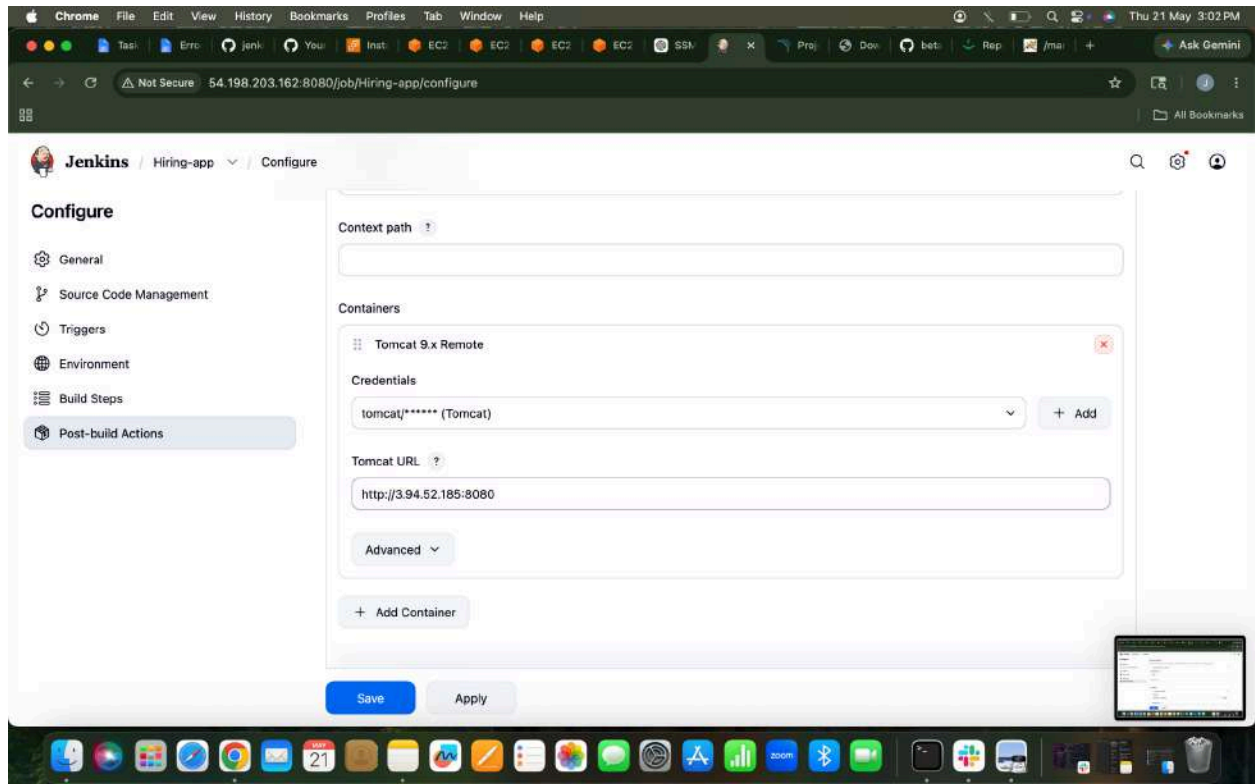
The bottom screenshot shows the 'Download progress' page. The progress is as follows:

- Preparation
 - Checking internet connectivity
 - Checking update center connectivity
 - Success
- Deploy to container: ✓ Success
- Loading plugin extensions: ✓ Success

Below the progress list, there are two instructions:

- [Go back to the top page](#) (you can start using the installed plugins right away)
- ☐ Restart Jenkins when installation is complete and no jobs are running

Now we need to goto our Jenkins Job→Configure→Post Build Action→Deploy to container→WAR/EAR files(**/*war) →Containers(tomcat 9→which ever version of tomcat we are downloading on our server) → credentials→Tomacat URL(Ipaddress:8080) →



Now build the job or run the job
Success

 **Jenkins** / Hiring-app

Status

Changes

Workspace

Build Now

Configure

Delete Project

SonarQube

Rename

 **Hiring-app**

Hiring App

 **SonarQube**

SonarQube Quality Gate

Test **Passed**

server-side processing: **Success**

Permalinks

- Last build (#2), 11 min ago
- Last stable build (#2), 11 min ago
- Last successful build (#2), 11 min ago
- Last completed build (#2), 11 min ago

Builds >

Filter

/

Today

 #3 09:32

 <

 #2 09:20

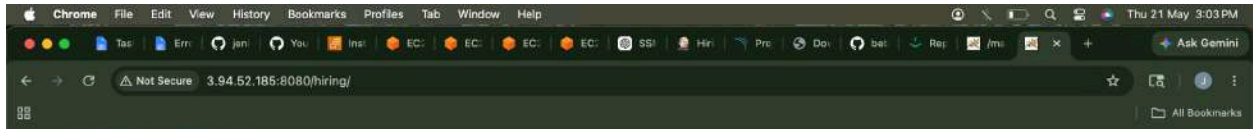
 <

 #1 09:12

 <

App

Now lets check the output in tomcat server url⇒**ipaddress:8080/hiring**



Welcome to Techie Horizon

Our Deployment is Successfull

2) Setup a Jenkins CICD pipeline using Declarative pipeline using feature-1.1 branch.

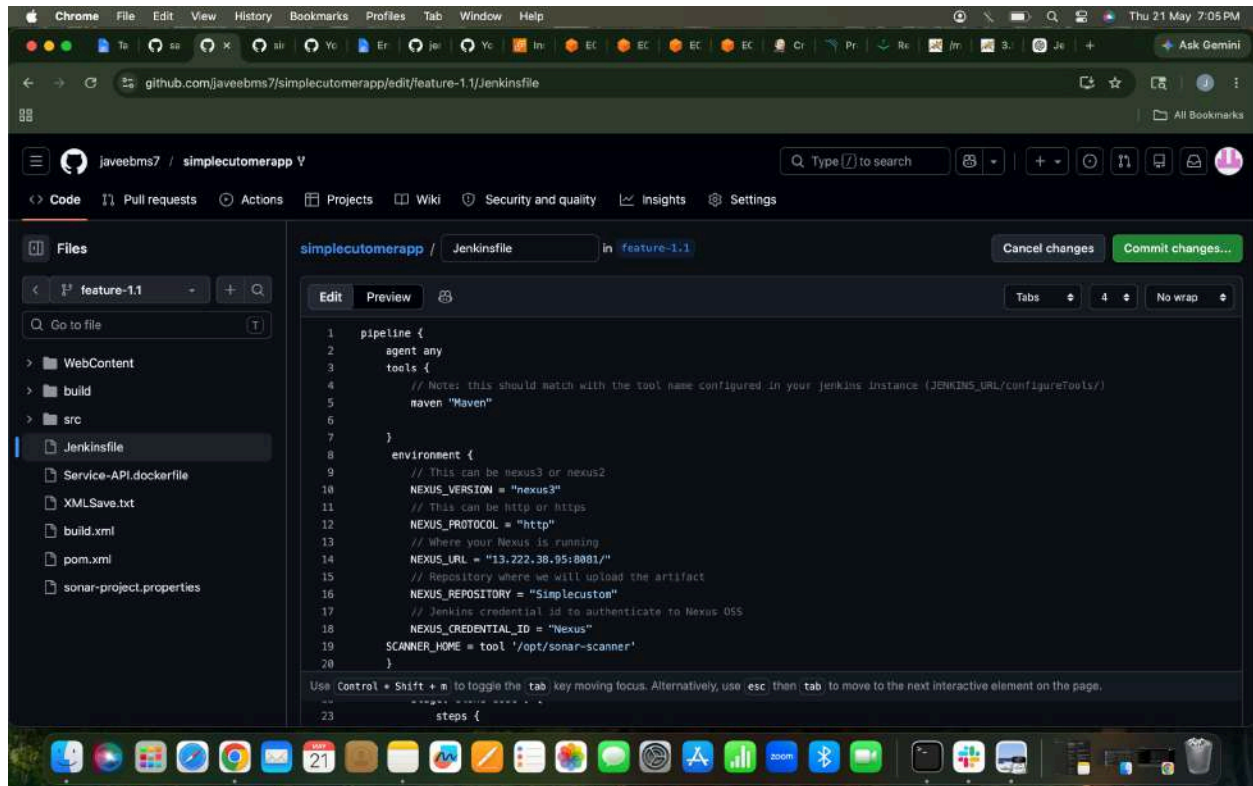
Source Code: https://github.com/betawins/sabear_simplecutomerapp/tree/feature-1.1

stages:

- Git Clone
- SonarQube Integration
- Maven Compilation
- Nexus Artifactory
- Slack Notification
- Deploy On tomcat

We already have all the setup from the above task.

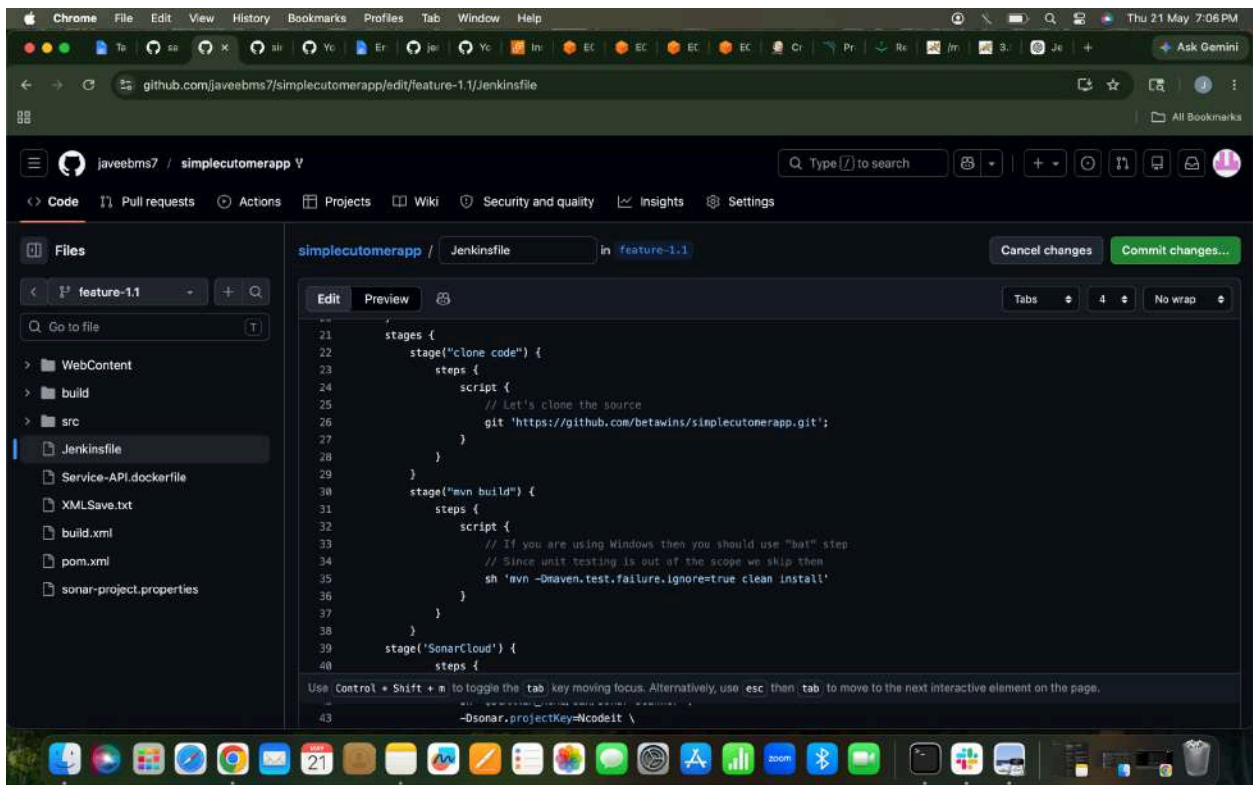
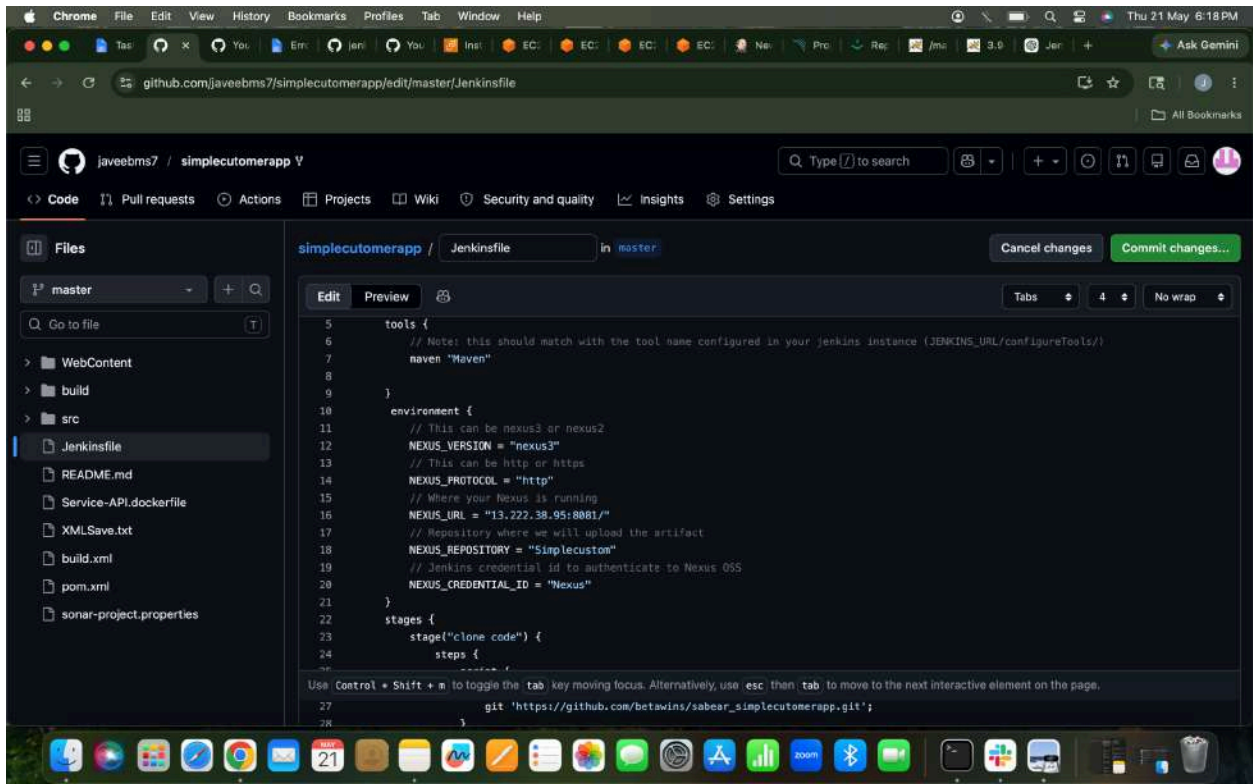
So lets for the fork the git hub repo first– Main repo link→fork→with other branches →fork
Below is the pipeline we have in the forked repo
For pipeline goto→repo→jenkins file→we can find the pipeline
Now using this pipe file we will do necessary changes for our use and accordingly as the
environment variable have been changed for previous fork

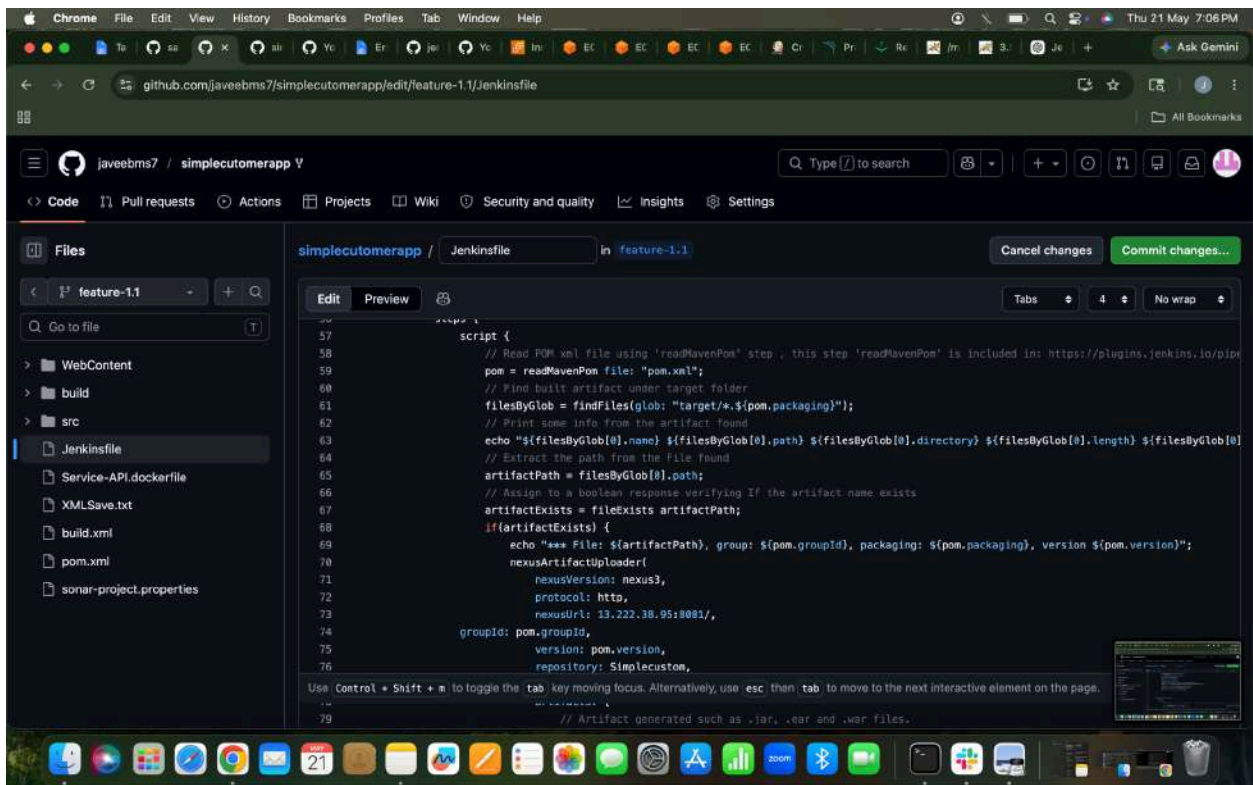
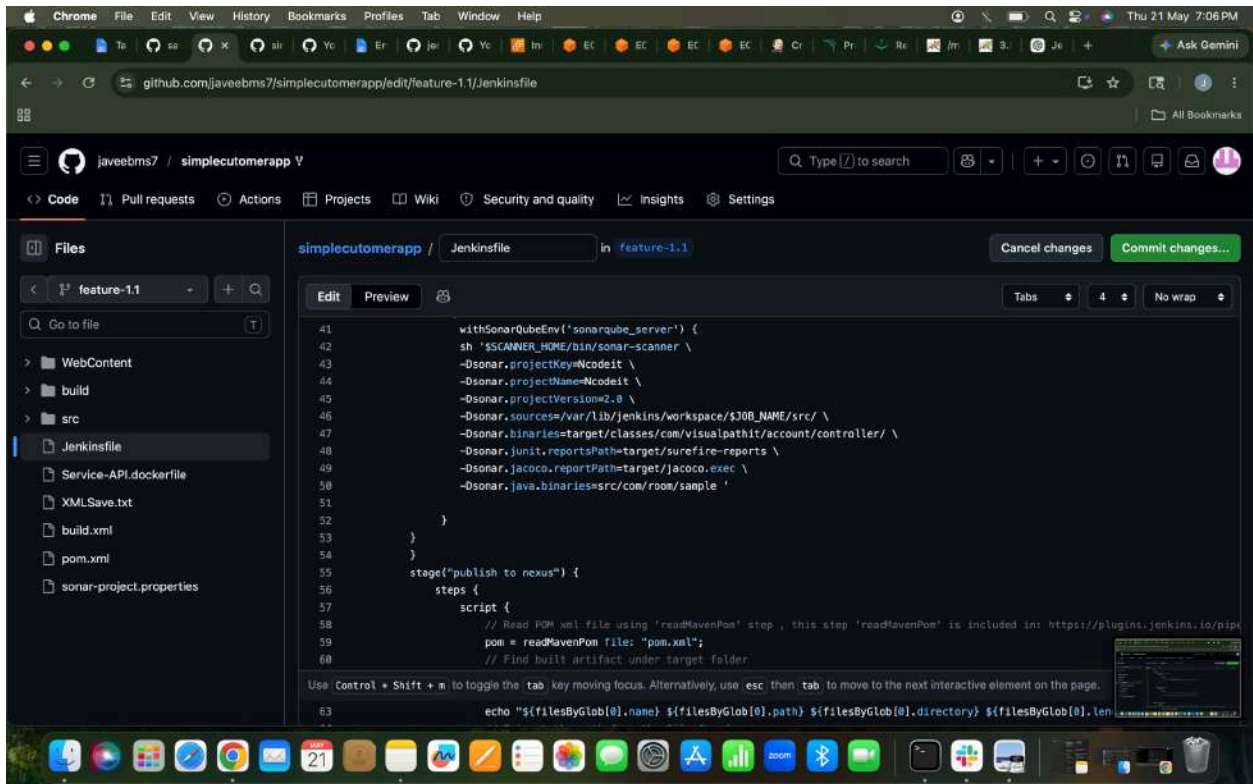


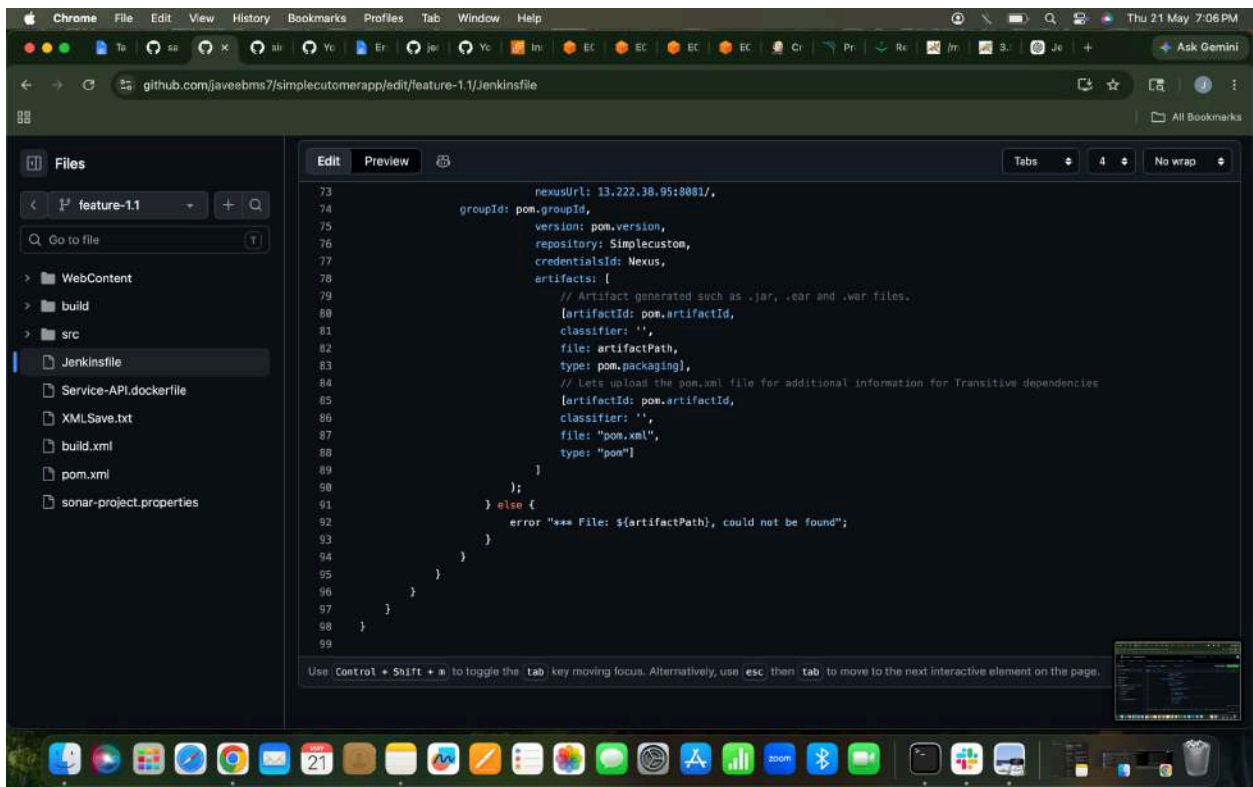
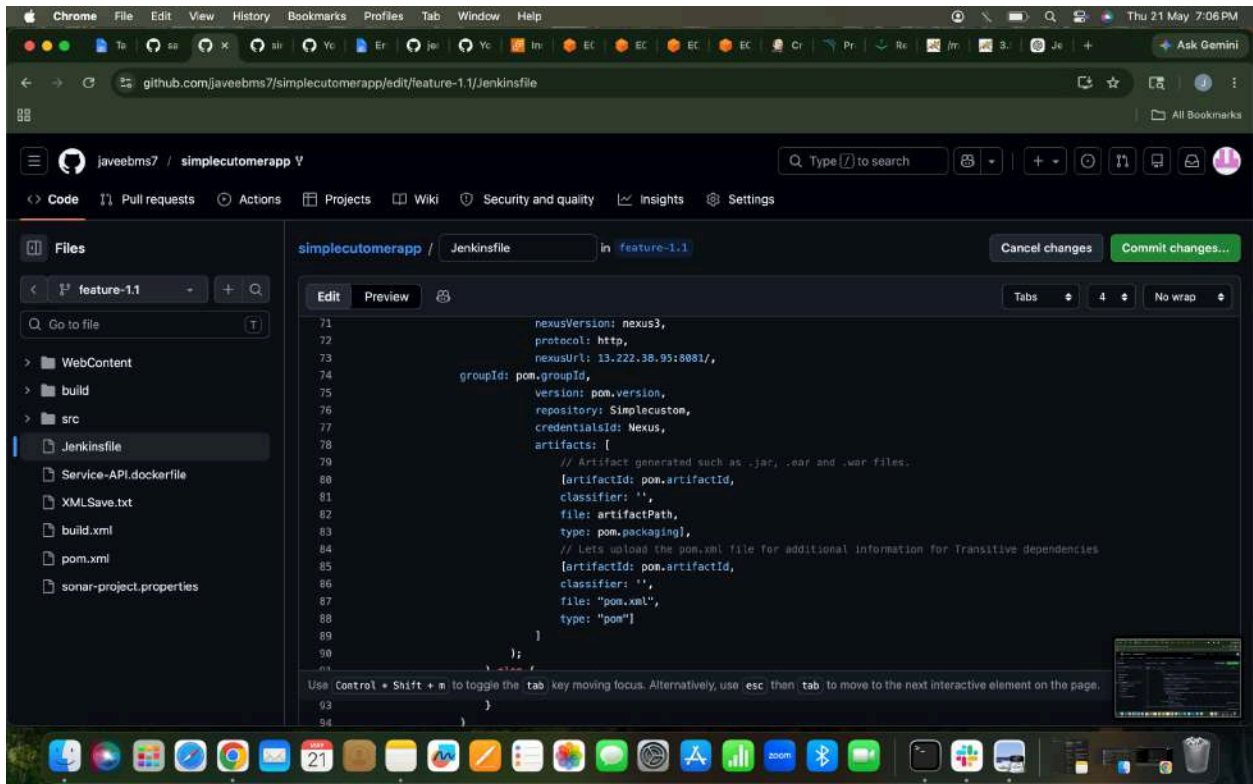
The screenshot shows a web browser displaying a GitHub repository page for 'javeebms7 / simplecutomerapp'. The repository is in a forked state, and the 'feature-1.1' branch is selected. The 'Jenkinsfile' is open in the 'Edit' view. The file contains a Jenkins pipeline configuration with the following structure:

```
1 pipeline {
2   agent any
3   tools {
4     // Notes: this should match with the tool name configured in your jenkins instance (JENKINS_URL/configureTools/)
5     maven "Maven"
6   }
7 }
8 environment {
9   // This can be nexus3 or nexus2
10  NEXUS_VERSION = "nexus3"
11  // This can be http or https
12  NEXUS_PROTOCOL = "http"
13  // Where your Nexus is running
14  NEXUS_URL = "13.222.38.95:8081/"
15  // Repository where we will upload the artifact
16  NEXUS_REPOSITORY = "Simplecutoston"
17  // Jenkins credential id to authenticate to Nexus OSS
18  NEXUS_CREDENTIAL_ID = "Nexus"
19  SCANNER_HOME = tool '/opt/sonar-scanner'
20 }
21
22
23 steps {
```

The interface includes a sidebar with a file explorer showing the repository structure: 'WebContent', 'build', 'src', and '.Jenkinsfile'. The main editor area has tabs for 'Edit' and 'Preview', and buttons for 'Cancel changes' and 'Commit changes...'. The bottom of the screen shows a macOS dock with various application icons.

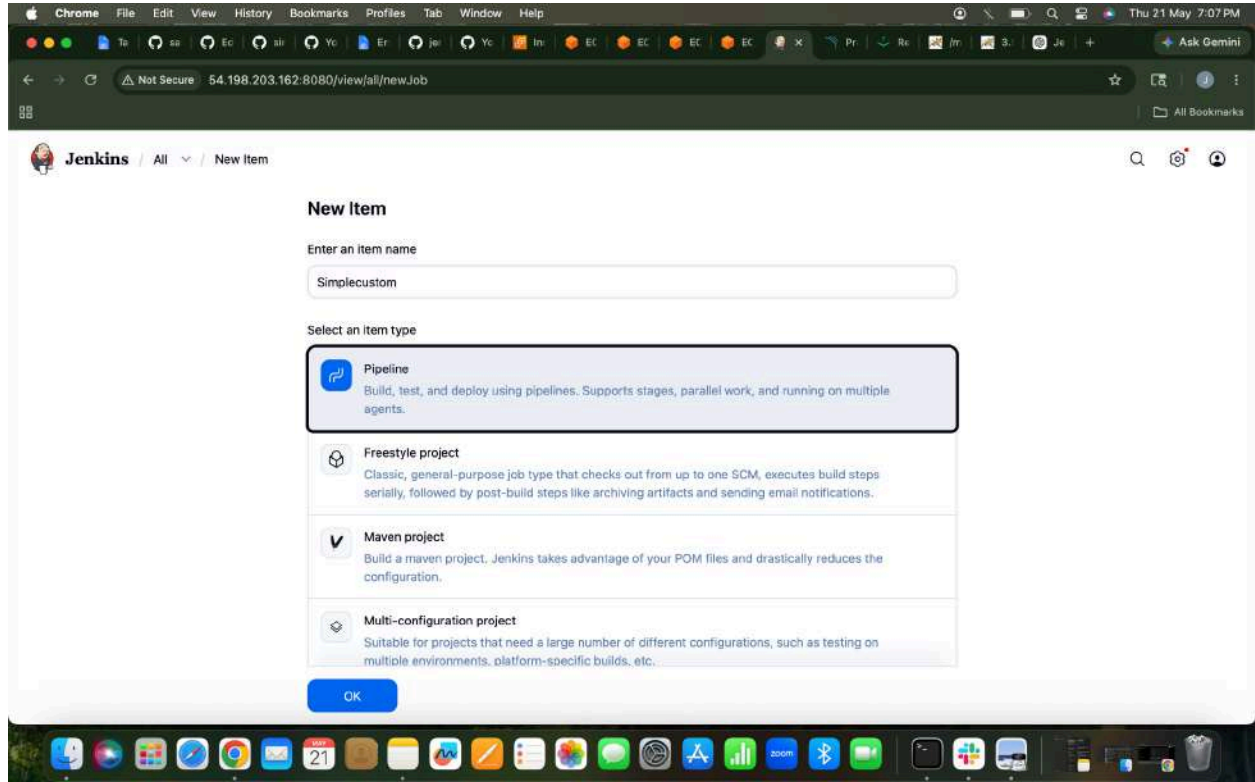


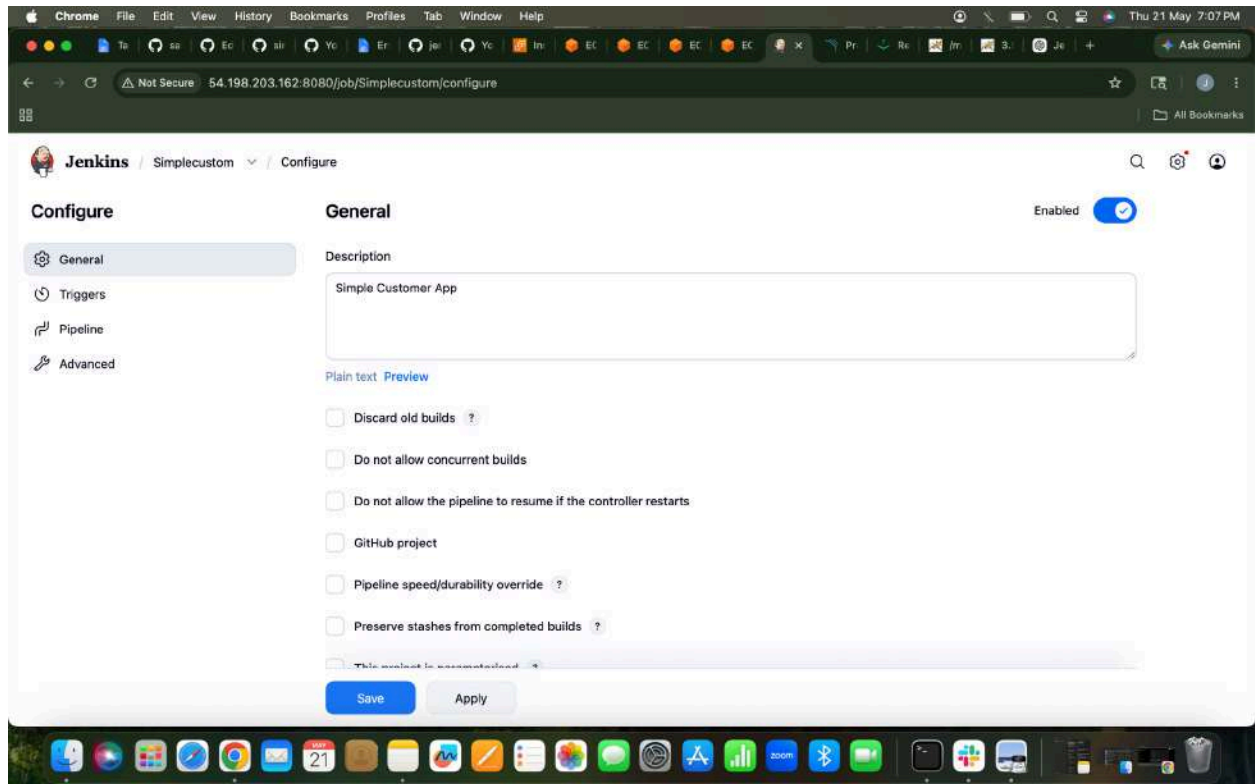




Create a new Job→Pipeline→ Declarative Pipe Line→In Pipe Line code→Git Clone→Add SonarQube Analysis→Maven→script to get war file and pus to repo→Deployments→sucess notification

Creating a new job with pipeline





Now lets start writing the pipe line

*******Pipeline Started*******

```
pipeline {
  agent any
```

*******Tools*******

```
tools {
  maven "Maven"
}
```

*******Enviroinment Variables*******

```
environment {
  NEXUS_VERSION = "nexus3"
  NEXUS_PROTOCOL = "http"
  NEXUS_URL = "13.222.38.95:8081"
  NEXUS_REPOSITORY = "Simplecustom"
  NEXUS_CREDENTIAL_ID = "Nexus"
  SCANNER_HOME = tool('new-sonar')
  BUILD_VERSION = "${BUILD_NUMBER}-SNAPSHOT"
}
```

*******Git Clone*******

```
stages {

  stage("clone code") {
```

```

        steps {
            git 'https://github.com/javeebms7/simplecutomerapp.git'
        }
    }
}

*****Maven Build*****
stage("mvn build") {
    steps {
        sh 'mvn clean install -Dmaven.test.failure.ignore=true'
    }
}

*****SonarQube Analysis*****
stage('SonarQube Analysis') {
    steps {
        withSonarQubeEnv('new-sonar') {
            sh """
            ${SCANNER_HOME}/bin/sonar-scanner \
            -Dsonar.projectKey=Ncodeit \
            -Dsonar.projectName=Ncodeit \
            -Dsonar.projectVersion=2.0 \
            -Dsonar.sources=src \
            -Dsonar.java.binaries=target/classes
            """
        }
    }
}

*****Nexus*****
stage("publish to nexus") {
    steps {
        nexusArtifactUploader(
            nexusVersion: 'nexus3',
            protocol: 'http',
            nexusUrl: '13.222.38.95:8081',
            repository: 'Simplecustom',
            credentialsId: 'Nexus',
            groupId: 'com.javatpoint',
            version: "${BUILD_VERSION}",
            artifacts: [
                [
                    artifactId: 'SimpleCustomerApp',
                    classifier: "",
                    file: "target/SimpleCustomerApp-${BUILD_VERSION}.war",
                    type: 'war'
                ]
            ]
        )
    }
}

```

```

    )
  }
}
*****Tomcat Deployment*****

```

```

stage('Deploy On Tomcat') {
    sh 'ls target/*.war | head -1 > warfile.txt'
    def warFile = readFile('warfile.txt').trim()
    echo "Deploying WAR: ${warFile}"
    deploy adapters: [
        tomcat9(
            credentialsId: 'tomcat',
            path: "",
            url: 'http://54.204.140.94:8080/manager/text'
        )
    ],
    contextPath: 'Simplecustomapp',
    war: warFile
}

```

```

*****Success Slack Notification*****

```

```

    slackSend(
        tokenCredentialId: 'Slack',
        channel: 'C0B5EQ639SQ',
        color: 'good',
        message: "SUCCESS: Job=${env.JOB_NAME} Build=#${env.BUILD_NUMBER} |
${env.BUILD_URL}"
    )

```

```

    } catch (Exception e) {

```

```

*****Failure Slack Notification*****

```

```

        slackSend(
            tokenCredentialId: 'Slack',
            channel: 'C0B5EQ639SQ',
            color: 'danger',
            message: "FAILED: Job=${env.JOB_NAME} Build=#${env.BUILD_NUMBER} |
${env.BUILD_URL}"
        )
        throw e // rethrow so Jenkins marks build as FAILED
    }
}
}
}
}
}

```

Chrome File Edit View History Bookmarks Profiles Tab Window Help Thu 21 May 7:07 PM

Not Secure 54.198.203.162:8080/job/Simplecustom/configure

Jenkins / Simplecustom / Configure

Configure

- General
- Triggers
- Pipeline
- Advanced

Script

```
1 pipeline {
2   agent any
3   tools {
4     // Note: this should match with the tool name configured in your jenkins instance (JENKINS_UC
5     maven "Maven"
6   }
7   environment {
8     // This can be nexus3 or nexus2
9     NEXUS_VERSION = "nexus3"
10    // This can be http or https
11    NEXUS_PROTOCOL = "http"
12    // Where your Nexus is running
13    NEXUS_URL = "13.222.38.95:8081/"
14    // Repository where we will upload the artifact
15  }
```

☒ Use Groovy Sandbox

Pipeline Syntax

Save Apply

Jenkins / Simplecustom / Configure

Configure

- General
- Triggers
- Pipeline
- Advanced

Definition

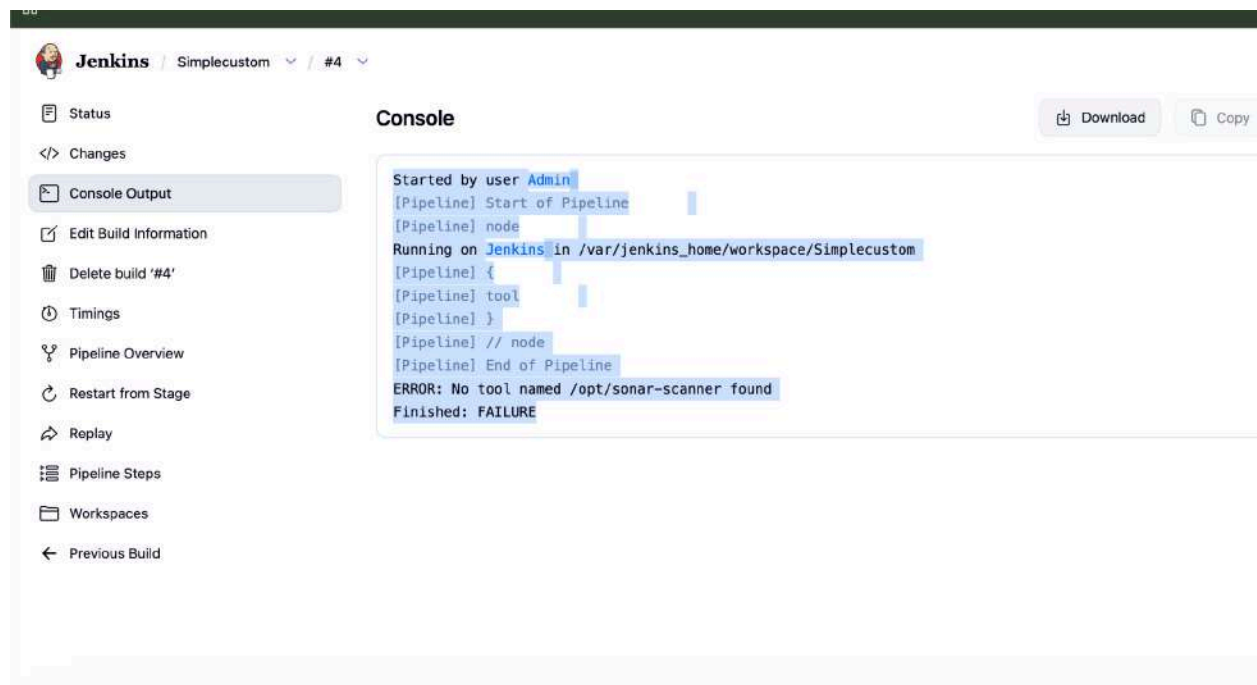
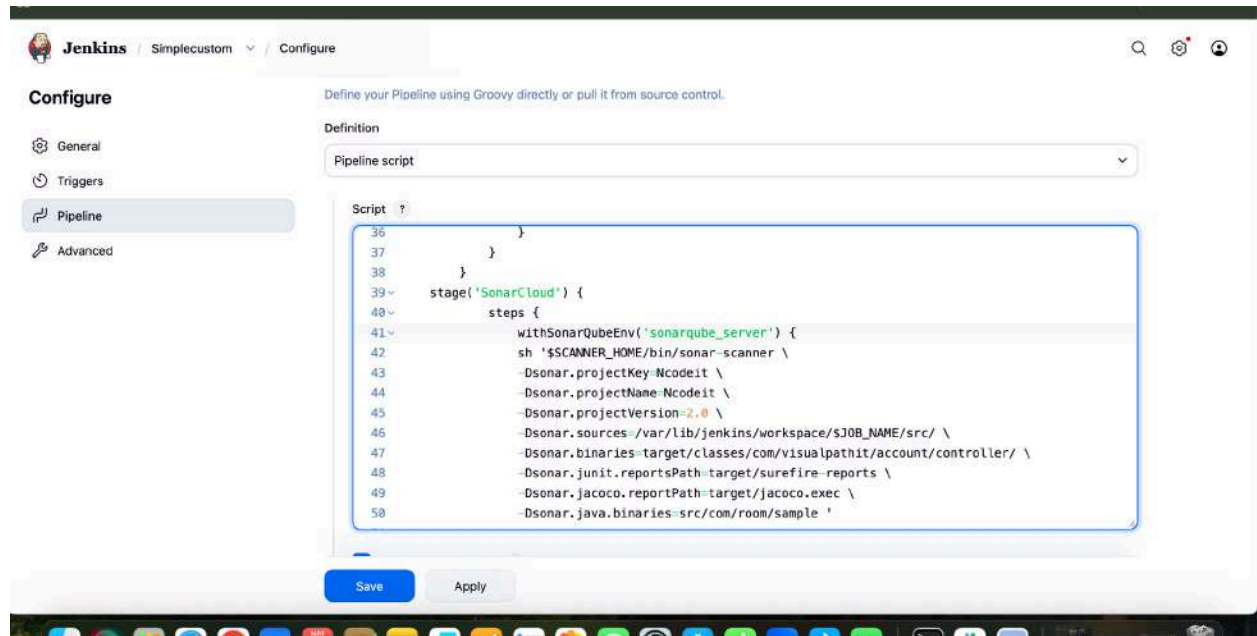
Pipeline script

Script

```
11 // This can be http or https
12 NEXUS_PROTOCOL = "http"
13 // Where your Nexus is running
14 NEXUS_URL = "http://13.222.38.95:8081/"
15 // Repository where we will upload the artifact
16 NEXUS_REPOSITORY = "Simplecustom"
17 // Jenkins credential id to authenticate to Nexus OSS
18 NEXUS_CREDENTIAL_ID = "Nexus"
19 SCANNER_HOME = tool 'new-sonar'
20 }
21 stages {
22   stage("clone code") {
23     steps {
24       script {
25         // Let's clone the source
26       }
27     }
28   }
29 }
```

☒ Use Groovy Sandbox

Save Apply



Changed to new-sonar

Jenkins / Simplecustom / Configure

Configure

General
Triggers
Pipeline
Advanced

Define your Pipeline using Groovy directly or pull it from source control.

Definition
Pipeline script

Script ?

```
36      }
37    }
38  }
39  stage('SonarCloud') {
40    steps {
41      withSonarQubeEnv('new-sonar') {
42        sh 'SSCANNER_HOME/bin/sonar-scanner \
43          -Dsonar.projectKey=Ncodeit \
44          -Dsonar.projectName=Ncodeit \
45          -Dsonar.projectVersion=2.0 \
46          -Dsonar.sources=/var/lib/jenkins/workspace/$JOB_NAME/src/ \
47          -Dsonar.binaries=target/classes/com/visualpathit/account/controller/ \
48          -Dsonar.junit.reportsPath=target/surefire-reports \
49          -Dsonar.jacoco.reportPath=target/jacoco.exec \
50          -Dsonar.java.binaries=src/com/room/sample '
```

Save Apply

Jenkins / Simplecustom / Configure

Configure

General
Triggers
Pipeline
Advanced

Define your Pipeline using Groovy directly or pull it from source control.

Definition
Pipeline script

Script ?

```
65      artifactPath = filesByGlob[0].path;
66      // Assign to a boolean response verifying if the artifact name exists
67      artifactExists = fileExists artifactPath;
68      if(artifactExists) {
69        echo "*** File: ${artifactPath}, group: ${pom.groupId}, packaging: ${pom.packag
70      }
71      nexusArtifactUploader(
72        nexusVersion: nexus3,
73        protocol: http,
74        nexusUrl: "http://13.222.38.95:8081/",
75        groupId: pom.groupId,
76        version: pom.version,
77        repository: Simplecustom,
78        credentialsId: Nexus,
```

Save Apply

Publish to nexus

Net Secure 54.198.203.162:8080/job/Simplecustom/configure

Jenkins / Simplecustom / Configure

Configure

- General
- Triggers
- Pipeline
- Advanced

```
51 }
52 }
53 }
54 }
55 stage("publish to nexus") {
56   steps {
57     script {
58       // Read POM xml file using 'readMavenPom' step , this step 'readMavenPom' is incl
59       //pom = readMavenPom file: "pom.xml";
60       // Find built artifact under target folder
61       filesByGlob = findFiles(glob: "target/*.${pom.packaging}");
62       // Print some info from the artifact found
63       echo "${filesByGlob[0].name} ${filesByGlob[0].path} ${filesByGlob[0].directory} $
64       // Extract the path from the File found
```

☒ Use Groovy Sandbox ?

Pipeline Syntax

Advanced

Advanced

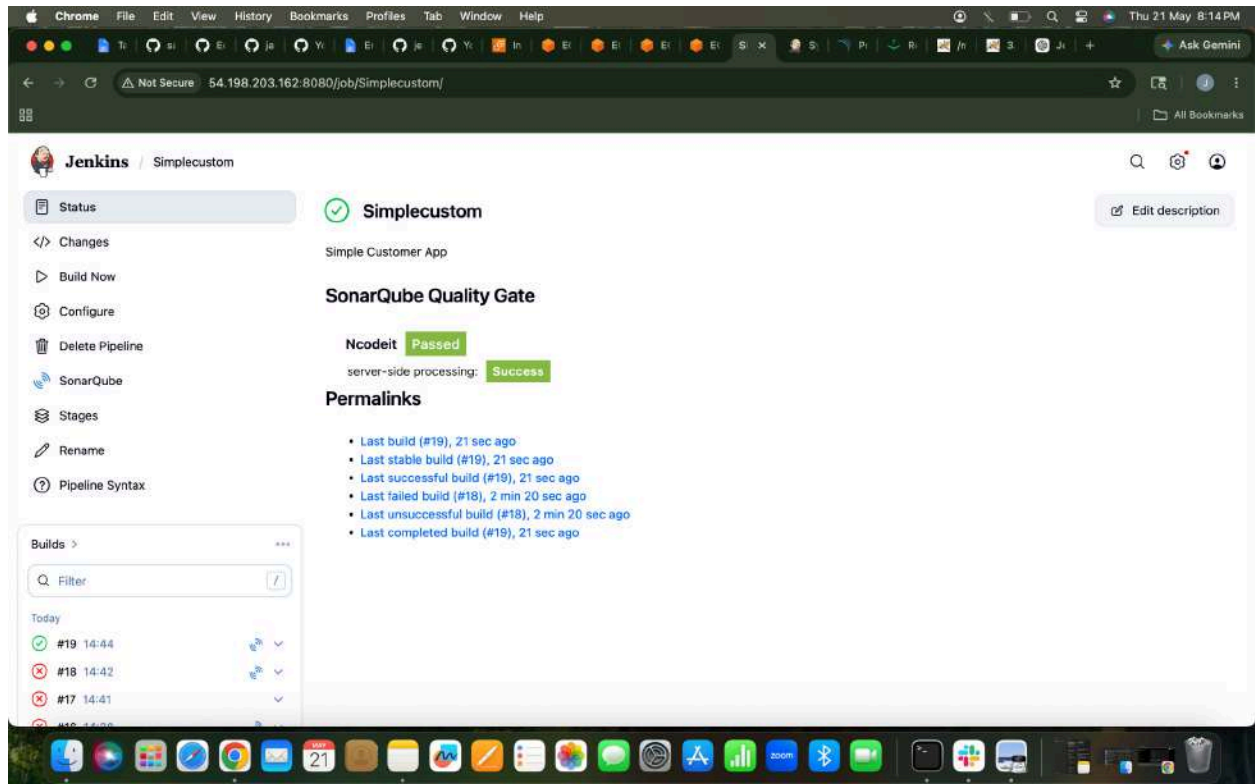
Script ?

```
54 }
55 stage("publish to nexus") {
56   steps {
57     script {
58       // Read POM xml file using 'readMavenPom' step , this step 'readMavenPom' is incl
59       //pom = readMavenPom file: "pom.xml";
60       // Find built artifact under target folder
61       // filesByGlob = findFiles(glob: "target/*.${pom.packaging}");
62       // Print some info from the artifact found
63       echo "${filesByGlob[0].name} ${filesByGlob[0].path} ${filesByGlob[0].directory} $
64       // Extract the path from the File found
65       artifactPath = filesByGlob[0].path;
66       // Assign to a boolean response verifying If the artifact name exists
67       artifactExists = fileExists artifactPath;
68       if(artifactExists) {
```

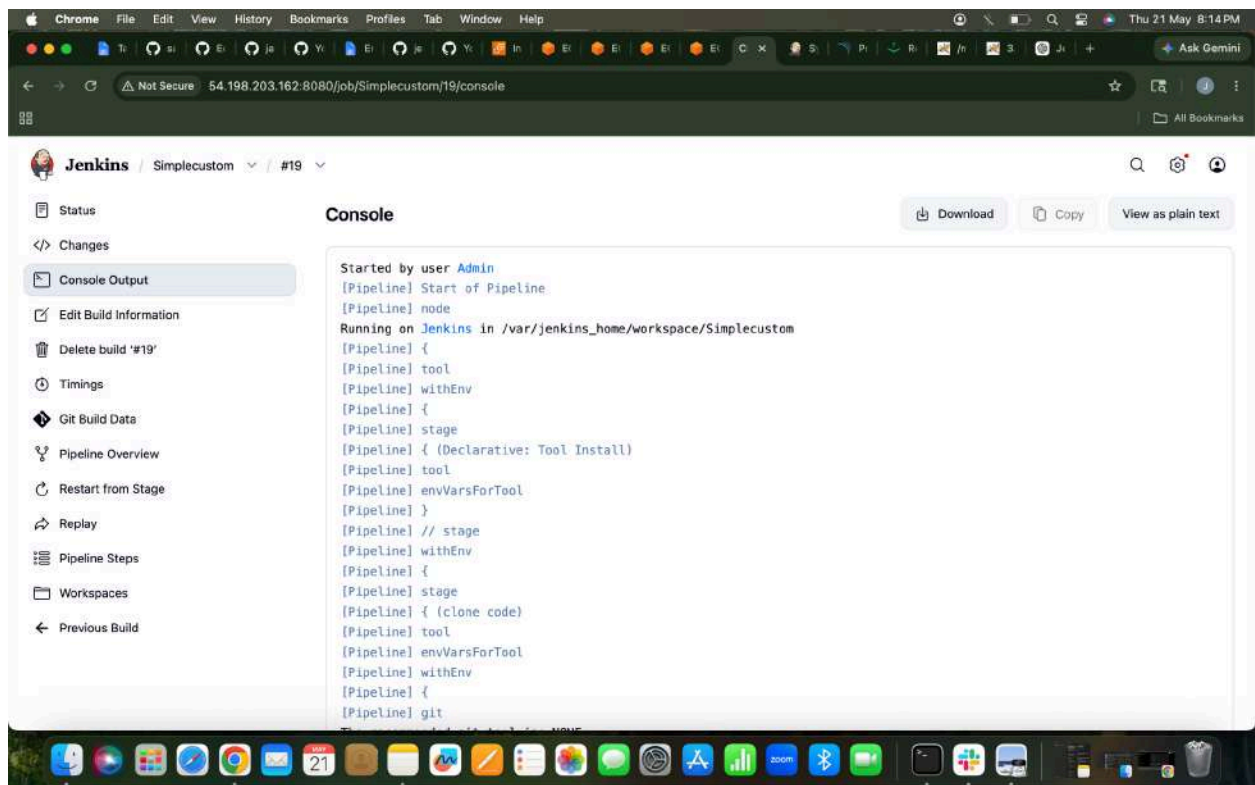
☒ Use Groovy Sandbox ?

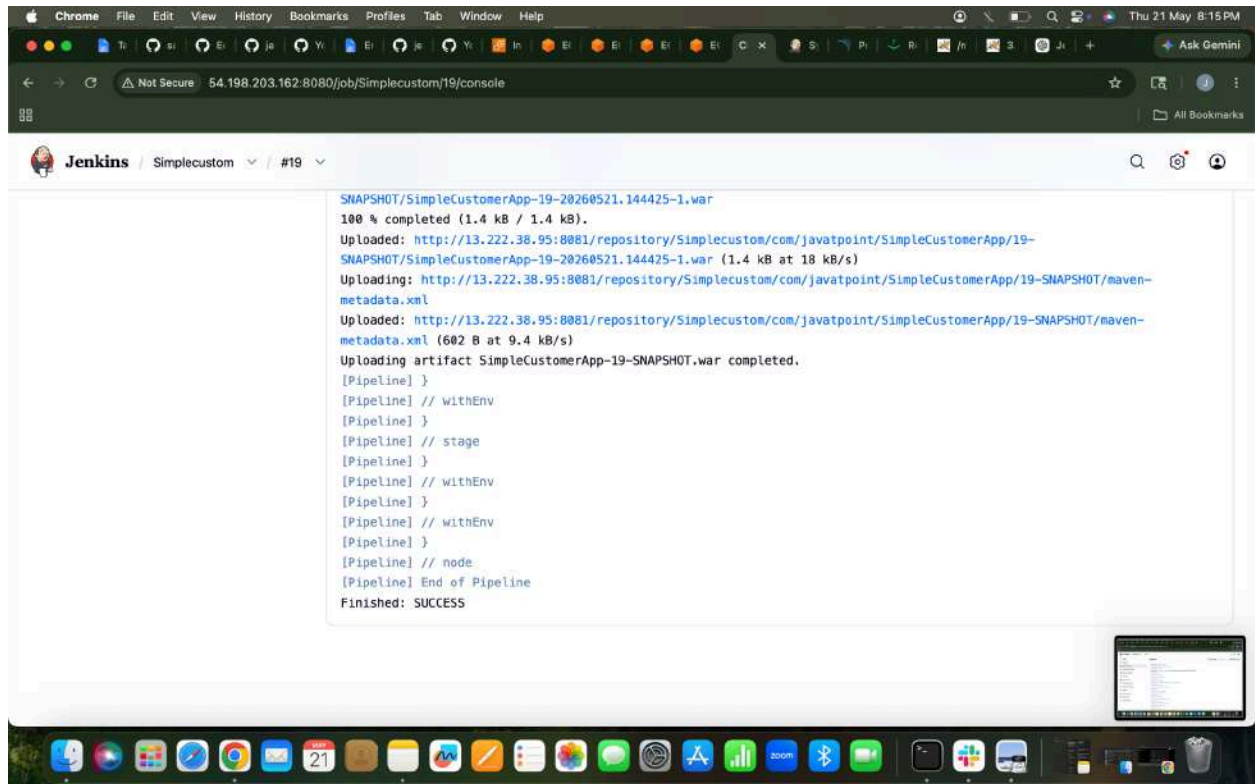
Pipeline Syntax

Build was successful

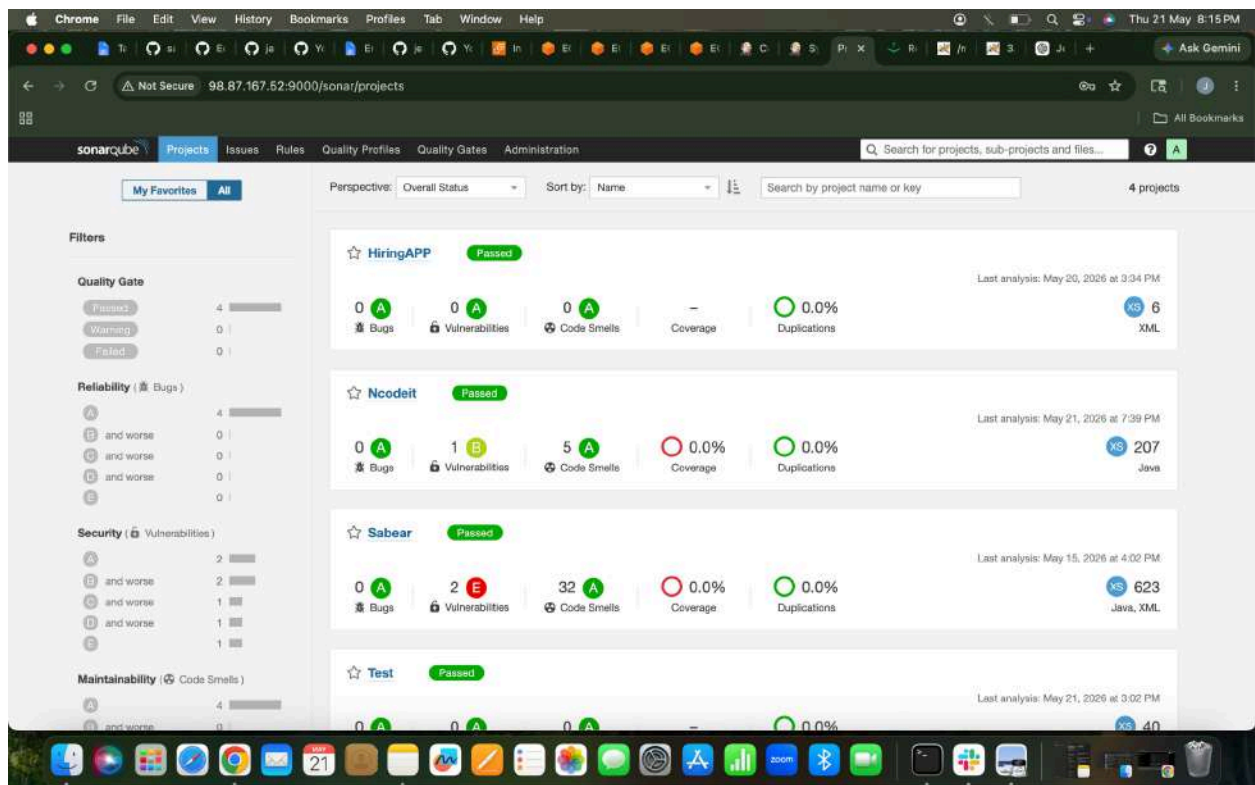


Console Output

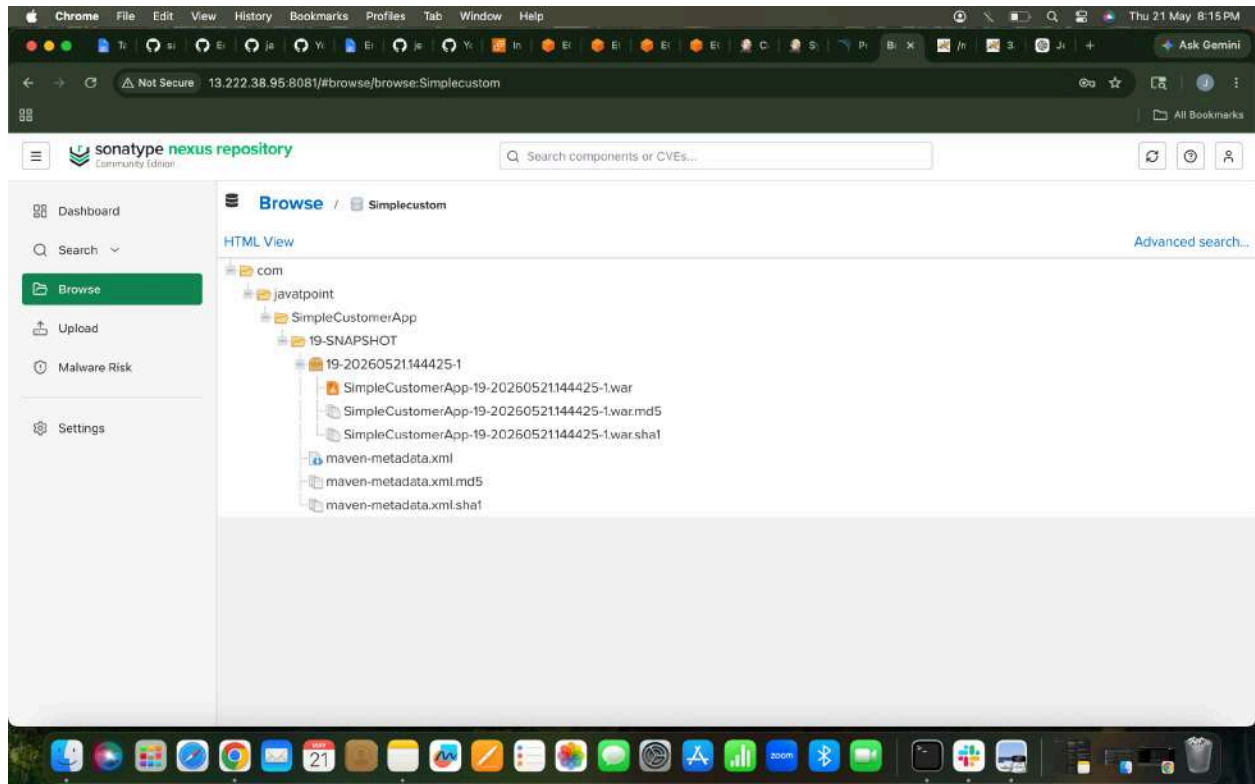




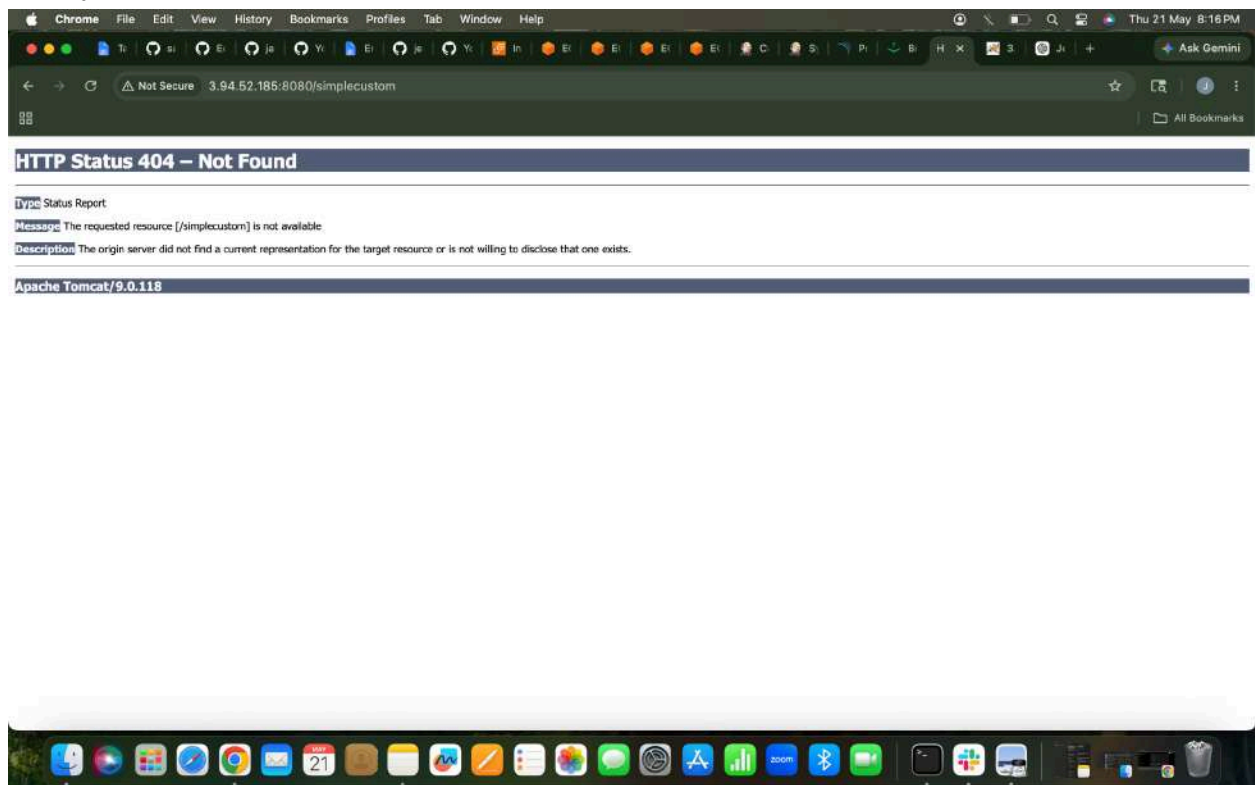
Output in Sonarqube→ncodeit



Nexus Repository Created



Deployment



3) Setup a Jenkins CICD pipeline using Scripted pipeline using feature-1.1 branch.

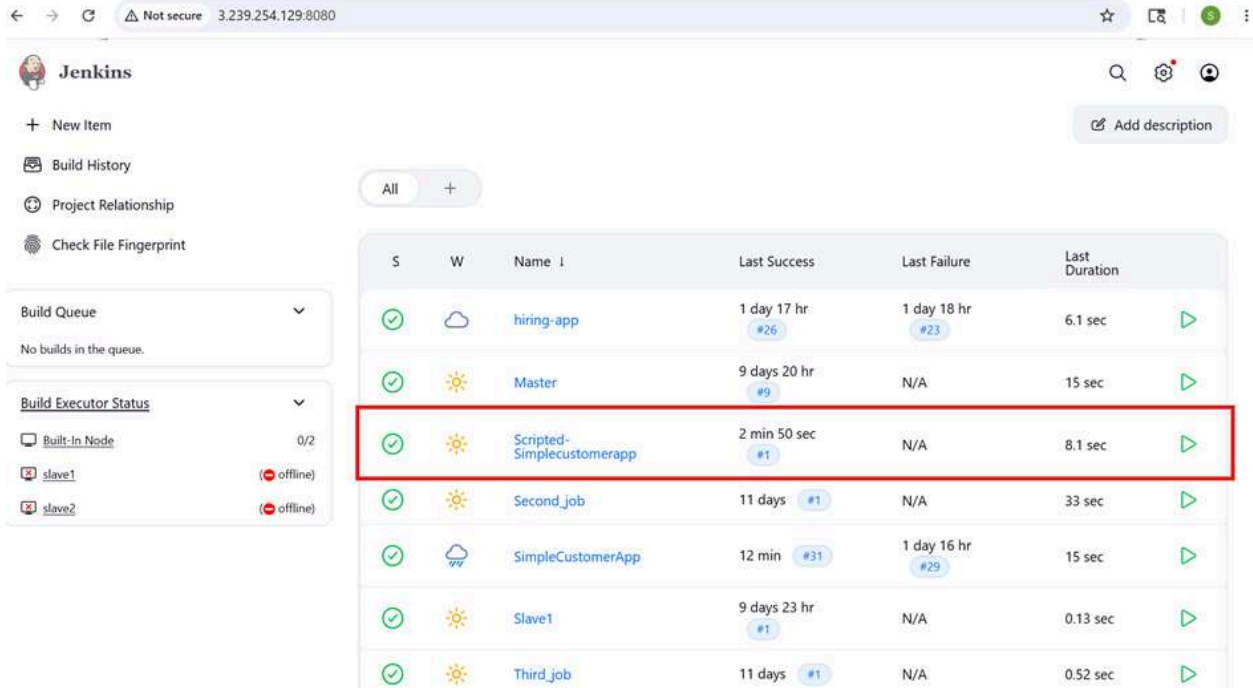
Source Code: https://github.com/betawins/sabear_simplecutomerapp/tree/feature-1.1

stages:

- Git Clone
- SonarQube Integration
- Maven Compilation
- Nexus Artifactory
- Slack Notification

- Deploy On tomcat

Created a new job and added the scripted



The screenshot shows the Jenkins dashboard with a table of recent builds. The build 'Scripted-Simplecustomerapp' is highlighted with a red box. The table has columns for Status (S), Webhook (W), Name, Last Success, Last Failure, and Last Duration.

S	W	Name	Last Success	Last Failure	Last Duration
✓	☁	hiring-app	1 day 17 hr #26	1 day 18 hr #23	6.1 sec
✓	☀	Master	9 days 20 hr #9	N/A	15 sec
✓	☀	Scripted-Simplecustomerapp	2 min 50 sec #1	N/A	8.1 sec
✓	☀	Second_job	11 days #1	N/A	33 sec
✓	☁	SimpleCustomerApp	12 min #31	1 day 16 hr #29	15 sec
✓	☀	Slave1	9 days 23 hr #1	N/A	0.13 sec
✓	☀	Third_job	11 days #1	N/A	0.52 sec

Added the below Script:

```
node {
  def mvnHome = tool 'Maven'
  env.PATH = "${mvnHome}/bin:${env.PATH}"

  try {
    stage('Clean Workspace') {
      cleanWs()
    }

    stage('Clone Code') {
      git branch: 'master',
      url: 'https://github.com/javeebms7/simplecutomerapp.git'
    }

    stage('Build using Maven') {
      sh 'mvn clean install'
    }

    stage('Publish to Nexus') {
      def pom = readMavenPom file: 'pom.xml'
```

```

def releaseVersion = pom.version.replace("-SNAPSHOT", "")
def artifactPath = "target/${pom.artifactId}-${pom.version}.war"
echo "Uploading Artifact: ${artifactPath}"
nexusArtifactUploader(
    nexusVersion: 'nexus3',
    protocol: 'http',
    nexusUrl: '13.218.240.82:8081',
    repository: 'Simplecustomerapp',
    groupId: pom.groupId,
    version: releaseVersion,
    credentialsId: 'nexus',
    artifacts: [
        [
            artifactId: pom.artifactId,
            classifier: "",
            file: artifactPath,
            type: 'war'
        ]
    ]
)
}

stage('Deploy On Tomcat') {
    sh 'ls target/*.war | head -1 > warfile.txt'
    def warFile = readFile('warfile.txt').trim()
    echo "Deploying WAR: ${warFile}"
    deploy adapters: [
        tomcat9(
            credentialsId: 'tomcat',
            path: "",
            url: 'http://54.204.140.94:8080/manager/text'
        )
    ],
    contextPath: "",
    war: warFile
}

// Success Slack Notification
slackSend(
    tokenCredentialId: 'Slack',
    channel: 'C0B5EQ639SQ',
    color: 'good',
    message: "SUCCESS: Job=${env.JOB_NAME} Build=#${env.BUILD_NUMBER} |
${env.BUILD_URL}"

```

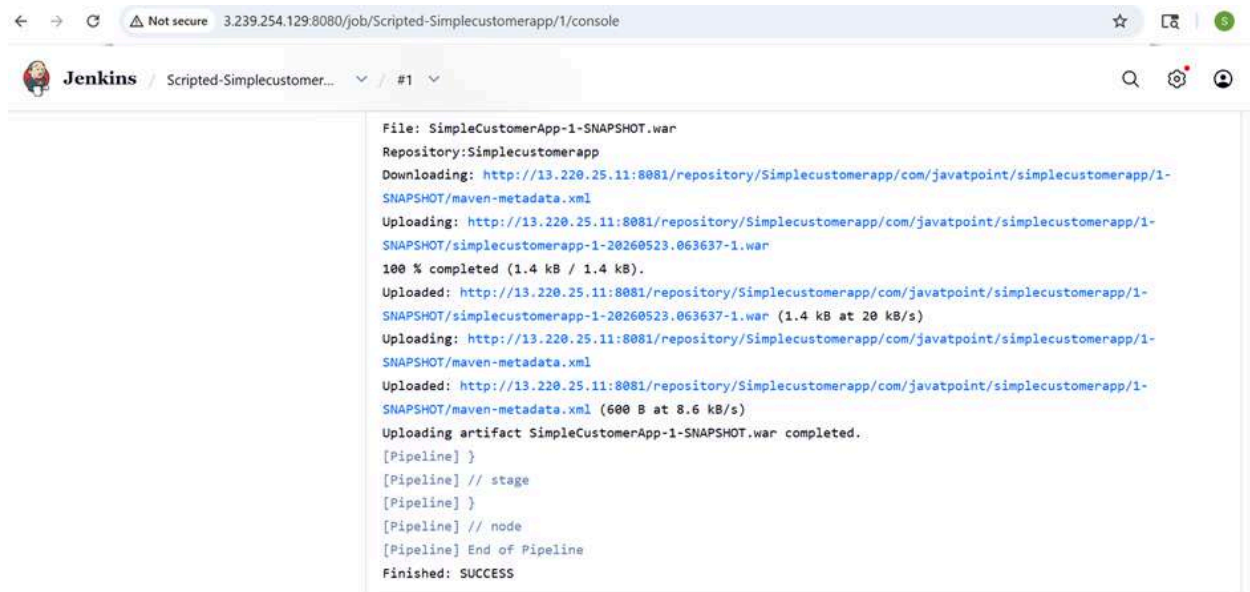
```

    )

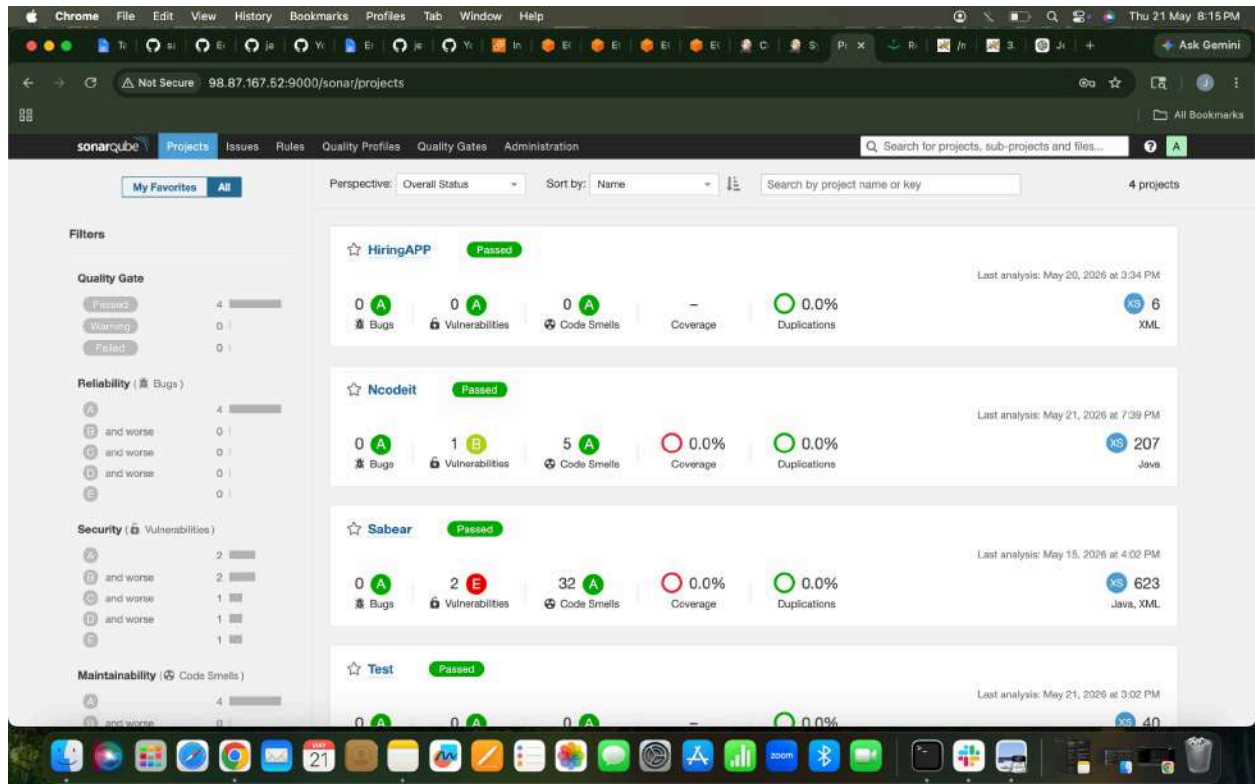
} catch (Exception e) {
    // Failure Slack Notification
    slackSend(
        tokenCredentialId: 'Slack',
        channel: 'C0B5EQ639SQ',
        color: 'danger',
        message: "FAILED: Job=${env.JOB_NAME} Build=#${env.BUILD_NUMBER} |
${env.BUILD_URL}"
    )
    throw e // rethrow so Jenkins marks build as FAILED
}
}
}

```

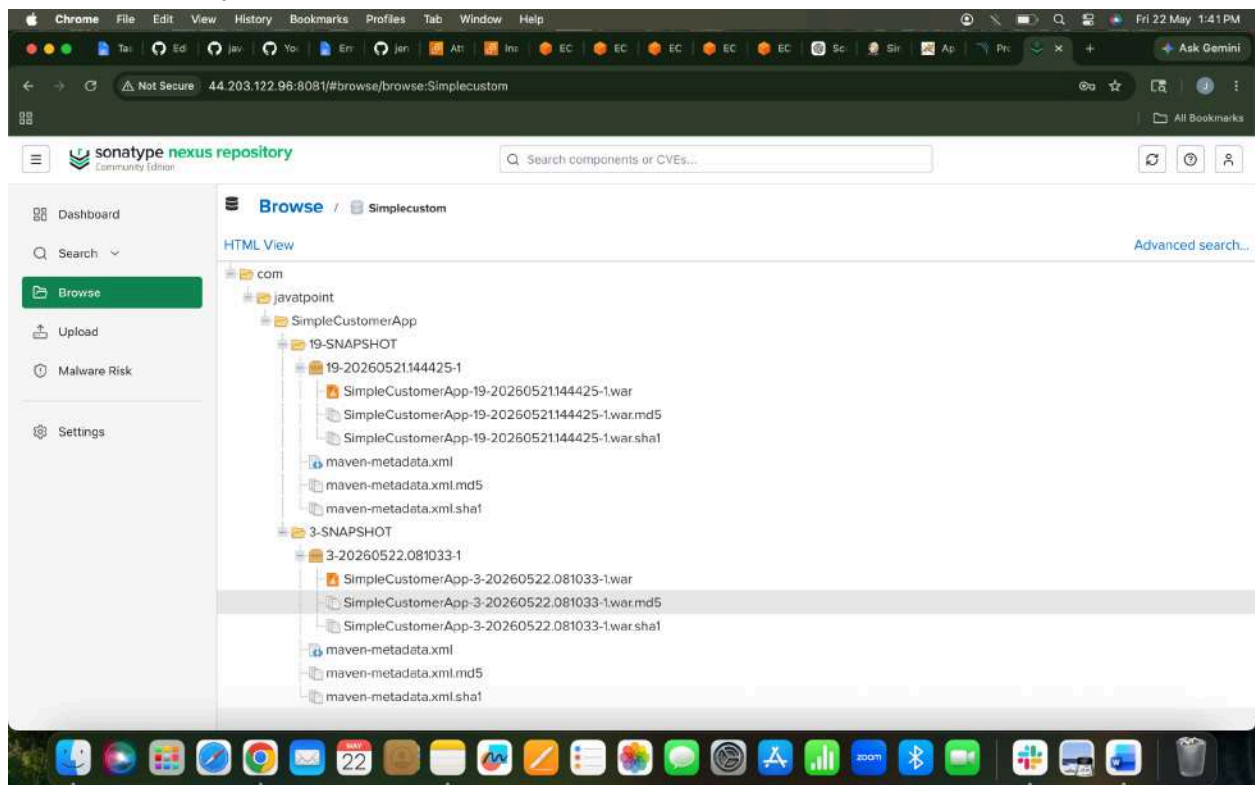
Build Run Successfully



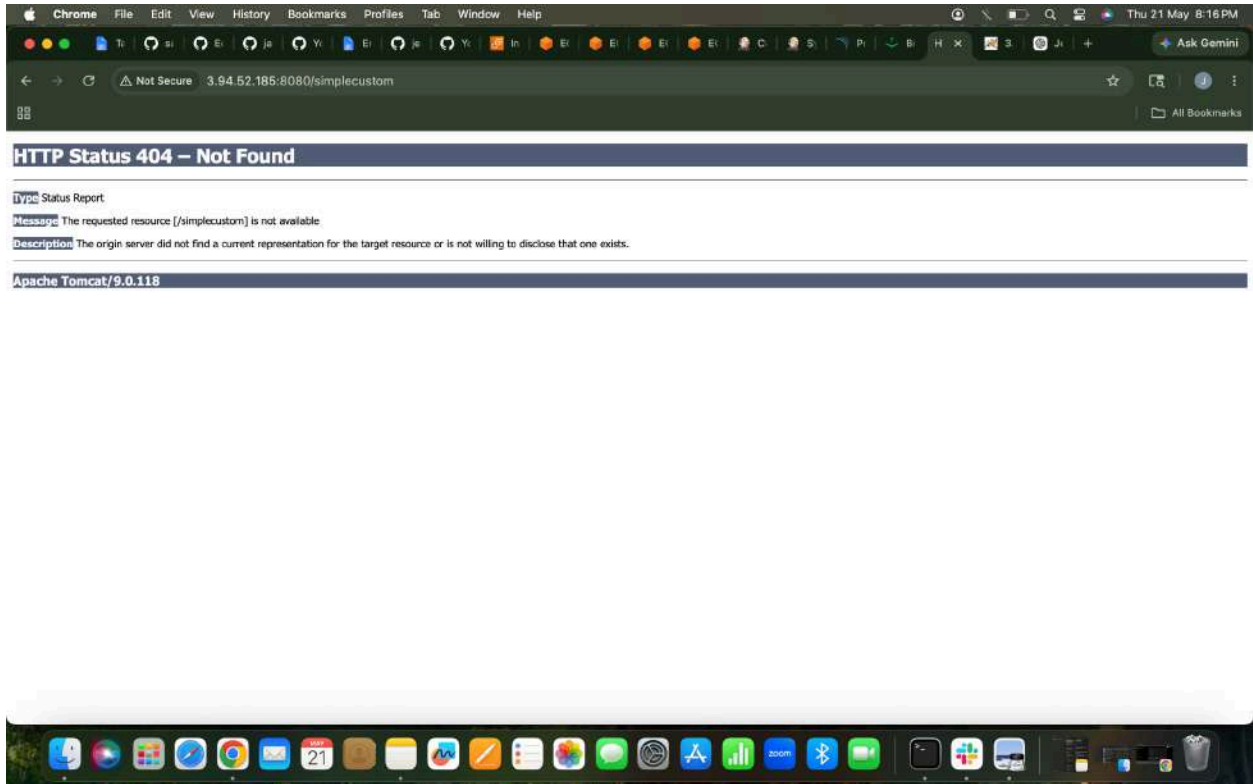
In SonarQube we can see



Nexus Repository is Created



Deploy to Tomcat



4) Write sample skeleton of pipelines.

Common Pipeline Sections Used

- agent → where the job runs (any, label, docker, node)
- tools → JDK, Maven, SonarQube
- environment → variables
- stages → actual CI/CD process
- steps → commands inside each stage
- post → success/failure actions

Basic Declarative Pipeline: Declarative Pipeline is the most commonly used Jenkins pipeline because it is simple and easy to read. It starts with pipeline and agent any, which tells Jenkins to run the job on any available machine. The tools section defines which Java (JDK) and Maven versions Jenkins should use. Then, the pipeline moves through different stages like Checkout, where Jenkins downloads code from GitHub, Build, where Maven compiles the application and creates a package, Test, where automated tests are executed, and Deploy, where the application is pushed to a server or environment. Finally, the post section shows whether the build was successful or failed.

```
pipeline {  
    agent any
```



```
tools {
  maven 'MVN_HOME'
  jdk 'JDK17'
}

environment {
  APP_NAME = "MyApp"
}

stages {

  stage('Checkout') {
    steps {
      echo 'Checking out code...'
      git branch: 'main',
      url: 'https://github.com/our-repo.git'
    }
  }

  stage('Build') {
    steps {
      echo 'Building application...'
      sh 'mvn clean package'
    }
  }

  stage('Test') {
    steps {
      echo 'Running tests...'
      sh 'mvn test'
    }
  }

  stage('Code Quality') {
    steps {
      echo 'Running SonarQube scan...'
      sh 'mvn sonar:sonar'
    }
  }

  stage('Deploy') {
    steps {
      echo 'Deploying application...'
    }
  }
}
```

```

    }
}

post {
    always {
        echo 'Pipeline execution completed'
    }

    success {
        echo 'Build Successful'
    }

    failure {
        echo 'Build Failed'
    }
}
}

```

Basic Scripted pipeline: Scripted Pipeline is a more flexible way of writing Jenkins automation using Groovy scripting. Unlike Declarative Pipelines, it uses `node` instead of `pipeline` and gives developers more control over logic and conditions. It contains stages like Checkout to download code, Build to compile the project, Test to run test cases, and Deploy to release the application. Scripted Pipelines are generally used for more complex workflows because they allow advanced customization, but they can be harder to understand for beginners.

```

node {

    stage('Checkout') {
        echo 'Checking out code...'
        git branch: 'main',
        url: 'https://github.com/our-repo.git'
    }

    stage('Build') {
        echo 'Building...'
        sh 'mvn clean package'
    }

    stage('Test') {
        echo 'Running tests...'
        sh 'mvn test'
    }
}

```

```

    stage('Deploy') {
        echo 'Deploying...'
    }
}

```

Multibranch Pipeline Skeleton: Multibranch Pipeline is used when a project has multiple Git branches such as main, dev, or feature branches. Jenkins automatically detects each branch and creates separate jobs for them. Instead of manually specifying the repository and branch, it uses checkout scm, which automatically pulls code from the branch being built. This type of pipeline helps teams work on multiple features at the same time because Jenkins can automatically build and test every branch independently without creating separate jobs manually.

```

pipeline {
    agent any

    stages {

        stage('Checkout') {
            steps {
                checkout scm
            }
        }

        stage('Build') {
            steps {
                sh 'mvn clean install'
            }
        }

        stage('Test') {
            steps {
                sh 'mvn test'
            }
        }
    }

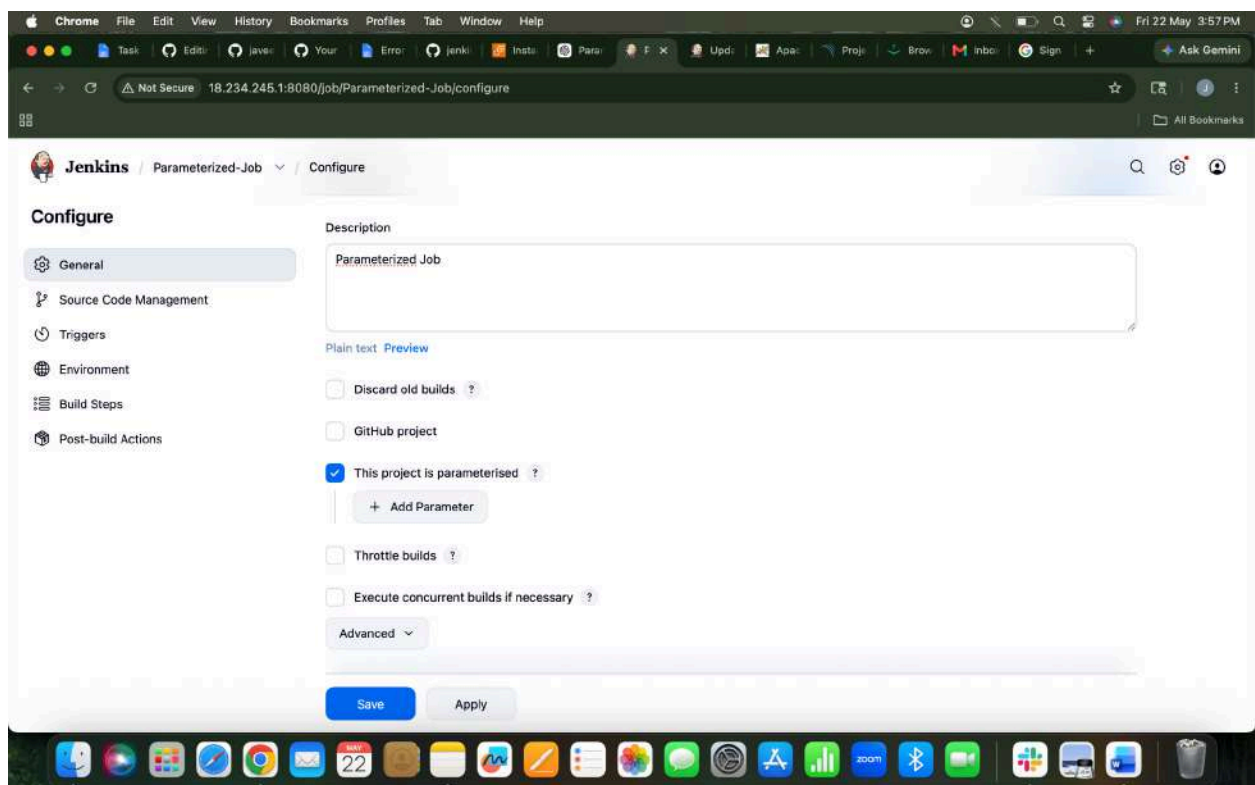
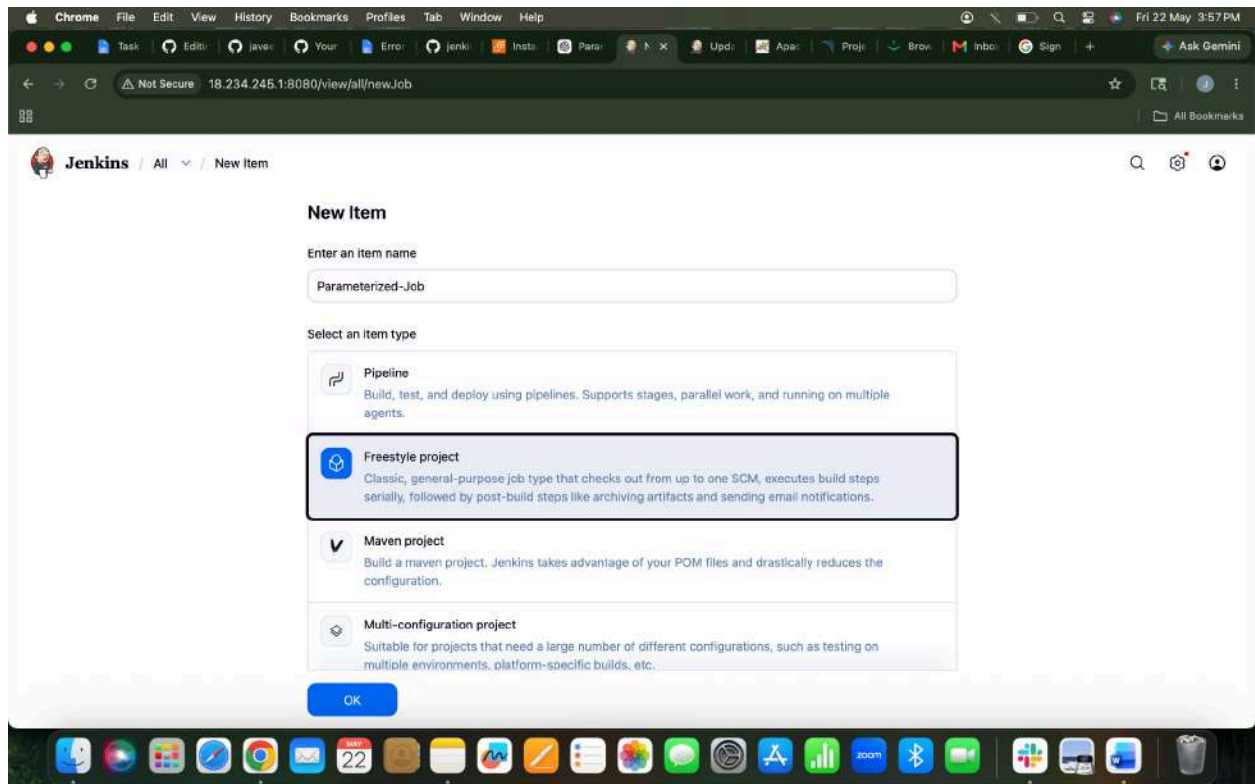
    post {
        success {
            echo "Build successful for branch: ${env.BRANCH_NAME}"
        }
    }
}

```

5) Create a parameterized job in Jenkins.

Source Code: <https://github.com/betawins/spring3-mvc-maven-xml-hello-world-1.git>

Creating a New Job→Freestyle→select this project is parameterized→select add parameter→



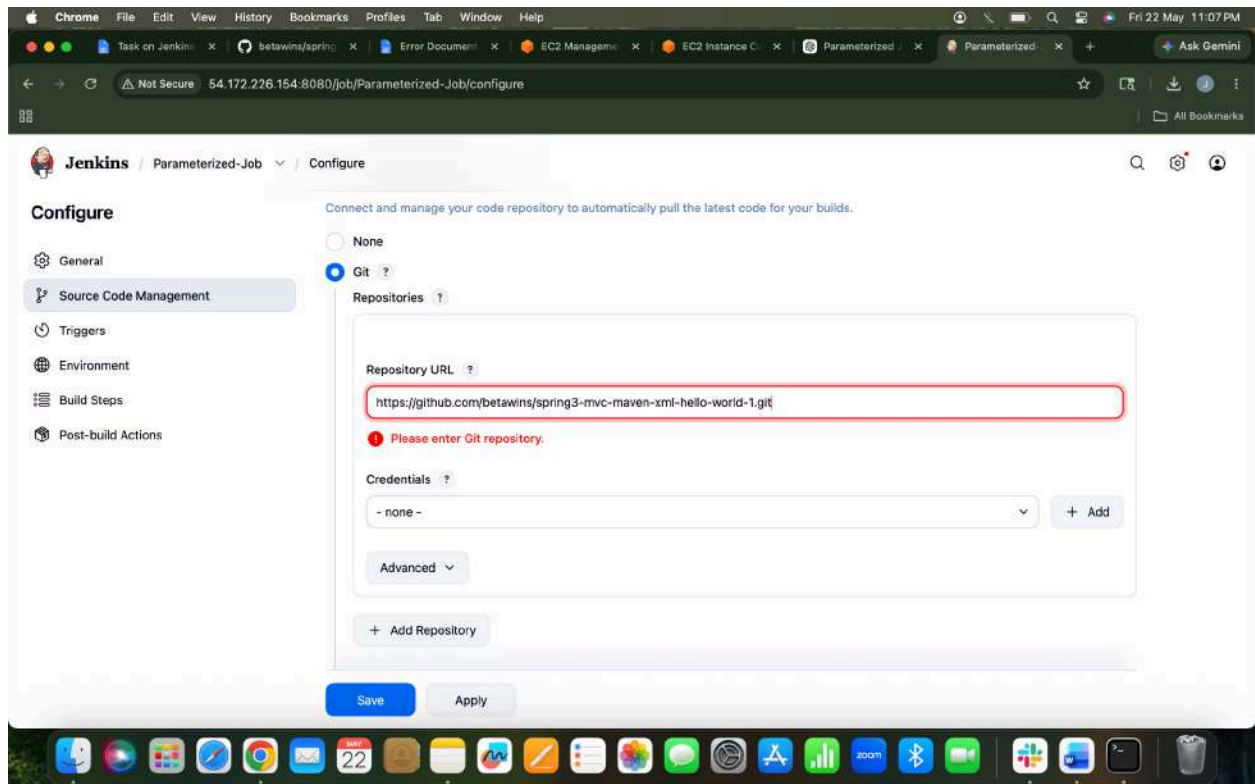
This Parameter is using for Github⇒ RepoString Parameter→Name(BRANCH) →Default Value(master) → Description

The screenshot shows the Jenkins web interface for configuring a parameterized job. The left sidebar contains a 'Configure' menu with options: General, Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions. The main area is titled 'Configure' and shows a checkbox 'This project is parameterised' which is checked. Below this, a 'String Parameter' configuration box is visible. It has three fields: 'Name' with the value 'BRANCH', 'Default Value' with the value 'master', and 'Description' with the placeholder text 'Enter Git branch name'. There are also links for 'Plain text' and 'Preview', and a checkbox for 'Trim the string' which is unchecked. At the bottom of the configuration box are 'Save' and 'Apply' buttons. The browser's address bar shows the URL '54.172.226.154:8080/job/Parameterized-Job/configure'.

This Parameter is using for Maven
RepoString Parameter→Name(Maven_Goal) →Default Value(clean install) → Description
⇒This lets you choose which Maven command Jenkins should execute.

The screenshot shows the Jenkins web interface for configuring a parameterized job. The left sidebar contains a 'Configure' menu with options: General, Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions. The main area is titled 'Configure' and shows a checkbox 'This project is parameterised' which is checked. Below this, a 'String Parameter' configuration box is visible. It has three fields: 'Name' with the value 'MAVEN_GOAL', 'Default Value' with the value 'clean package clean install clean verify', and 'Description' with the placeholder text 'Select Maven build goal'. There are also links for 'Plain text' and 'Preview', and a checkbox for 'Trim the string' which is unchecked. At the bottom of the configuration box are 'Save' and 'Apply' buttons. The browser's address bar shows the URL '54.172.226.154:8080/job/Parameterized-Job/configure'.

In SCM ⇒ Select Git⇒add git repo link→Branch(master)

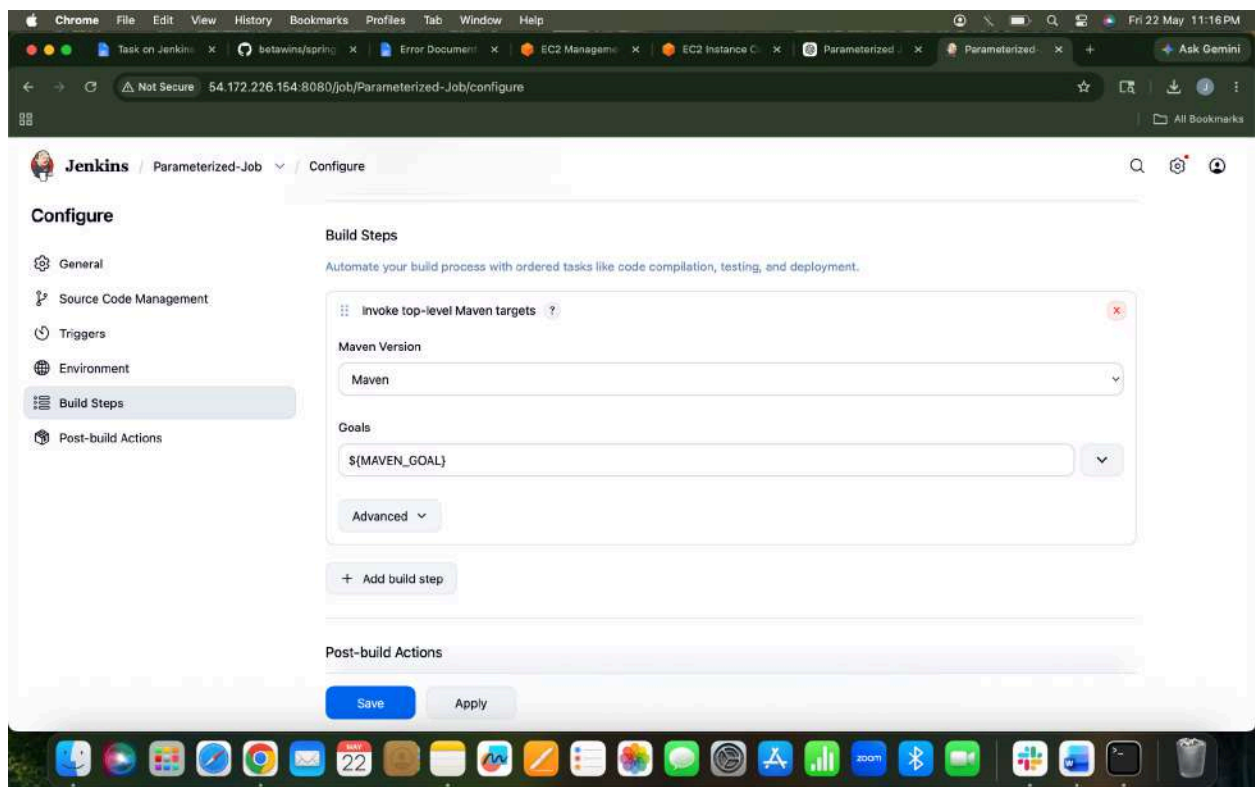
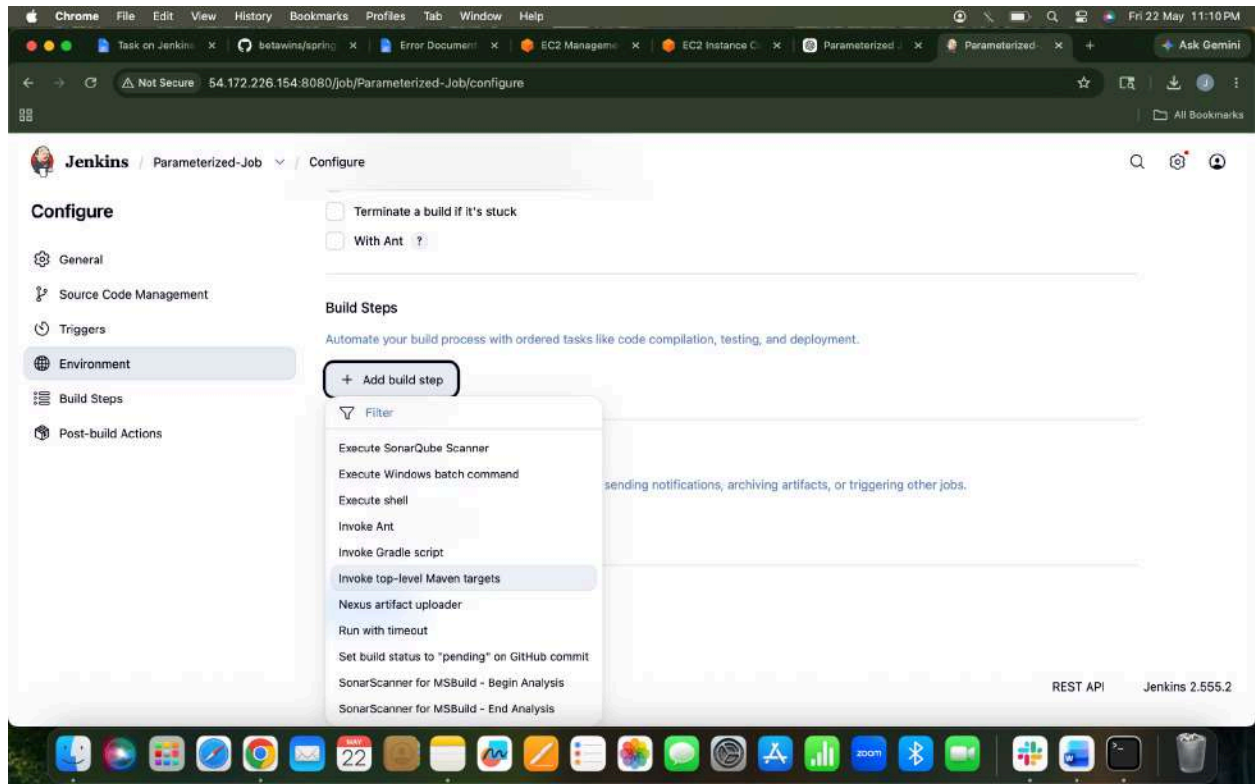


Configure Build Step⇒In Build section→Click Add Build Step→Choose Invoke top-level Maven targets

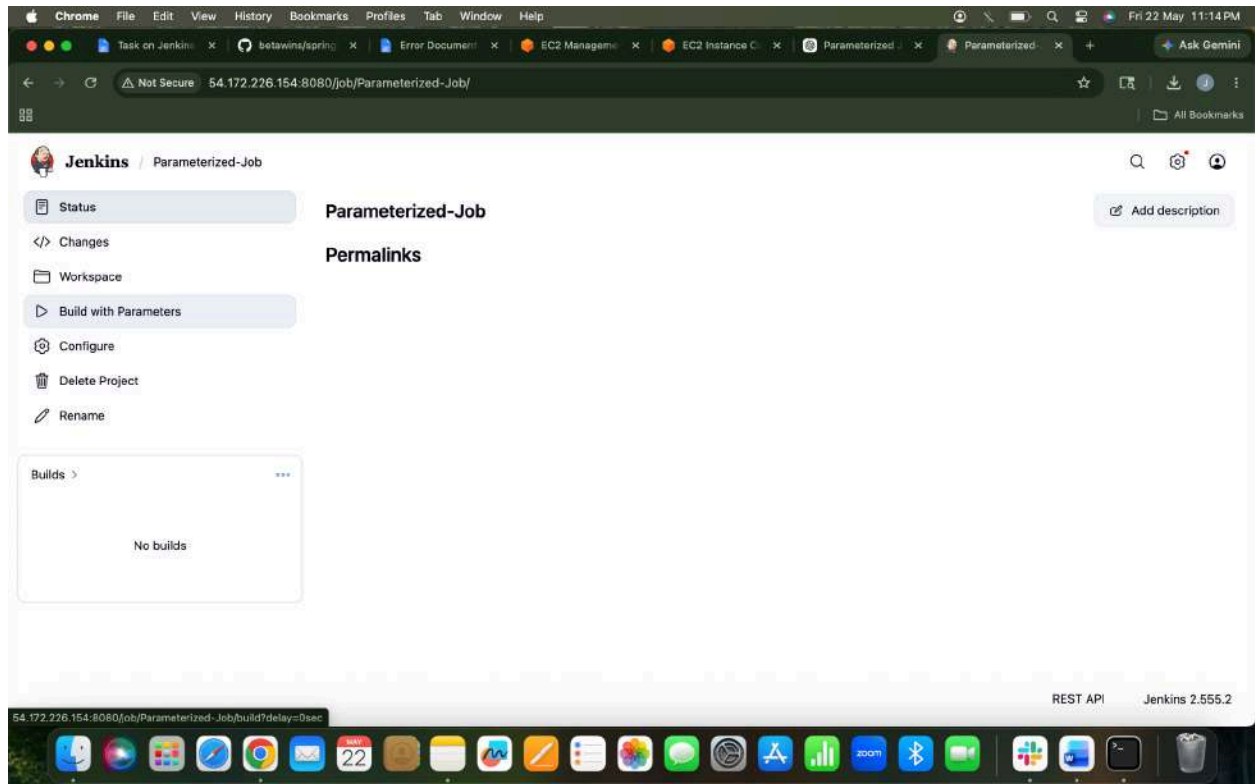
As Maven is configured in Jenkins, select Maven installation.

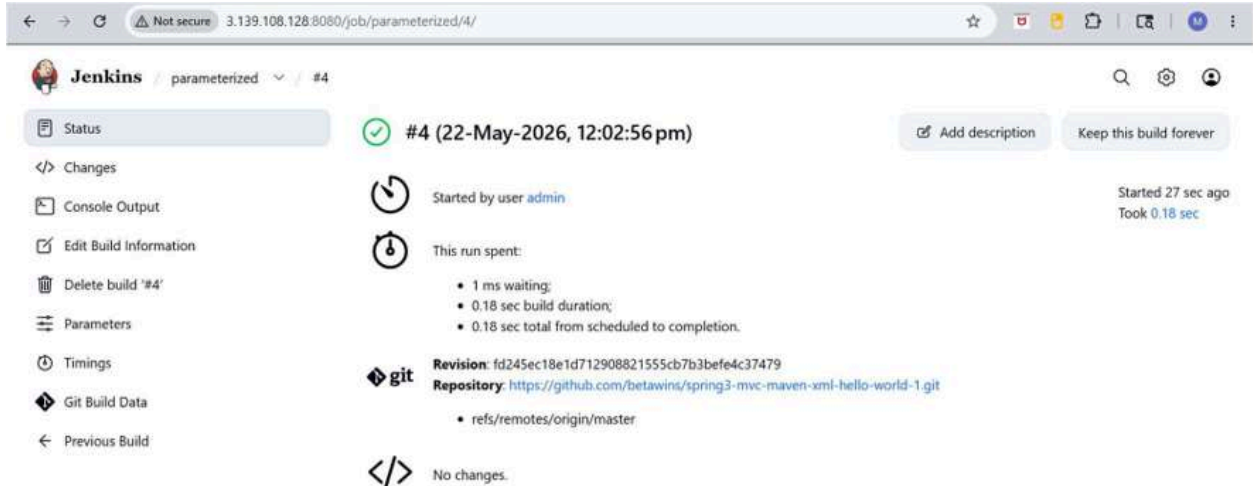
Goals: \${MAVEN_GOAL}

This makes the Maven command parameterized.

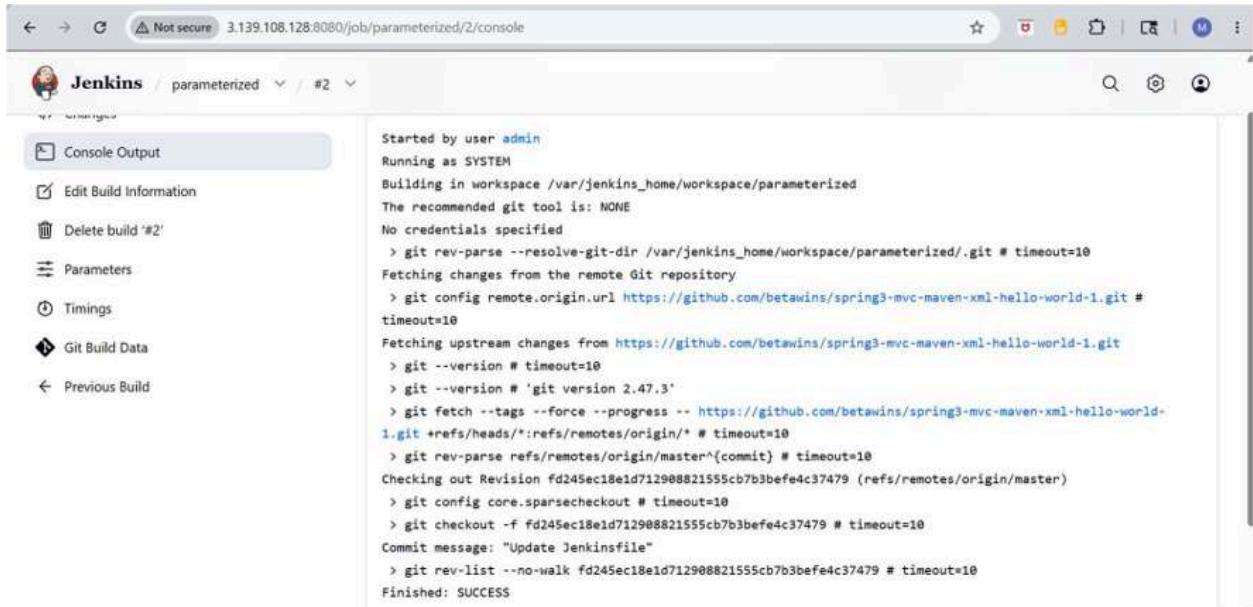


Lets Build The job with parameters now





The screenshot shows the Jenkins web interface for job #4. The browser address bar indicates the URL is 3.139.108.128:8080/job/parameterized/4/. The Jenkins logo and job name 'parameterized' are at the top. A sidebar on the left contains links: Status, Changes, Console Output, Edit Build Information, Delete build '#4', Parameters, Timings, Git Build Data, and Previous Build. The main content area shows a green checkmark icon and the text '#4 (22-May-2026, 12:02:56 pm)'. Below this, it says 'Started by user admin' and 'This run spent: 1 ms waiting, 0.18 sec build duration, 0.18 sec total from scheduled to completion.' It also shows the 'Revision' as 'fd245ec18e1d712908821555cb7b3befe4c37479' and the 'Repository' as 'https://github.com/betawins/spring3-mvc-maven-xml-hello-world-1.git'. At the bottom, it says 'No changes.'



The screenshot shows the Jenkins web interface for job #2, specifically the 'Console Output' tab. The browser address bar indicates the URL is 3.139.108.128:8080/job/parameterized/2/console. The Jenkins logo and job name 'parameterized' are at the top. The sidebar on the left contains links: Console Output, Edit Build Information, Delete build '#2', Parameters, Timings, Git Build Data, and Previous Build. The main content area displays the console output for the build, which includes the following text:

```
Started by user admin
Running as SYSTEM
Building in workspace /var/jenkins_home/workspace/parameterized
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /var/jenkins_home/workspace/parameterized/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/betawins/spring3-mvc-maven-xml-hello-world-1.git #
timeout=10
Fetching upstream changes from https://github.com/betawins/spring3-mvc-maven-xml-hello-world-1.git
> git --version # timeout=10
> git --version # 'git version 2.47.3'
> git fetch --tags --force --progress -- https://github.com/betawins/spring3-mvc-maven-xml-hello-world-
1.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/master^{commit} # timeout=10
Checking out Revision fd245ec18e1d712908821555cb7b3befe4c37479 (refs/remotes/origin/master)
> git config core.sparsecheckout # timeout=10
> git checkout -f fd245ec18e1d712908821555cb7b3befe4c37479 # timeout=10
Commit message: "Update Jenkinsfile"
> git rev-list --no-walk fd245ec18e1d712908821555cb7b3befe4c37479 # timeout=10
Finished: SUCCESS
```

The Job ran successfully

6) Setup one slave machine for Jenkins.

Main Thing to be taken into consideration before create a Master→ Slave is the JAVA
→versions should be same

Create a new instance and Install Java and create ssh-keygen

```
aws [Option+S] Ask Amazon Q

tracker-3.7.3-3.amzn2023.0.1.x86_64
upower-1.90.6-139.amzn2023.x86_64
webrtc-audio-processing-0.3.1-6.amzn2023.0.2.x86_64
wireplumber-libs-0.5.7-1.amzn2023.0.1.x86_64
xdg-desktop-portal-gtk-1.15.1-61.amzn2023.x86_64
xml-common-0.6.3-56.amzn2023.0.2.noarch

Complete!
[root@ip-172-31-1-51 ~]# ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_rsa
Your public key has been saved in /root/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:09wjzChRLxkERW7k0v8t8seQ8zd0nJXeO0mcbyOdfw4 root@ip-172-31-1-51.ec2.internal
The key's randomart image is:
+---[RSA 3072]-----+
|      .+=          |
|      *            |
|      . B          |
|      + =          |
|      . o S . .++  |
|      . * ++. =O+   |
|      . . O ==.+E+. |
|      .   = O+.*=O  |
|      . . ..=B     |
+---[SHA256]-----+
[root@ip-172-31-1-51 ~]#
```

i-02a9fe1ab41f58afc (Jenkins-Slave)

PublicIPs: 44.202.93.225 PrivateIPs: 172.31.1.51

CloudShell Feedback Console Mobile App

Now we need to copy the newly created ssh-keygen to authorized keys and give it necessary permissions


```
aws | [AWS Logo] | [Search] | [Terminal] | [Notifications] | [Help] | [Settings] | United States (N. Virginia) ▼

cat: id_rsa.pub: No such file or directory
ec2-user@ip-172-31-1-51 .ssh]$ cd
ec2-user@ip-172-31-1-51 ~]$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/ec2-user/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ec2-user/.ssh/id_rsa
Your public key has been saved in /home/ec2-user/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:HwrO9T5V+1LUpf1AyTEEm1bBNvpt3CrBoBf3OhVA8UY ec2-user@ip-172-
The key's randomart image is:
---[RSA 3072]---+
      o**Eo |
      +=o+.. |
      +o +o+. |
      .+ ..* o |
      . S..* B +. |
      o o.+..* B o |
      o ..o. B . |
      .. = o |
      .. + |
-----[SHA256]-----+
ec2-user@ip-172-31-1-51 ~]$ cd /home/ec2-user/.ssh/
ec2-user@ip-172-31-1-51 .ssh]$ cat id_rsa.pub > authorized_keys
ec2-user@ip-172-31-1-51 .ssh]$ chmod 700 authorized_keys
ec2-user@ip-172-31-1-51 .ssh]$
```

i-02a9fe1ab41f58afc (Jenkins-Slave)

PublicIPs: 44.202.93.225 PrivateIPs: 172.31.1.51

First verify PEM exists on master ⇒ ls -ltr ⇒ If not present, upload/copy by creating a new file `test.pem` to master. From master as we are using docker we need to copy our file to docker container. Command used to copy → `docker cp slave.pem jenkins:/root/`

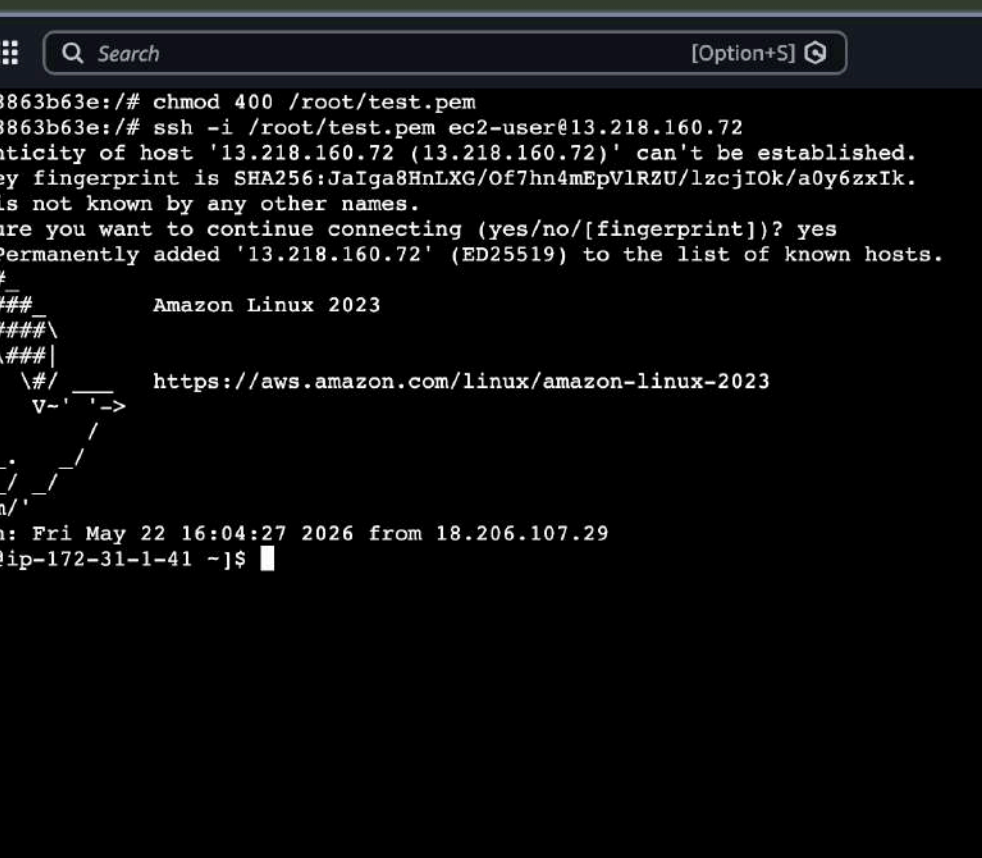
```
ec2-user@ip-172-31-0-39 ~]$ ls
test.pem
ec2-user@ip-172-31-0-39 ~]$ docker cp test.pem jenkins:/root/
no such directory
ec2-user@ip-172-31-0-39 ~]$ docker images
permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Head "http://%2Fvar%2Frun%2Fdocker.sock/_ping": dial uni
x /var/run/docker.sock: connect: permission denied
ec2-user@ip-172-31-0-39 ~]$ sudo docker cp test.pem jenkins:/root/
Successfully copied 3.58kB to jenkins:/root/
ec2-user@ip-172-31-0-39 ~]$
```

i-02b46c38bfb019c3 (Jenkins-Docker)

PublicIPs: 54.172.226.154 PrivateIPs: 172.31.0.39

Next⇒Test SSH⇒ssh -i /root/slave.pem ec2-user@44.202.93.225

Do not continue until this works. If this step fails, Jenkins agent will never work.

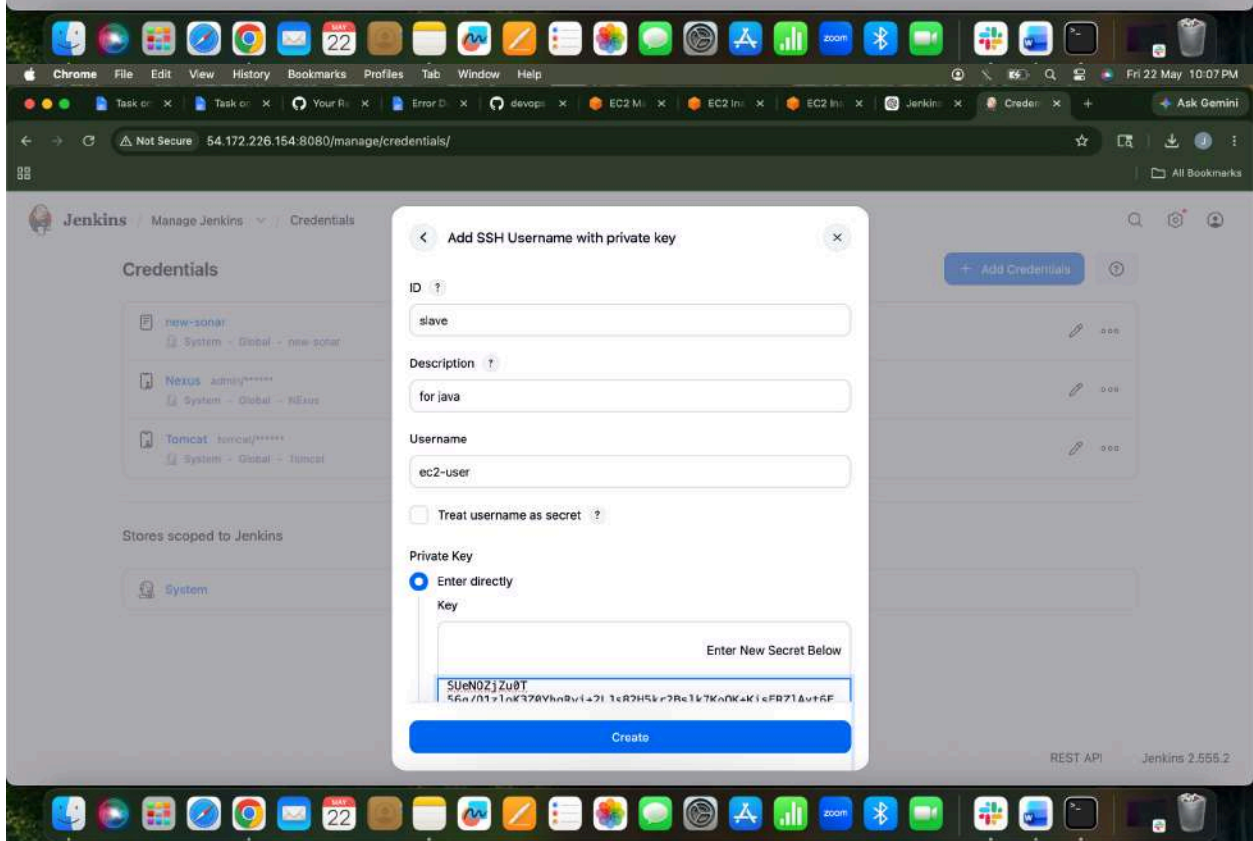
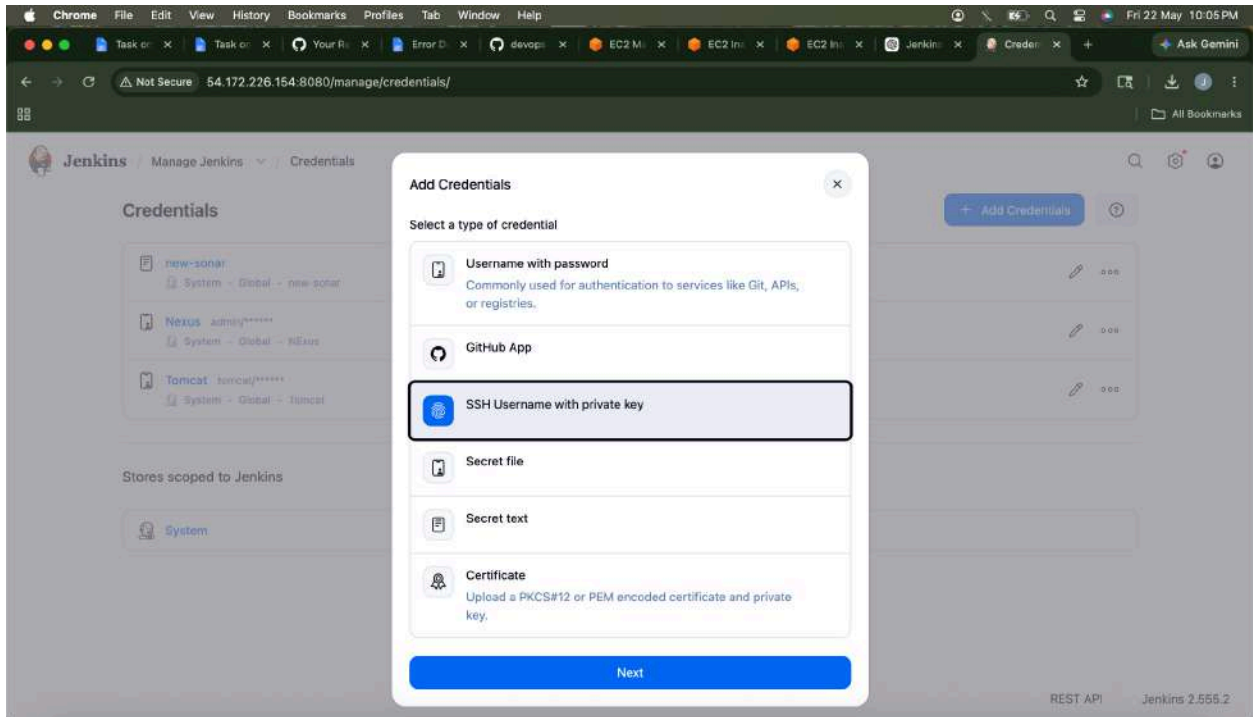


```
root@2c288863b63e:/# chmod 400 /root/test.pem
root@2c288863b63e:/# ssh -i /root/test.pem ec2-user@13.218.160.72
The authenticity of host '13.218.160.72 (13.218.160.72)' can't be established.
ED25519 key fingerprint is SHA256:JaIga8HnLXG/Of7hn4mEpVlRZU/lzcjIOk/a0y6zxIk.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '13.218.160.72' (ED25519) to the list of known hosts.

      _
     _\##### Amazon Linux 2023
    _\#####\
   _\#####\
  _\#####\
 _\#####\
V- ' ' -> https://aws.amazon.com/linux/amazon-linux-2023

Last login: Fri May 22 16:04:27 2026 from 18.206.107.29
[ec2-user@ip-172-31-1-41 ~]$
```

Now lets go to Jenkins GUI => create credentials using the pem key

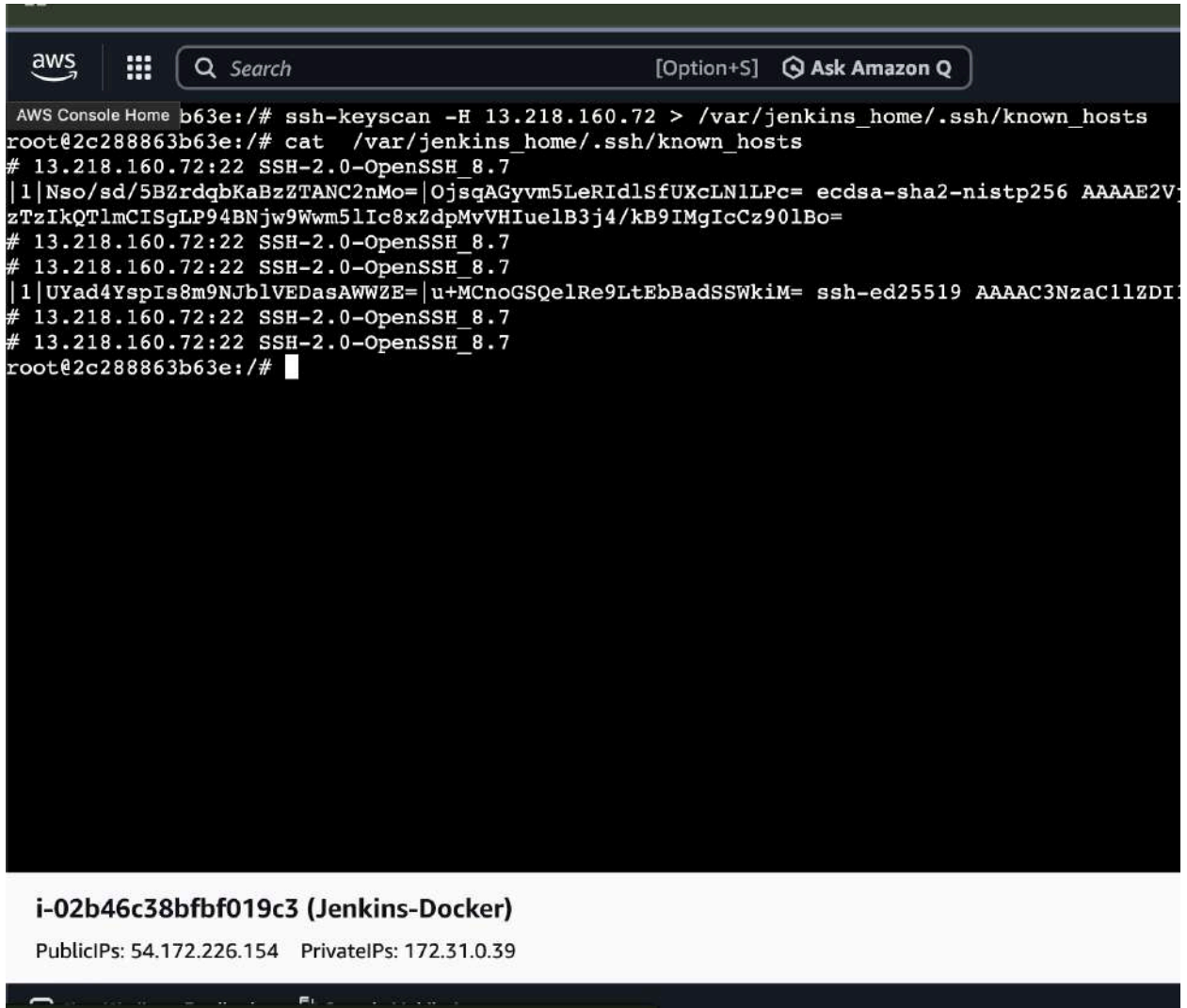


```
Connection to 13.218.160.72 closed.
root@2c288863b63e:/# cat /root/test.pem
-----BEGIN RSA PRIVATE KEY-----
MIIEowIBAAKCAQEAX9WAEPLeZmtPZToEMCaTEuIK6anOhJLxr9dntRTHqIHI2mrP
SqFX8xBn4dW/0IofHBs1gr5sniG4zVWiAlRbAL1PXiIHda9ney8hPeaypaXetSUj
0oVpW0v5BeTQOOR/TVzIK+ls9G5zK39Is8xy7CjMT7rXXFROMZJ78z+4wcZAiRBY
tJfIaGcfmFqk80gE5hhAzEDzkie5b1C6uoPK1m23Yheerb2KDT1E5G7zvWuM/zum
gcKkDaaZx7v9tKeXwEdjH4gP8JTxolT8w7Gb8QE0boACdeYkzX5fyMk60/iTpETX
x2pWWXLYBjj1Ue3ZMiaCoGD1BcxmySUGdXEugwIDAQABAOIBAApa4dv++c9M2Dxi
mTz1gsGDvRUGm5D7HFMEUndrsDznHVMUq5EucyRBQWD0drYTsJD+p/MycuZu0Oiy
+0YroNuU87vo/4H1qOMVOh+/CJRu0YmJdyW9jAvzjl/jz3ZjQMV9fVHpMqAn6Qfj
/Jx0hlCwdh/birVaGQMwMAOu1aAU0mI4OgHNT1BHUFKZDqDGDnfqIbAhxQXVCEMk
YQzSEW8NJcTe6MISqwcF0Mi7jqv7OjyfMo05PwiJZYdZr1U9K99NAU95qfDqSA3d
PTOpov4v8Q+7TektUfLElyGVRQsLp5y48ijXWkK8WeknuexZVgCe+YTuBMJLy1VN
pIb6saECgYEA6VYv4g+zIBEwwLuaCBG4M2x1eQgG2z1BQ1ASw1GN0hwv1NAfA/oN
0U+Yhco0x6D6nw0N+gn7qvI13kcQ+1Uc1mBHWBdf0Qn55tfl4AkV7movhntpXEZg
33cKIod29FG411EmG9SKNZ/7gtgDbMLbkD0jsFPW1axrH8XvCmFdOe8CgYEA2z5I
qcBajCXwq4KcW3PKYcKvBbYYicDGCykk7cQW+F3yiiURNQMISfGUFs6qZnQKMGG
FauuZV+d00RryncF/3RwKq6Ly4V+7TrOxJ01ZmEHaj88N3zFQus4F2ZhJ21kBkRN
jkh118kRx/NJgQPR4a+VzrAUZeI71w/gg2tHeK0CgYAwBqZKYDLdKcygDrzwZj1E
kFFJfQZ0Pr+Y14RU7JZFJGOEPKX1mpUXfMuG1gTev9plvCkhSsa686YprvTmoXgT
A7QLg0Wd7/C4u707tIMIFGD9zA7uiCnONOEoJDuuo9dofUFbvw5HZUt0Lgnea+jn
Mj5TP55bUmamHoZJ26pljwKBgQDDxb0q35m7VdGvxSSLwTyxyR9ykSUeNOZjZu0T
56q/Q1zloK3Z0YhgRvj+2LJs82H5kr2Bs1k7KoOK+KjsFRZ1Avt6EXzBUsaBXabv
CIgyAGiWMSKZpa+bzuC6fHA6vcK4mdtLNziIrkJTctpg+xc6WQkD1FXSmI7ovsOz
dmNOzQKBgGIst8dlRSmTpU4926/+E7/qGAzhEY36bFI0oFtWN0o+kwdXLJ6T9jeJ
FypI2HwcEnkfZveJDaBa2BueKXsP5hmBL5gh7cEQhQORUNX2orDrQR0IhHhuWoX6
S2A4NOzZFnzSFgy4rEzWjcLHM8AjgcCfnLjW2+bYKcmB/42Goj4q
-----END RSA PRIVATE KEY-----
root@2c288863b63e:/#
```

i-02b46c38bfbf019c3 (Jenkins-Docker)

PublicIPs: 54.172.226.154 PrivateIPs: 172.31.0.39

Now we need to go to Master Jenkins and create a directory and in that we need to create a new file and add ssh-keyscan both slave machines in to it

The image shows a screenshot of the AWS CloudShell interface. At the top, there is a navigation bar with the AWS logo, a search bar, and a button labeled 'Ask Amazon Q'. Below the navigation bar, the terminal window displays the following commands and output:

```
AWS Console Home b63e:/# ssh-keyscan -H 13.218.160.72 > /var/jenkins_home/.ssh/known_hosts
root@2c288863b63e:/# cat /var/jenkins_home/.ssh/known_hosts
# 13.218.160.72:22 SSH-2.0-OpenSSH_8.7
|1|Nso/sd/5BZrdqbKaBzZTANC2nMo=|OjsqAGyvm5LeRIdlSfUXcLN1LPc= ecdsa-sha2-nistp256 AAAAE2V
zTzIkQTlmCISgLP94BNjw9Wwm5lIc8xZdpMvVHIue1B3j4/kB9IMgIcCz90lBo=
# 13.218.160.72:22 SSH-2.0-OpenSSH_8.7
# 13.218.160.72:22 SSH-2.0-OpenSSH_8.7
|1|UYad4YspIs8m9NJblVEDasAWWZE=|u+MCnoGSQelRe9LtEbBadSSWkiM= ssh-ed25519 AAAAC3NzaC1lZDI
# 13.218.160.72:22 SSH-2.0-OpenSSH_8.7
# 13.218.160.72:22 SSH-2.0-OpenSSH_8.7
root@2c288863b63e:/#
```

Below the terminal window, there is a section titled 'i-02b46c38bfbf019c3 (Jenkins-Docker)' with the following information: 'PublicIPs: 54.172.226.154 PrivateIPs: 172.31.0.39'.

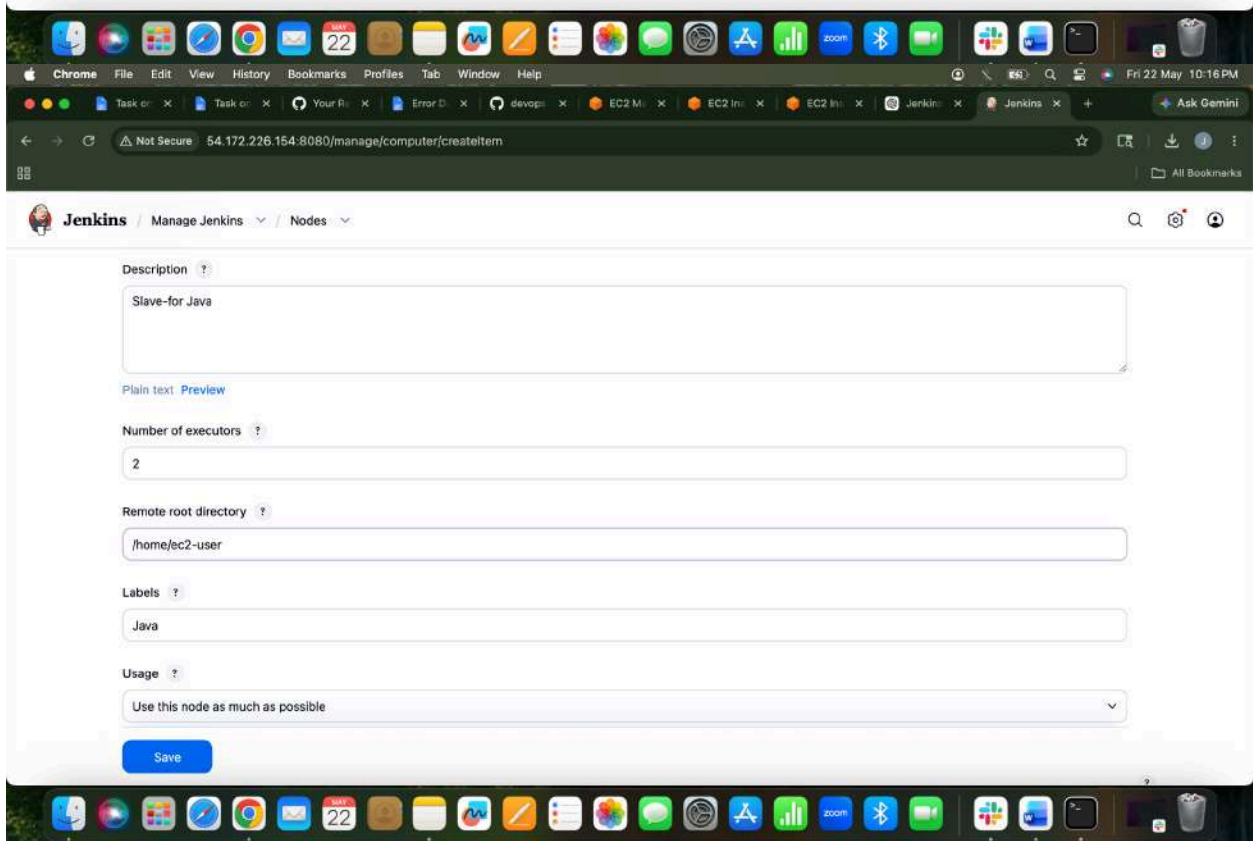
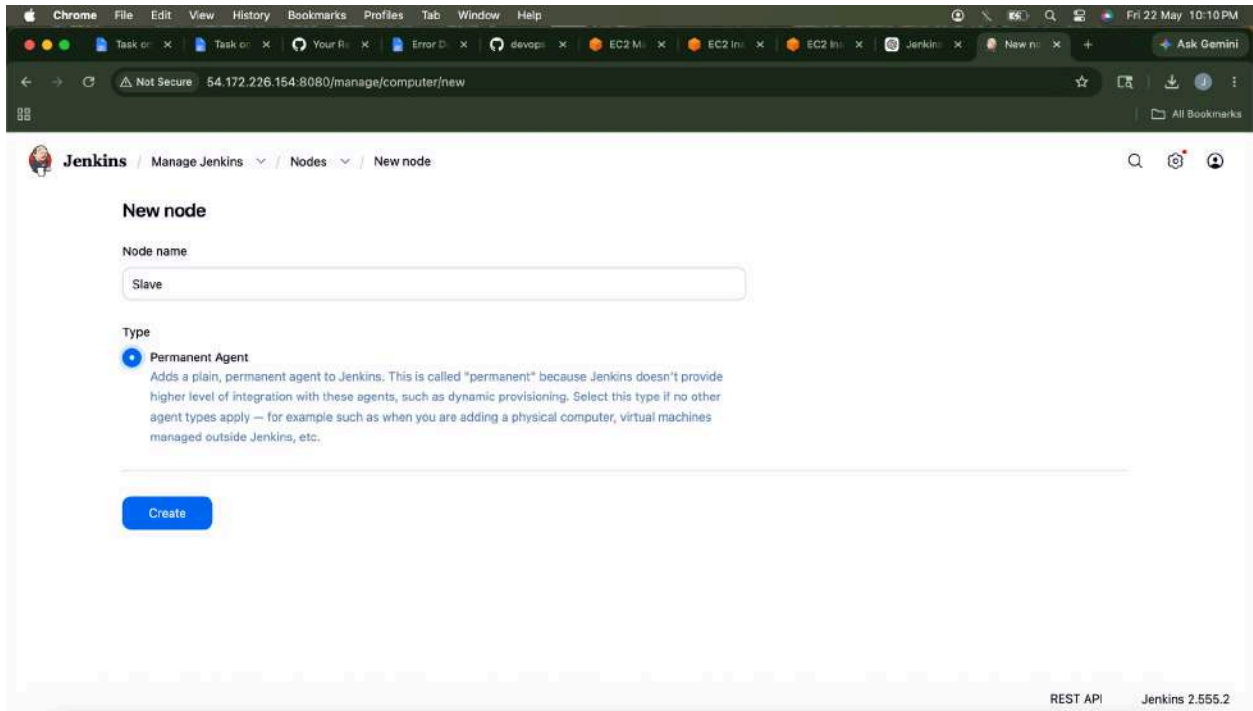
Now change owner and user using chown jenkins:jenkins, the reason for doing this is default users of jenkins will be jenkins and Give the necessary permissions to know host file

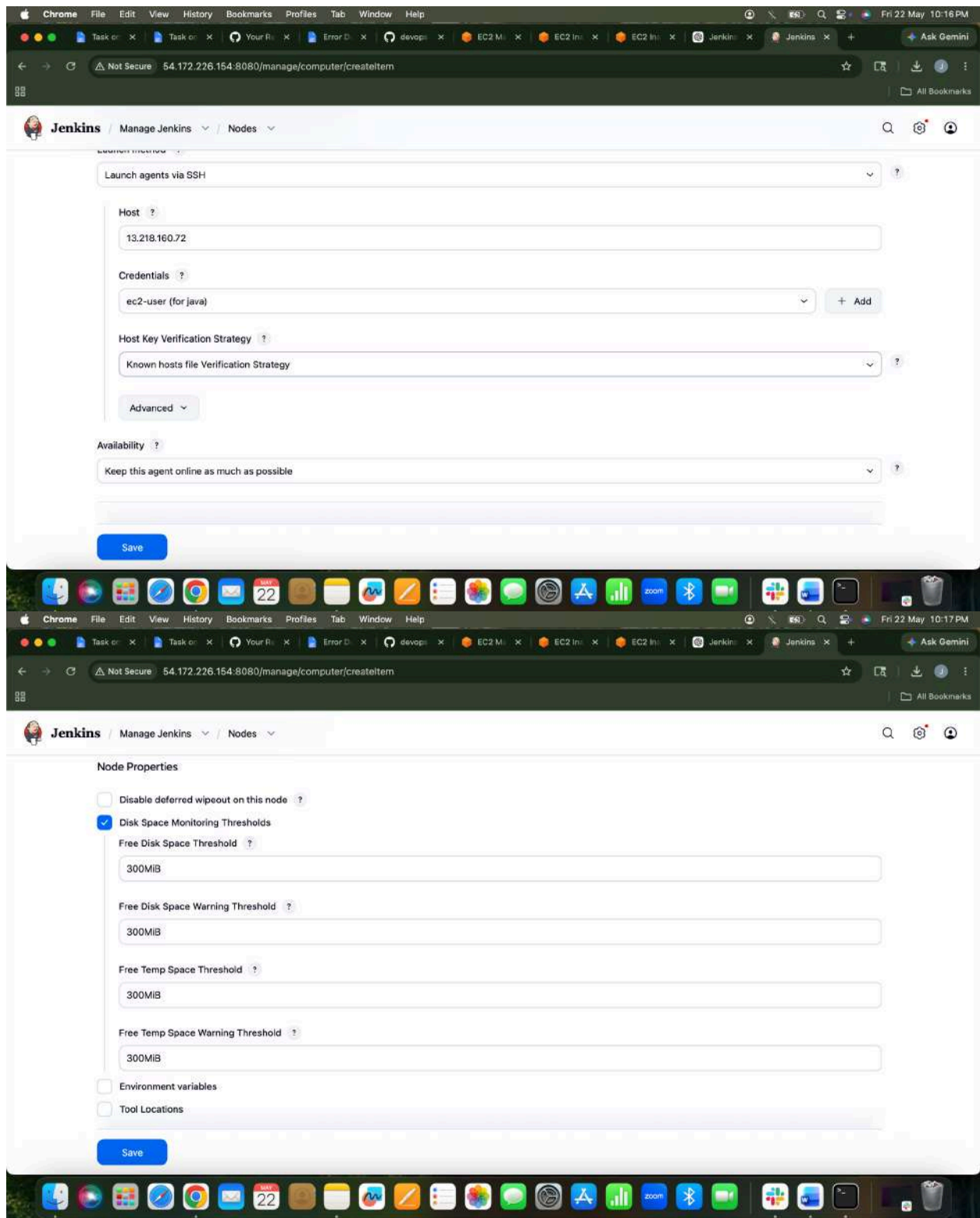

```
aws [Option+S] Ask Amazon Q
root@2c288863b63e:/# ssh-keyscan -H 13.218.160.72 > /var/jenkins_home/.ssh/known_hosts
root@2c288863b63e:/# cat /var/jenkins_home/.ssh/known_hosts
# 13.218.160.72:22 SSH-2.0-OpenSSH_8.7
|1|Nso/sd/5BZrdqbKaBz2TANC2nMo=|OjsqAGyvm5LeRIdlSfUXcLN1LPc= ecdsa-sha2-nistp256 AAAAE2Vj2HNhLX
zTzIkQTlmCISgLP94BNjw9Wwm51Ic8xzdpMvVHIuelB3j4/kB9IMgIcCz901Bo=
# 13.218.160.72:22 SSH-2.0-OpenSSH_8.7
# 13.218.160.72:22 SSH-2.0-OpenSSH_8.7
|1|UYad4YspIs8m9NJblVEDasAWWZE=|u+MCnoGSQelRe9LtEbBadSSWkiM= ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AA
# 13.218.160.72:22 SSH-2.0-OpenSSH_8.7
# 13.218.160.72:22 SSH-2.0-OpenSSH_8.7
root@2c288863b63e:/# cd /var/jenkins_home/.ssh/known_hosts
bash: cd: /var/jenkins_home/.ssh/known_hosts: Not a directory
root@2c288863b63e:/# cd /var/jenkins_home/.ssh
root@2c288863b63e:/var/jenkins_home/.ssh# ls -ltr
total 4
-rwx-----. 1 jenkins jenkins 559 May 22 16:58 known_hosts
root@2c288863b63e:/var/jenkins_home/.ssh#
```

i-02b46c38bfbf019c3 (Jenkins-Docker)
PublicIPs: 54.172.226.154 PrivateIPs: 172.31.0.39

CloudShell Feedback Console Mobile App

Now lets create a Node⇒ Here things to be noted ⇒root directory path should be on the **slave (agent) machine, not the master**.Jenkins, when you connect to the slave EC2, use this folder to store workspace files, builds, and agent data. Host address⇒ Slave IP address





Run the node if everything configured successfully then it will be success

```
<===[JENKINS REMOTING CAPACITY]===>channel started
```

```
Remoting version: 3355.v388858a_47b_33
```

```
Launcher: SSHLauncher
```

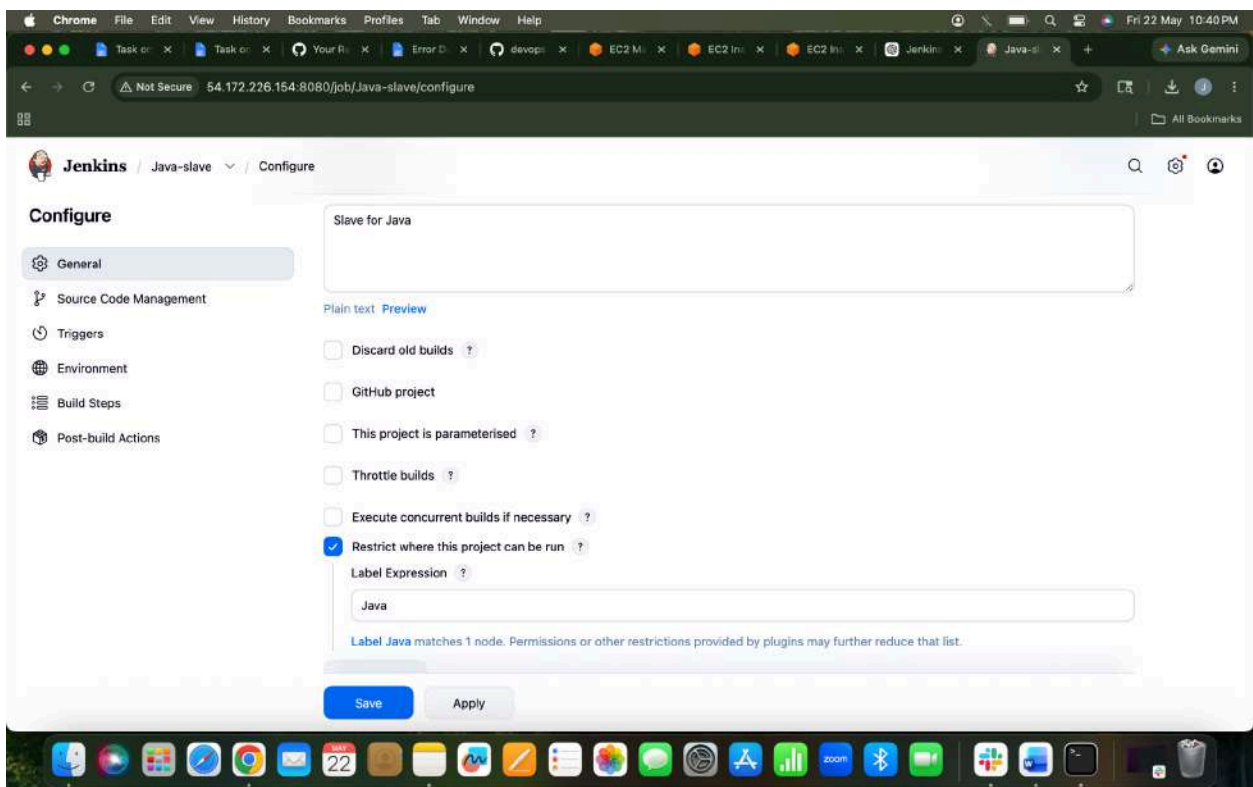
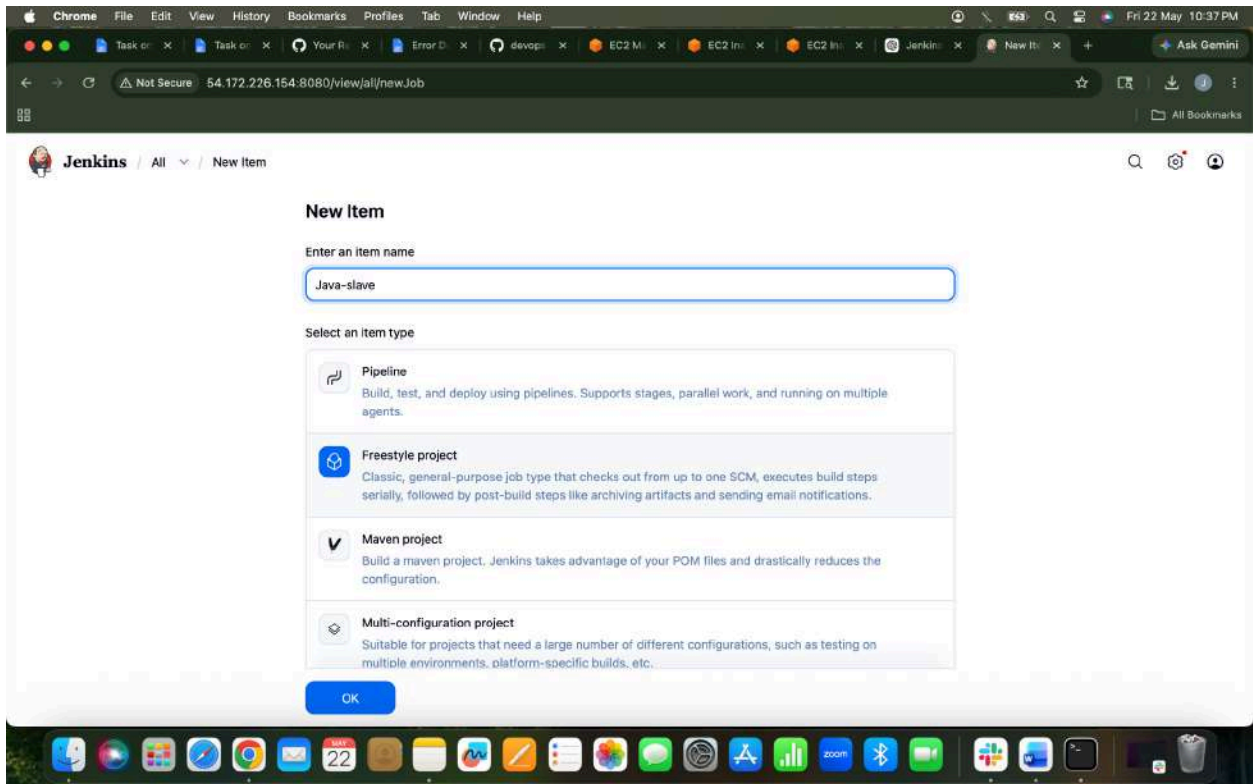
```
Communication Protocol: Standard in/out
```

```
This is a Unix agent
```

```
Agent successfully connected and online
```



Now lets create a new job and check if the node we create is working or not.



The newly created job ran successfully on the slave we created.



Jenkins Official Documentation

Recommended

- Pipeline: <https://www.jenkins.io/doc/book/pipeline/>
- Environment Variables: <https://www.jenkins.io/doc/book/pipeline/jenkinsfile/#using-environment-variables>
- Parameters: <https://www.jenkins.io/doc/book/pipeline/syntax/#parameters>
- Handling credentials in Pipeline: <https://www.jenkins.io/doc/book/pipeline/jenkinsfile/#handling-credentials>